

Aspect-Based Sentiment Analysis of Code-Switched Consumer Reviews for Local Coffee and Matcha Businesses in Davao City

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A.Z.F.
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Abstract

Davao City's expanding coffee and matcha market has produced a massive volume of online reviews. Analyzing this feedback is notoriously difficult because local consumers constantly switch between Taglish and Bislish. Standard sentiment tools like English lexicons struggle to process these regional syntax quirks and miss mixed opinions entirely. To address this, this study developed an Aspect-Based Sentiment Analysis pipeline using a fine-tuned XLM-RoBERTa model to accurately classify aspect-level sentiments.

Data collection yielded 19,586 raw Google Maps reviews across 400 local establishments using a custom DOM scraper. Filtering out rating-only entries resulted in 12,571 usable text instances. Taxonomy induction via Gemini 3 categorized the feedback into six primary dimensions: Product Quality, Customer Service, Ambiance, Facilities, Operations, and Price. Decomposing the reviews into sentence-aspect pairs revealed severe class imbalances. Combining active learning with targeted synthetic data generation addressed these deficits while maintaining a manageable manual annotation workload.

Human annotators labeled 500 informative instances, achieving substantial inter-rater reliability with an initial $\kappa = 0.6550$ on the seed batch, which improved to $\kappa = 0.7428$ during active learning. These authentic records were merged with 214 TF-IDF collision-filtered synthetic instances. Synthetic representation was strictly capped at 29.97% to prevent model collapse, establishing a 714-row training baseline. By freezing the lower encoder layers and applying inverse class weights, the fine-tuned model achieved 79.75% accuracy and a 0.5967 Macro-F1 score on a 400-instance out-of-sample test set. This architecture significantly outperformed both a VADER lexicon baseline and a tuned TF-IDF Support Vector Machine. McNemar's test confirmed the statistical significance of these gains ($p = 2.2967 \times 10^{-5}$ against VADER; $p = 1.1524 \times 10^{-3}$ against SVM).

While the model effectively isolates conflicting sentiments within individual reviews, parsing regional Bisaya double negatives, blunt short complaints, and sparse neutral observations remain persistent challenges. Across the broader corpus, Product Quality dominated consumer discussions (47.88%), and Ambiance earned the highest overall satisfaction (4.46 mean rating). Customer Service proved to be the most polarizing dimension (4.00 mean rating, SD = 1.444). Integrating these granular predictions into a directory-style interface replaces vague aggregate star ratings with aspect-specific satisfaction scores, providing actionable market intelligence for local establishments and consumers.

Keywords: Aspect-Based Sentiment Analysis (ABSA), Code-Switching, XLM-RoBERTa, Active Learning, Synthetic Data Generation, Consumer Reviews, Natural

Language Processing.

Chapter 1

Introduction

1.1 Background of the Study

Cafe culture, particularly matcha and coffee shop hopping, became increasingly popular in the Philippines. This trend led to a significant rise in demand for Matcha and Coffee, prompting multiple shops to open in the city. In urban areas such as Davao City, this rapid increase has led to a highly competitive and saturated market [20]. Consequently, consumers are given a hard time efficiently comparing cafés. This concern becomes even more significant as people from different parts of the country travel to Davao City specifically to explore its growing coffee and matcha scene.

Despite the availability of online reviews on platforms such as Google Maps, Facebook, and Instagram, these reviews are often unstructured, not normalized, fragmented, and overwhelming in volume. The presence of many positive reviews and high rankings for a business or its products may make it more likely that a potential consumer will feel confident about making a purchase [29]. However, the large volume of reviews requires consumers to manually sift through them, making decision-making time-consuming and inefficient. Furthermore, some of these reviews may contain biased, misleading, or even irrelevant information, which can negatively affect consumer choices and business reputations [30].

This major limitation stems from the inability of existing systems to effectively organize, analyze, and interpret large volumes of unstructured review data. Although thousands of reviews are available across platforms, consumers must manually read them in raw, unstructured text, which is both time-consuming and inefficient. Moreover, this issue is further complicated by code-switching, informal language, and inconsistent expressions, making it difficult, and in some cases impossible, for traditional analysis methods to accurately capture sentiment and meaning [37][31].

To address these challenges, prior studies have used automated data collection methods, such as Application Programming Interfaces (APIs), to convert raw reviews into structured datasets for analysis, thereby improving scalability and data reliability [40]. Sentiment analysis, on the other hand, relied on lexicon-based techniques such as

VADER, most especially in the earlier years [8]. Although it was useful in English-written texts, the techniques were found to be limited in handling multilingual and code-switched texts commonly found in Philippine contexts. As a result, more advanced learning-based models, particularly multilingual transformer models such as XLM-RoBERTa, were introduced to better interpret complex language patterns and improve classification performance [37]. Multiple studies have also utilized aspect-level analysis through Aspect-based Sentiment Analysis (ABSA) to extract more detailed insights by evaluating specific attributes [58]. Extensive manual annotation was also greatly reduced by using hybrid strategies, including Active learning, which has been found to improve performance [62]. The aforementioned approaches offer several advantages for addressing the limitations of traditional text and review analysis, particularly when integrated into a unified process. While standard Application Programming Interfaces (APIs) offer structured data, they often impose severe extraction limits. Thus, customizing an automated web scraping process ensures greater scalability and sufficient data volume when handling large datasets.

Furthermore, the application of Aspect-Based Sentiment Analysis (ABSA) enables the decomposition of reviews into distinct attributes, yielding more precise insights. To improve efficiency and reduce reliance on large labeled datasets, techniques such as Active Learning prioritize the most informative samples for annotation, significantly reducing labeling effort while maintaining performance [44]. Lastly, data augmentation strategies, such as regulated synthetic data generation, help address issues of data scarcity and class imbalance [22]. By using these approaches, a framework is created that transforms unstructured, multilingual, and context-rich texts into more meaningful insights.

Though the Davao City cafe and matcha shop scene is rapidly expanding, there is a distinct lack of research utilizing advanced Natural Language Processing (NLP) techniques to evaluate consumer preferences. The study fills the gap by implementing XLM-RoBERTa a Aspect-Based Sentiment Analysis (ABSA) framework, enhanced by Active Learning, specifically for Taglish and Bislish reviews.

1.2 Problem Statement

This study investigates the challenge of effectively analyzing unstructured, multilingual, and context-rich consumer reviews, specifically those that contain code-switching and informal language. Existing platforms generate large volumes of raw, unstructured textual data that is challenging to interpret using traditional methods, hindering the extraction of meaningful, reliable results. Specifically, this study aims to answer the following questions:

1. What aspect categories characterize the multilingual consumer reviews of local coffee and matcha businesses in Davao City, and how does the code-switched nature of the text affect aspect-level sentiment classification?
2. How does the proposed Multilingual Transformer model perform compared to lexicon-based approaches on code-switched and informal review texts?
3. What are the aspect categories that emerged from the consumer reviews of Davao City Coffee and matcha establishments, and what are the prevailing consumer sentiments associated with these discovered dimensions?

4. How can active learning and synthetic data generation improve model performance and reduce annotation effort?
5. How can the proposed model be evaluated and integrated into a system that supports consumer decision-making and business analysis?

1.3 Objectives

The main objective of this study is to develop an XLM-RoBERTa-based system for aspect-based sentiment analysis that efficiently handles multilingual and code-switched consumer reviews. The study also wishes to accomplish the following objectives:

1. To identify the aspect categories present in multilingual consumer reviews of local coffee and matcha businesses in Davao City and examine how code-switching affects aspect-level sentiment classification.
2. To compare the performance of the proposed XLM-RoBERTa-based model against lexicon-based approaches on code-switched and informal review texts.
3. To identify the aspect categories that emerged from consumer reviews of Davao City coffee and matcha establishments and determine the prevailing consumer sentiments associated with each discovered dimension.
4. To implement active learning strategies and synthetic data generation to reduce annotation effort while maintaining classification accuracy.
5. To evaluate the proposed model and integrate it into a system that supports consumer decision-making and business analysis.

1.4 Significance

This study presents significant contributions to various stakeholders by addressing the challenge of analyzing large volumes of unstructured, multilingual, and context-rich consumer reviews.

1. **For Consumers:** The rapid growth of cafe culture in Davao City has led to a highly competitive and saturated market [20]. Consequently, consumers face difficulties in efficiently comparing different establishments. An Aspect-Based Sentiment Analysis (ABSA) system alleviates the burden of manually processing thousands of reviews. Rather than reading extensive texts, consumers are provided with concise, data-driven summaries of specific aspects they find most valuable.
2. **For Local Coffee and Matcha Businesses:** Aggregated star ratings often fail to provide actionable insights. For instance, a single review may contain positive sentiments regarding a product's taste but negative sentiments toward the establishment's customer service or ambiance. Decomposing reviews into specific features enables businesses to accurately pinpoint operational strengths and identify specific areas requiring improvement.

3. **For Academic Research:** This study fine-tunes XLM-RoBERTa on Taglish and Bislish, code-switched dialects prevalent in the Philippines. Data scarcity constrains model performance; this work employs two complementary techniques to mitigate it. Active Learning identifies the most informative unlabeled examples for manual annotation based on model uncertainty [62]. Synthetic data generation then augments the training set, reducing the annotation burden while maintaining dataset quality.

1.5 Scope and Limitations

This study evaluates consumer feedback for coffee and matcha shops across Davao City. To build a robust dataset, every available public review from establishment inception up to the scraping date was collected without imposing arbitrary volume caps. Because the Google Places API restricts bulk extraction, a custom HTML scraper secured the necessary raw text volume, saving the output into separate CSV files for coffee and matcha venues. The corpus is inherently messy, consisting of multilingual code-switching across English, Tagalog, and Bisaya. An XLM-RoBERTa model handles aspect-based sentiment classification across predefined categories, backed by active learning to surface uncertain samples for labeling and LLM-driven synthetic generation to pad underrepresented classes. Risks tied to synthetic data, including linguistic homogenization and model collapse, are mitigated by strictly capping synthetic records below the 30% stabilization threshold and applying a TF-IDF cosine similarity filter (≥ 0.85) to preserve lexical diversity, rather than running recursive generative simulations. Synthetic inputs only supplement sparse classes, leaving human annotations as the core training foundation. Web scraping introduced standard technical hurdles, including dynamic DOM updates and IP throttling. Finally, while the final model feeds into an integrated platform, the underlying machine learning pipeline processes text exclusively, ignoring attached review photos and videos.

1.6 Definition of Terms

1. **Active Learning (AL):** A machine learning framework that reduces manual annotation effort by selecting only the most informative or uncertain samples for labeling, rather than processing an entire dataset.
2. **Application Programming Interface (API):** A set of protocols that automates the extraction of large volumes of online consumer reviews into structured formats, such as CSV files, for further analysis.
3. **Aspect-Based Sentiment Analysis (ABSA):** A natural language processing technique that identifies specific attributes of a subject in text, such as taste, price, service, and ambiance, and determines the sentiment expressed toward each.
4. **Code-Switching:** The practice of alternating between two or more languages within a single text, as seen in "Taglish" (Tagalog-English) and "Bislish" (Bisaya-English), the dominant mixed-language variants in the dataset.

5. **Lexicon-Based Approaches:** Sentiment analysis methods that classify text using predefined word lists where each entry carries a positive or negative polarity score, with VADER as a representative example.
6. **Model Collapse:** A sustained decline in model performance caused by training predominantly on synthetic rather than human-generated text, resulting in reduced linguistic diversity over successive iterations.
7. **Multilingual BERT (XLM-RoBERTa):** A pre-trained transformer model that processes text across multiple languages in context, capable of interpreting code-mixed reviews written in Filipino dialects and English.
8. **Synthetic Data Generation:** The process of producing artificial training examples from a small seed set of real reviews using large language models, applied to address data scarcity and class imbalance.
9. **Unstructured Data:** Raw text lacking a predefined schema or organization, such as user-generated online reviews, which resists systematic processing without specialized tools.
10. **Web Scraping:** An automated technique that extracts data from websites by parsing HTML to gather raw online reviews at scale.
11. **Fleiss' Kappa (κ):** A statistical measure used to evaluate inter-rater reliability and determine labeling consistency when annotations involve three or more independent raters [19].
12. **Macro-F1 Score:** The primary evaluation metric for the imbalanced dataset, which computes the F1-score independently for each class and calculates the unweighted mean to prevent the model from defaulting to the dominant class.
13. **Natural Language Inference (NLI):** A task structure that reformulates aspect extraction by pairing review sentences with candidate aspect-sentiment expressions to better capture contextual relationships.
14. **Support Vector Machine (SVM):** A traditional machine learning algorithm utilized alongside TF-IDF feature extraction as a baseline aspect-conditioned ternary classifier to compare against the fine-tuned transformer model.
15. **Term Frequency-Inverse Document Frequency (TF-IDF):** A vectorization method used to convert text into sparse feature vectors, allowing the system to mathematically measure lexical overlap and discard redundant synthetic instances.
16. **Zero-Shot Learning:** A transfer learning approach in which a pre-trained model classifies unseen data without requiring domain-specific fine-tuning.

Chapter 2

Review of Related Literature

2.1 Local Coffee and Matcha Market

The rapid increase of cafe culture in the Philippines has significantly transformed local beverage shops [20]. Urban cities like Davao City have experienced exponential growth in coffee and matcha, creating a highly saturated, competitive environment. Local consumers, particularly the Gen Z demographic, are a key driver of this market.

1. *Market Competition:* The local coffee shop industry in Davao City has risen, evolving into a competitive service environment. When new establishments open, the market is quick to absorb them. However, sustaining the business requires distinct differentiations. Local cafe owners are leveraging locally sourced products (Mt. Apo coffee and award-winning Davao cacao) to establish their own unique market. This stimulates demand for both local and national brands and local competitors [3].

2. *Influencers:* Consumer decisions in Davao City are dependent on digital social proof [20] [4]. Local consumers usually conduct online research before making purchasing decisions. Hence, consumer choices are heavily influenced by online reviews, social media recommendations, and peer opinions, making reputation management and digital influence necessary for market success in the region.

3. *Digital Engagement:* Gabato et al. [20] investigated the preferred digital marketing approaches among Gen Z college students in Davao City. They approached the problem by understanding how different digital attributes drive consumer interaction. Using a conjoint analysis approach through the PAPRIKA method, their study assessed attributes including social media, publicity types, and content types. They demonstrated that in the specific context of Davao City's Gen Z student market, content type (26.7%) and platform choice (24.7%) are the most influential factors. This proves that this demographic favors engaging, interactive content over traditional online advertisements [20]

4. *Consumer Decision-Making:* Gen Z consumers in Davao City often consider their experience and the physical ambiance of a coffee shop before they return. According to a study by Alcraz et al. [4], they modeled consumer behavior using the "Theory

of Planned Behavior" to assess Gen Z tourists' attitudes toward local coffee shops. They found a correlation between experiences and visit intention (0.693) and a correlation of (0.614) between attitudes and visit intention [4].

2.2 Text Analysis of Consumer Reviews

Consumer reviews have become a primary source of public sentiment in digital markets, and processing them effectively requires both categorizing the type of feedback being expressed and understanding its structure [34] [24]. Lehto et al. [34] addressed this by analyzing unstructured consumer feedback from virtual travel agencies and categorizing reviews into three categories (complaints, compliments, and satisfaction). Classifying text along these polarities enables businesses to isolate specific problem areas and convert large volumes of feedback into actionable insights for managing consumer loyalty.

Structuring the text itself requires further decomposition. Herhausen et al. [24]noted that business researchers often lack coherent frameworks for processing digital stakeholder communication, and proposed applying a message design logic lens to break reviews into three components (background, situation, and intention). Organizing unstructured language into these elements provides researchers with a replicable method for translating consumer-generated text into measurable insights [24].

However, the explosion of user-generated content (UGC) on platforms like Google Maps has turned local market analysis into a massive "Big Data" challenge. While consumer reviews provide valuable feedback, the huge volume of unstructured text renders manual analysis employing traditional qualitative methods impractical [66]. Almuayqil et al. (2022) [5] confronted the challenge of extracting reliable sentiment from large bodies of redundant and inconsistent review data. Pairing sentiment analysis algorithms with a preprocessing pipeline that included tokenization, lowercasing, and punctuation removal, they found that rigorous data normalization substantially improved a classifier's ability to handle noisy text and detect sentiment polarity accurately [5].

Extracting meaningful insights from thousands of Davao City coffee and matcha shop reviews cannot rely solely on manual reading [30]. Natural language at scale also introduces problems of sparsity, dimensionality, and ambiguity that rule-based models and simple frequency counts are ill-equipped to handle [2]. Automated preprocessing pipelines paired with advanced sentiment analysis are, therefore, necessary to produce reliable, localized findings from this volume of consumer-generated [49] [37].

Relying purely on raw star ratings introduces a major statistical vulnerability. Inherent behavioral biases heavily skew the data. Hu et al. [26] identified this phenomenon as the "J-shaped distribution" of online product reviews. Consumer ratings rarely follow a normal distribution. They heavily skew toward extreme positive 5-star endorsements. A smaller secondary peak appears at extreme negative 1-star complaints. Moderate or neutral reviews remain severely underrepresented.

Taking a raw sample from scraped platforms guarantees a dataset dominated by positive sentiment. This imbalance actively prevents machine learning classifiers from accurately learning the boundaries of minority classes. This study uses stratified sampling

to counteract the inherent skew in the Davao City corpus. The pipeline enforces strict stratified sampling across both target aspects and proxy ratings. This structural control ensures minority sentiments are sufficiently preserved for transformer fine-tuning.

2.3 Code-Switching

Code-switching presents consistent uses in examining reviews from the Philippines. This refers to how writers or speakers switch between English, Tagalog, and other regional languages within the same text. [46] [14]. Mixed-language varieties such as Taglish (Tagalog-English) and Bislish (Bisaya-English) present particular difficulties for NLP models trained on monolingual English corpora [25]. A Davao-based review, for instance, might use the Bisaya term *lami* (meaning "delicious" [70]) alongside English or Tagalog commentary about service. A standard English lexicon would register no positive sentiment in such a sentence at all [46].

There are other issues where code-switching involves the simultaneous blending of distinct grammatical rules and syntactic structures, which standard computational models are not designed to handle [46] [68]. Patil et al. [49] noted that most sentiment analysis tools in big data contexts are built around monolingual English, leaving them poorly suited for low-resource languages, particularly when multiple languages appear within a single post. Conventional feature extraction methods tend to treat hybridized or region-specific expressions as out-of-vocabulary errors, discarding the semantic content that gives those expressions their meaning.

Other works on Philippine consumer data confirm this. Cosme and De Leon [14] discovered that lexicon-based tools perform poorly on Filipino-English reviews because single-language systems cannot reliably determine polarity when inputs shift between languages mid-sentence. Ang Ngo Ching et al. [8] reached a similar conclusion. Observing that English-only dictionaries lack the syntactic coverage necessary to process Taglish text without significant accuracy loss.

Through these limitations, techniques that can tolerate inconsistencies in grammar, vocabulary, and context across languages are essential. Maceda et al. [37] make this point in the context of code-mixed social data, where the input rarely conforms to the clean, monolingual text that traditional models assume. Multilingual transformers are suited to this environment, as their shared subword vocabularies and self-attention mechanisms enable fine-tuning on localized, code-mixed corpora, enabling sentiment classification across blended language boundaries [68] [37].

2.4 Lexicon-Based Approaches

Sentiment analysis has long relied on tools such as VADER, developed by Hutto and Gilbert in 2014 [28]. The model combines a curated lexicon with grammatical rules for negation and degree modifiers, and it was designed specifically for the messiness of social media: slang, emojis, and punctuation stacking. On standard English data, it performs well, achieving an F1-score of 0.96 [28]. That performance held in practice as

Mathayomchan and Taecharungroj (2020) [40] used VADER to analyze 935,000 Google Maps reviews from 5,010 UK restaurants. Mapping lexical signals across food, service, atmosphere, and value, they found that food quality is the main driver of five-star ratings, and poor service accounts for most one and two-star reviews, which is clean and useful for restaurant owners.

The limitations of this approach become apparent, however, when the input deviates from standard English. Ang Ngo Ching et al. [8] examined this problem in the context of Philippine online reviews, where Filipino-English code-switching and code-mixing are pervasive features of written discourse rather than exceptional occurrences. The researchers evaluated six sentiment analysis tools, including VADER, against an annotated dataset of 1,000 reviews sourced from Shopee and Google Maps, assessing performance at both the sentence and review levels. VADER recorded a macro-F1 score of 0.30, representing a 35% decline from its performance on standard English text. This is due to the model's English-only lexicon, which lacks the vocabulary and syntactic coverage necessary to process Taglish, the hybrid register in which Filipino speakers routinely alternate between Filipino and English [8]. The findings indicate that lexicon-based methods are insufficient as primary classification tools for multilingual, code-switched text in localized contexts such as the Philippines [8].

Furthermore, when compared to learning-based models, lexicon methods are particularly ineffective for highly localized data, such as consumer reviews in Davao City. Standard English lexicons completely miss positive sentiment when encountering unique regional syntax and Bisaya terms, such as *lami* (Bisaya adjective meaning delicious [70]). Ortega [46]addressed this linguistic barrier by developing a neural network-based sentiment analyzer trained specifically on Bisaya-language text. The study demonstrated that both Bisaya and mixed Tagalog-English syntax consistently avoid standard dictionary mappings, leading to severe misclassification of consumer opinions when processed with conventional lexicon-based tools [46]. These findings reinforce a conclusion consistent with prior literature. While lexicon-based algorithms such as VADER offer a useful baseline, their dependence on monolingual English dictionaries makes them inadequate for text characterized by regional code-switching. This limitation, specifically the failure of English lexicons to account for Bisaya and Taglish syntax, is the reason this study favors learning-based models over traditional approaches.

2.5 Learning-Based Approaches

2.5.1 Machine Learning

Where lexicon-based approaches fail against ambiguity and context dependency, machine learning methods offer greater adaptability. Pratmanto et al. [53] applied Naive Bayes classification to Indonesian e-commerce reviews scraped from the Google Play Store, focusing on user feedback for the Shopee platform. Before classification, the text data was preprocessed through tokenization, stop-word removal, and stemming, then partitioned into training and testing subsets. The model achieved 96.67% accuracy, indicating that probabilistic classifiers can handle large volumes of user-generated reviews without the

need for manually constructed sentiment lexicons [53].

Pakpahan et al. [47] addressed the challenge of granular, aspect-level analysis by evaluating four machine learning algorithms, which are Naive Bayes, SVM, Random Forest, and K-Nearest Neighbors, on 20,000 Indonesian reviews from Shopee and Tokopedia. Word2Vec was used for feature extraction to capture semantic relationships between terms. Of the four models, SVM performed best, reaching 91% accuracy in aspect extraction and 90% in sentiment polarity classification [47].

Ulhaq and Suprayogi [64] took a similar comparative approach, applying SVM, Random Forest, and Multilayer Perceptron (MLP) architectures to over 10,000 preprocessed Tokopedia reviews vectorized using TF-IDF. The MLP model achieved the highest accuracy of 87.21%, narrowly surpassing SVM, indicating that neural classifiers offer a modest but consistent advantage over traditional methods on preprocessed consumer text [64].

However, while traditional machine learning algorithms demonstrate high performance on standardized datasets, their reliance on static feature extraction methods (such as TF-IDF) limits their ability to capture deep sequential context and complex semantic relationships [49]. Since these traditional methods treat text as isolated tokens or static vectors, they lack the representational capacity required to process severe linguistic complexity and code-switching effectively [68]. Consequently, these traditional architectures are insufficient for handling the highly nuanced, code-switched syntax (Taglish and Bislish) present in localized Davao City reviews. This critical limitation in processing dynamic, mixed-language structures naturally necessitates the deployment of advanced learning-based approaches, such as Multilingual Transformers, which are explicitly designed to capture these complex contextual relationships [37].

2.5.2 Multilingual Transformers

Where lexicon-based and traditional machine learning methods fall short in handling linguistic complexity, the literature points toward deep learning as the more capable alternative. Multilingual transformer models, particularly mBERT [15] and XLM-RoBERTa [13], have gained considerable traction in code-switched text analysis. Their effectiveness stems from transfer learning in both models, which are pre-trained on large multilingual corpora [68] [49], which allows them to recognize sentiment across mixed-language patterns such as Taglish and Bislish before being adapted to a specific classification task [37].

The core difficulty in processing code-switched text is that most NLP models are trained on monolingual data, leaving them poorly equipped for sentences that blend two or more languages within a single utterance. Winata et al. [68] examined this problem by evaluating mBERT and XLM-RoBERTa across several code-mixed datasets. Their findings confirmed that while both models carry strong cross-lingual capabilities through shared subword vocabularies and self-attention mechanisms, fine-tuning on task-specific, code-switched data produces substantial gains in classification performance, establishing these architectures as well-suited backbones for mixed-language tasks [68].

The necessity of localized fine-tuning was directly taken up in the Philippine context by Maceda et al. [37], who analyzed public sentiment on higher education policy using social media data rich in Filipino-English code-mixing. They adapted mBERT's pre-trained linguistic knowledge to the syntactic patterns of their Taglish corpus, ultimately achieving an F1-score of 81.14% and outperforming Support Vector Machine baselines by a considerable margin [37]. The result reinforced the practical value of domain-specific fine-tuning when the target language environment departs substantially from standard monolingual text.

Furthermore, the sheer scale and variability of social media data present obstacles. Patil et al. [49] assessed sentiment analysis frameworks against unstructured data from social media platforms, where textual noise, informal jargon, and inconsistent grammar routinely disrupt conventional feature extraction. Their comparative evaluation found that transformer-based models consistently outperformed traditional machine learning methods in accuracy and robustness across datasets with high linguistic variability [49].

Salve and Tubil [55] demonstrated that XLM-RoBERTa performs well on Southeast Asian language contexts, specifically in handling the kind of rapid language switching and non-standard syntax that appears in bilingual, low-resource settings. In the Philippines, consumer-generated text mixes Tagalog and English in which creates problems for most natural language processing models. Informal spellings, slang, and unstable sentence structures produce ambiguities that some models struggle to resolve. To address this, Salve and Tubil [55] fine-tuned XLM-RoBERTa Large on 10,000 Taglish reviews, using a custom SentencePiece tokenizer and explicit ambiguity tagging at both the token and sentence levels. The model outperformed both its Base variant and mBERT, reaching a testing precision of 97.62%, an F1-score of 95.77%, and an ROC-AUC of 97.50%. These results suggest that XLM-RoBERTa's pre-training across more than 100 languages gives it a meaningful advantage when classifying mixed-language text, which is directly relevant to the code-switched review data collected for this study [55].

These considerations bear directly on the analytical demands of this study. Consumer reviews from Davao City coffee and matcha shops are written in Bisaya-English and Tagalog-English [46] [20]. These varieties use region-specific vocabulary, such as *lami* (a Bisaya adjective meaning "delicious" [70]), that standard lexicon-based tools are unable to interpret. Thus, while this research is theoretically grounded in the Aspect-Based Sentiment Analysis (ABSA) framework utilized by Maceda et al. [37]. The study will implement XLM-RoBERTa as the primary classification engine. By substituting the baseline mBERT architecture with XLM-RoBERTa, the study retains the proven ABSA methodology for localized Philippine texts while significantly optimizing the model's ability to process the complex, informal code-switching of Davao City consumers, as supported by the recent performance benchmarks of Salve and Tubil [55].

Multilingual transformers offer superior representational capacity. Adapting these massive architectures to small, specialized domains introduces severe risks. The models face catastrophic forgetting and rapid overfitting. Conneau et al. [13] and Malik et al. [38] demonstrated exactly how fully fine-tuning a transformer on limited datasets overwrites its pre-trained linguistic knowledge. The architecture memorizes the training noise rather than actually generalizing.

To negate the instability, Loshchilov and Hutter [36] introduced AdamW. Their work proved that decoupling weight decay from the gradient update effectively penalizes extreme parameter drift during optimization. Regularization alone remains insufficient for models containing hundreds of millions of parameters. Malik et al. [38] found a structural solution. Physically freezing the lower encoder layers preserves the foundational cross-lingual representations. These specific layers process universal syntactic structures. Locking them strictly limits the trainable parameter space.

This study applies these exact mechanics to the localized Davao City corpus. The pipeline freezes the bottom six encoder layers and applies decoupled weight decay to the XLM-RoBERTa architecture. This configuration ensures the model adapts to the specific semantic nuances of Bislish and Taglish consumer reviews. The combined constraints physically prevent the network from collapsing under the sparsity of a highly constrained training set.

2.6 Web Scraping

Web scraping refers to the automated extraction of unstructured data from websites and its conversion into machine-readable formats [63][12]. In sentiment analysis research, it enables large-scale retrieval of user-generated content, such as consumer reviews, that would otherwise be impractical to collect manually [12][17].

Digital data collection methods generally fall into three categories: manual gathering, direct API access, and custom HTML or DOM scraping. API access is preferred where available, as it returns clean, structured data. Mathayomchan and Taecharungroj [40], for instance, drew on Google Maps data to analyze customer experiences across 5,000 businesses, extracting over 935,000 reviews to measure attributes such as food quality, service, atmosphere, and value. However, it has become increasingly difficult to replicate through official channels [40].

Official APIs have grown substantially more restrictive. Chen et al. [12] documented this shift, comparing eight data collection methods and finding that while APIs are straightforward to use, their export limits and associated costs make them impractical for large-scale research. Custom scraping scripts, by contrast, were able to gather sufficient data volume without those constraints [12].

This broader restriction has been described as the "APIcalypse" by Trezza [63]. Examined how platforms banning or severely capping free API access have disrupted large-scale digital research, finding through a survey of digital researchers that API limitations have pushed much of the field toward web scraping as the primary means of maintaining data volume and research viability.

Dongo et al. [17] compared official APIs against custom HTML scraping for real-time social media data collection and found that scraping was considerably faster and more flexible for high-volume retrieval. Crucially, when the scraped text was properly normalized, the two methods produced equivalent analytical results, suggesting that data quality need not be sacrificed for scale [17].

In the context of this study, the standard Google Places API limits extraction to 5 reviews per business, far below the volume needed to fine-tune a deep learning model. Following the approaches documented by Chen et al. [12], Trezza [63], and Dongo et al. [17], this study employs custom HTML scraping to work around that ceiling. The textual noise inherent to scraped data will be addressed through normalization and the preprocessing pipeline described by Maceda et al. [37], ensuring the model receives sufficiently clean input for fine-tuning.

2.7 Synthetic Data Generation

Synthetic text generation creates artificial data using either code-generated data from algorithmic/mathematical models or Large Language Models (LLMs) trained on small seed datasets of real data [67][22]. To deal with the problem of data scarcity, researchers have increasingly looked to LLM driven generation as a workaround. Wang et al. [67] tackled the issue of language models being too dependent on human written instruction data, which is both expensive and limited in quantity and diversity. Their solution was the "Self-Instruct" framework, an iterative setup where a vanilla GPT-3 model was prompted to essentially bootstrap its own dataset by generating new instructions, identifying task types, and creating corresponding input-output instances. After filtering and using over 52,000 purely synthetic instances to fine-tune the model, they saw a 33% absolute performance improvement. That is a pretty significant result, and it effectively matched models trained on large private human datasets [67].

Scaling synthetic generation is one thing, but applying it to structured tasks like ABSA introduces its own set of problems. Monfared et al. [42] found that raw LLM augmentation often struggles to maintain label consistency, meaning the generated sentences frequently fail to preserve the specific aspect level annotations that were actually requested. Their proposed fix was an agentic data augmentation method that uses iterative generation and verification to produce higher quality synthetic training examples. Under a ReAct style architecture, a generator agent first extracted a style policy to produce candidate sentences, while a separate evaluator agent checked that each generated review actually contained the right aspect terms and correct sentiment polarities before it got added to the dataset. Compared to a matched-prompting baseline across three ABSA subtasks, the agentic approach clearly outperformed raw prompting in label preservation [42]. When combined with real data, it also consistently gave stronger performance gains, particularly for models like T5-Base, which ended up performing at a level more comparable to its more heavily pretrained counterparts [42].

However, efficiency and scalability is considered costly [22]. Relying too heavily on synthetic text introduces what is often called "model collapse," and it is a real concern. Guo et al. [22] looked directly at this risk, asking whether training models on data generated by their predecessors might permanently degrade linguistic richness. They simulated recursive fine-tuning experiments across various natural language generation tasks and applied specialized metrics to track how text structure changed over successive iterations. What they found was a consistent and significant decline in linguistic diversity, specifically in lexical, syntactic, and semantic richness. Models trained excessively on

synthetic data eventually converge to uniform behavior and lose the "tails" of the original distribution [22].

Compared to traditional manual annotation, synthetic data generation is the more scalable option for dealing with data scarcity [22] [42]. As Wang et al. [67] showed, synthetic generation can rapidly produce massive datasets from small seed samples. Still, to avoid the linguistic homogenization that Guo et al. [22] warned about, synthetic generation has to be carefully balanced with authentic data rather than used as a straight replacement.

To operationalize this balance and prevent generative model collapse, recent research has established exact mathematical thresholds for mixing authentic and synthetic data. He et al. [23] investigated this boundary within a recursive learning framework by evaluating customized weighted training schemes to optimally integrate real and synthetic batches. Their methodology spanned three distinct statistical scenarios—exact finite-sample calculations for Gaussian distribution estimation, asymptotic analysis for Generalized Linear Models (GLMs), and non-parametric discrete distribution approximations—which were subsequently validated using Monte Carlo simulations. The findings demonstrated that naive data mixing is strictly suboptimal. Crucially, they derived a unified analytical expression revealing that when mixing batches of equal size, the optimal weight assigned to real data is exactly the reciprocal of the golden ratio (approximately 61.8%). This dictates a theoretical ceiling, indicating that synthetic data influence must be kept below approximately 38.2% to prevent performance degradation.

With these findings in mind, this study will use the Self-Instruct framework from Wang et al. [67] to augment training datasets, though model collapse as discussed by Guo et al. [22] remains a real boundary. Following the safety limit defined by He et al. [23], this study limits synthetic data at 30%. Synthetic generation serves purely as a supplement, not a replacement for actual data collection. The pipeline generates synthetic Bislish and Taglish reviews only for aspect-sentiment pairs left underrepresented after manual annotation. Keeping this ratio well below the 38.2% threshold gives the model a balanced class distribution while preserving the distinct linguistic nuances of Davao City consumer reviews.

2.7.1 Semantic Validation and Diversity Filtering

While limiting the scope of synthetic generation mitigates macro-level model collapse [22], batch-generated instances still suffer from micro-level homogenization, where LLMs produce lexically varied but semantically identical text (e.g., swapping basic synonyms without altering core sentence structure)[56]. To enforce genuine syntactic variance, modern methodologies employ vector-based semantic filtering before dataset integration.

Converting generated text into sparse feature vectors via TF-IDF allows researchers to mathematically measure lexical overlap [45]. By computing the cosine similarity between the new synthetic vectors and the authentic seed dataset, we can discard any instances that cross a strict similarity threshold. This filtering acts as a built-in regularization step. It prevents dataset dilution and forces the downstream transformer to

map actual linguistic boundaries, rather than just memorizing repetitive phrasing and spiraling into model collapse [72].

2.7.2 Empirical Sample Size Determination

Synthetic data generation mitigates class imbalance, but defining the exact data volume required for a model to learn a domain-specific category is a core problem in Aspect-Based Sentiment Analysis (ABSA) [51]. Traditional machine learning pipelines rely on arbitrary, fixed sample thresholds (forcing every class to equal 500 instances). This approach lacks empirical grounding and introduces unnecessary synthetic noise.

Modern data engineering uses progressive sampling techniques to establish mathematically defensible sizing criteria. Provost et al. [54] defined the foundational framework. They demonstrated that a model's accuracy scaling behavior is tracked efficiently by evaluating performance across incrementally increasing data subsets instead of training blindly on an entire corpus. Researchers construct empirical learning curves by measuring performance gains at strategic batch intervals.

Huang et al. [27] applied this logic to transformer architectures. They showed that tracking learning curve plateaus identifies the exact saturation point where adding training instances stops yielding meaningful improvements in F1-score. This ensures synthetic augmentation acts strictly as a targeted corrective measure. It triggers only until a class hits its empirically derived sufficiency threshold, optimizing data efficiency and preserving the authentic distribution of localized consumer reviews.

2.8 Aspect-Based Sentiment Analysis (ABSA)

General sentiment analysis gives a broad picture of consumer satisfaction, but it tends to fall short when reviews are complex, when a single sentence praises one thing and criticizes another in the same breath [43] [51] [16] [7]. Aspect-Based Sentiment Analysis (ABSA) addresses this by pulling out specific entities or topics and identifying the sentiment attached to each one individually [43] [51].

One of the bigger challenges in applying ABSA to regional service sectors is code-switching, which tends to quietly demolish the accuracy of standard NLP tools [16]. Dionela et al. [16] looked at this problem specifically within Taglish, a low resource, code-switched blend of Tagalog and English that shows up constantly in Filipino consumer reviews. They built a pipeline comparing a traditional rule-based system, a generative LLM, and fine-tuned Gemma-3 models for identifying and extracting both explicit and implicit aspects. The generative LLM came out ahead across all tasks, which suggests that advanced generative models handle the messy syntactic structures and informal phrasing of Philippine code-switched reviews better than the alternatives [16].

Getting the aspects out is only part of the problem though. Accurately classifying the sentiment and emotional nuance inside code-mixed text is its own challenge [58]. Suciati and Budi [58] tackled this in the context of code-mixed Indonesian reviews, pointing out that standard ternary sentiment, positive, negative, neutral, often lacks the depth

needed for nuanced consumer analysis. They compared traditional machine learning algorithms like Decision Trees, Random Forest, and Support Vector Machines against deep learning approaches like GRU and BiLSTM for multi-label classification across restaurant aspects such as food, price, service, and ambiance. The result was that traditional ensemble methods, specifically Random Forest and Extra Trees, actually outperformed the more complex deep learning models on certain tasks and achieved high F1 scores as high as 89.54% for food and 92.65% for service. The takeaway is that for highly localized, code-mixed datasets, model architecture must be matched carefully to the specific aspect being evaluated rather than just defaulting to whatever is most complex [58].

Furthermore, capturing the contextual meaning of localized slang and code-switching also requires more than standard text representation [7]. Amalia and Winarko [7] identified this gap with traditional word embeddings, which struggle to track how the meaning of a word shifts depending on context, particularly in unstructured Indonesian restaurant reviews. Their fix was a hybrid architecture combining a Convolutional Neural Network (CNN) for feature extraction with various contextualized word embeddings, comparing BERT, ELMo, and Word2vec across aspect classification tasks. The ELMo-CNN combination came out on top, which reinforced the idea that pairing deep contextualized embeddings with a solid neural classifier meaningfully improves how well a model reads the nuanced, localized semantics of Southeast Asian consumer reviews [7].

With these findings as a guide, this study will adopt the multi-label aspect framework validated by Suciati and Budi [58] to analyze the kind of unstructured, fragmented feedback that shows up regularly in Davao City café reviews. An XLM-RoBERTa architecture will handle the multi-label classification, isolating and evaluating specific aspect-sentiment pairs. The goal is to make sure that the large volume of complex, code-mixed reviews actually gets converted into structured, usable insights that reflect the operational realities of local coffee and matcha businesses rather than just producing generic sentiment scores.

2.8.1 Aspect Extraction

To identify specific attributes or features from opinionated texts, an Aspect-based sentiment analysis (ABSA) is conducted, where aspect extraction plays a key role. However, unlike traditional sentiment analysis methods, rather than assigning the entire text a single polarity, ABSA identifies a review's varying aspects along with their respective polarity. Since multiple sentiments can be expressed in a single review, ABSA provides a more detailed understanding of user feedback [43].

Mukherjee and Liu [43] conducted a study and identified the limitations of existing aspect extraction approaches, particularly those that either extract aspect terms without proper categorization or rely on unsupervised topic modeling, which may produce aspects that are not meaningful or aligned with user needs. Hence, aspect categorization is subjective, as different applications require different categories of the aspects. For that, they introduced a semi-supervised modeling approach for aspect extraction and categorization using user-given seed words. Their models, SAS, and ME-SAS also model sentiment jointly with extraction of aspect terms and grouping them into categories.

In contrast, Poria et al. [52] dealt with the problem of implicit aspects which are neglected by previous methods. They noted that implicit aspects are typically expressed through contextual clues rather than direct mentions, making them more difficult to detect. To this end, they proposed a rule-based approach, common-sense knowledge, and dependency parsing, which extract both explicit and implicit aspects. The implicit aspect clue is determined and mapped to an aspect category with linguistic rules and semantic information. Their research work showed that the detection accuracy of the developed approach is higher than existing techniques.

Suryabrata et al. [60] addressed the challenges of aspect extraction in the case of multilingual and code-mixed scenarios by proposing a transformer-based ABSA framework relying on fine-tuned multilingual models, including mBERT and XLM-RoBERTa. Their major innovation is the sentence-pair classification strategy, which reformulates aspect extraction as a Natural Language Inference (NLI) task. The review sentences were then paired with candidate aspect-sentiment expressions in the framework, enhancing the capture of contextual relations between aspects and sentiments. For evaluation, their samples set of 20,000 aspect-level instances were split into (70%) training, (15%) validation, and (15%) test sets using random sampling without stratification. The sample splits were reused for all models to observe a fair comparison between single-sentence classifiers and their proposed sentence-paired models.

Their results showed that the XLM-RoBERTa achieved an accuracy of 97.20% and an F1-score of 0.9729, while mBERT showed more stability in performance across the validation sets.

Among the reviewed studies, the work of Suryabrata et al. [60] best fits the proposed framework. This, combined with a sentence-pair (NLI) approach, allows for more accurate identification of implicit and explicit aspect–sentiment pairs in a multilingual–code switched environment with a fine-tuned version of XLM-RoBERTa. The approach fits with the study objectives since the information to be analyzed is unstructured, in different languages, and context-specific consumer reviews.

2.8.2 Aspect Taxonomy

Building an aspect taxonomy is one of the more consequential early decisions in any ABSA study because it shapes what the model can and cannot detect downstream [10][9]. Older approaches relied on fixed dictionaries, which tend to miss necessary attributes and break down quickly when the text is informal or syntactically irregular [16]. More recent work has moved toward using Large Language Models to induce taxonomies directly from raw data, without requiring predefined categories or domain-specific seed sets.

In a study, Brady and Islam [10] took this direction when analyzing political advertising content on social media in the 2024 U.S. presidential election. Since the content was highly subjective and shifted rapidly, they built an end-to-end framework that combined unsupervised clustering with prompt-based LLM inference. The model constructed a taxonomy iteratively, without needing seed sets or prior domain knowledge, and the resulting topic labels were more semantically coherent and interpretable than those produced by classical methods [10].

Boughanmi et al. [9] applied a similar approach to food and beverage feedback, working with 20,000 Yelp reviews of Starbucks locations. They tested eight prompt variants against manual human coding and found that the LLM consistently matched human annotators while processing each review. Beyond speed, the extracted attributes showed strong predictive validity for customer ratings, which suggests that generative extraction is not just efficient but substantively accurate [9].

The challenge becomes considerably harder in low-resource and code-switched settings. Dionela et al. (2026) [16] examined this directly using Filipino e-commerce reviews written in Taglish, where standard NLP tools and rigid rule-based systems tend to fail. Their pipeline compared rule-based extraction, a fine-tuned Gemma-3 model, and a generative LLM (Gemini 2.0 Flash). The generative model outperformed the other approaches across all extraction tasks, handling implicit aspects and informal syntax without relying on any predefined lexicon [16].

This study follows the framework used by Dionela et al. [16] for its taxonomy discovery phase. Reviews from Davao City coffee and matcha shops are written heavily in Bislish and Taglish, which means a standard industry taxonomy would likely miss locally relevant concerns like specific ambiance traits or parking availability. Deploying a generative LLM as an unsupervised aspect allows the taxonomy to emerge from the actual language consumers use, rather than from categories decided in advance [16].

2.9 Active Learning Frameworks

Fine-tuning data demands significant amounts of high quality labeled data. When it comes with large volumes of scraped consumer reviews, exhaustive manual annotation quickly becomes a serious bottleneck [44][62][39]. Tharwat and Schenck [62] examined this problem, noting that even with an abundance of free, unlabeled text from digital platforms, the time and cost of manual labeling remain among the biggest obstacles to building supervised learning models. Their survey pointed to Active Learning as a practical workaround. Rather than labeling an entire dataset blindly, AL uses specific query strategies to find the most informative and uncertain data points in the instance space. Focusing human effort on high-value samples achieves strong model generalizability at a fraction of the usual labeling cost [62].

Considering that the feedback is multi-dimensional and complex. Muralidhar and Hulipalled [44] addressed the time and cost constraints associated with using expert human annotators for ABSA on unstructured social media data. They built a Deep Active Learning framework designed to predict aspect-based ratings for smartphones, using uncertainty sampling to iteratively pick out only the most confusing or informative reviews for human annotation, while a lexicon-based filter removed uninformative neutral statements at the same time. The DAL model hit 92% accuracy, which is a strong result, and it showed that intelligently querying uncertain samples can drastically cut down annotation effort without hurting predictive performance [44].

The same logic extends to fine-tuning modern language models. Margatina [39] explored this in her work on data efficient language models, starting from the observa-

tion that not all training examples contribute equally to the learning process even when large amounts are available. By simulating various active learning algorithms, she showed that AL guides training more intelligently by selecting only the most informative unlabeled data points at each iteration. Optimizing data selection this way consistently outperformed random sampling, letting language models reach strong performance with far fewer human labeled instances [39].

These findings directly inform how this study will handle the annotation bottleneck for Davao City consumer reviews. Following the approach demonstrated by Muralidhar and Hulipalled [44], the system will use active learning to selectively surface only the most uncertain Bislish and Taglish samples for manual review. This keeps human effort manageable while ensuring the XLM-RoBERTa model is trained on the most informative data available, which is the central aim of bridging raw web-scraped text and a robust aspect-based classification system [58][16].

2.10 Manual Labeling in Sentiment Analysis

Active learning mechanisms are good at identifying the most uncertain or informative instances in a text corpus, but identifying them is only half the problem [35]. These algorithms cannot actually resolve the ambiguity on their own. What they need, and what every active learning pipeline fundamentally depends on, is a human annotator to provide the definitive ground truth for those flagged edge cases [35]. This is why manual labeling is not just a supplementary step but a necessary one in any sentiment analysis workflow [1][6].

In the Philippine context, where text in localized communications tends to be noisy and heavily context dependent, determining the situational validity of messages requires careful, extensive analysis. Ablazo [1] addressed this by using crowdsourcing through Google Forms, recruiting 250 undergraduate students from Ateneo de Davao University to serve as annotators. The dataset was split into batches, and each instance was independently labeled three times using predefined situational categories. A custom script then checked for annotator agreement, accepting a label only when at least two out of three annotators reached the same conclusion. From a total of 12,054 cleaned instances, the process produced 11,340 entries with a matched majority agreement. Instances where the annotators could not agree were discarded, so only labels that actually had consensus were passed on to the automated classifiers [1].

The question of how to direct that human effort more efficiently is what Li et al. [35] tried to answer. They note that deep learning models need large datasets, but relying on instance by instance manual labeling makes it hard to balance label quality with cost. Programmatic labeling can scale up, but it produces noisy results. Active learning, on the other hand, struggles with cold start problems when there is not enough labeled data to begin with. To address these limitations together, Li et al. [35] developed the DALL framework, which combines data programming, active learning, and large language models into one integrated labeling system. Rather than annotating random samples, DALL uses active learning strategies like uncertainty sampling, query by committee, and abstain sampling to flag the instances most likely to confuse or conflict with current model

predictions. These are then routed to human annotators, who correct the labels and refine the rules [35].

Fine-tuning pretrained language models for Aspect-Based Sentiment Analysis is another difficulty since the annotation needs to be both granular and large in scale. Alturayeif et al. [6] point out a specific problem with standard active learning in this context. Algorithms often query instances that are semantically similar to each other, thus, human annotators end up labeling the same kinds of phrasings repeatedly without actually adding new information. To fix this, Alturayeif et al. [6] proposed the EASE framework, which builds diversity and data augmentation into the active learning process. Their method calculates contextual similarity scores using Sentence BERT and averages those with model confidence scores, so that the samples flagged for human review are not only uncertain but also meaningfully different from what has already been labeled. After annotation, those edge cases go through Easy Data Augmentation to generate additional training samples synthetically. By filtering out redundant instances before they ever reached human annotators, the framework reduced the total number of manually labeled samples needed by somewhere between 88% and 94% across benchmark datasets [6].

Building on these findings, this study adopts the active learning and data augmentation principles from Alturayeif et al. [6] to guide its manual annotation phase. The dataset is made up of highly contextual, code-switched Taglish and Bislish reviews from local coffee and matcha establishments, so exhaustive manual labeling is simply not practical given the scale and variety of the data. By applying the framework's dual requirement of uncertainty and sample diversity, the pipeline makes sure human annotators are only asked to evaluate complex, non-redundant linguistic edge cases rather than running through similar reviews that do not add anything new [6].

2.10.1 Inter-Rater Reliability

Once the complex, non-redundant edge cases are identified for human review, keeping the resulting annotations consistent becomes its own challenge. Using a majority voting strategy, like the consensus approach Ablazo [1] demonstrated, provides a clear label for subjective and code-switched text, but relying only on simple percentage agreement is often not enough for machine learning applications that need more rigorous validation.

The importance of choosing the right IRR metric for qualitative annotations is a topic the literature has addressed fairly directly. El Dehaibi and MacDonald [18] looked at IRR measures for qualitative annotations in machine learning datasets and found that running an IRR check before training provides a necessary internal validity step. They explain that Cohen's Kappa corrects for chance agreement between exactly two raters on categorical items, but once a study involves three or more raters, Fleiss' Kappa becomes the appropriate metric.

Chamid et al. [11] show what Cohen's Kappa looks like applied to a more demanding annotation scenario. Their study evaluated labeling consistency for ABSA using 4,307 Indonesian marketplace reviews, where two expert linguists independently labeled aspect detection, sentiment classification, and aspect class. The resulting Kappa values were 0.909 for aspect detection, 0.893 for sentiment classification, and 0.971 for aspect

class, all falling in the almost perfect agreement range. These scores confirmed that the annotated dataset was reliable enough to actually train accurate classifiers on.

When annotation design scales up to three or more raters, Fleiss' Kappa is discussed. Krušić [32] illustrates this in building a sentiment annotated corpus of historical texts for NLP, where three semi-expert annotators categorized sentences into positive, negative, neutral, and mixed sentiment.

Fleiss' Kappa across the three raters came out to 0.405, which falls in the fair to moderate range. A majority vote requiring at least two out of three annotators to agree was used to finalize labels, and the overall majority vote agreement reached 92.5%. These results were considered sufficient to establish the annotations as a reliable enough corpus for fine-tuning machine learning models.

Following this established standard for multi-rater setups, this study utilizes Fleiss' Kappa (κ) [19] to validate the ground-truth annotations provided by the peer annotators. The metric is formally defined as:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e} \tag{2.1}$$

Where the observed agreement $\bar{P}$ and the expected chance agreement $\bar{P}_e$ are derived from the distribution of ratings across subjects and categories:

$$\bar{P} = \frac{1}{N}\sum_{i=1}^{N} P_i \quad \text{where} \quad P_i = \frac{1}{n(n-1)}\sum_{j=1}^{k} x_{ij}(x_{ij}-1) \tag{2.2}$$

$$\bar{P}_e = \sum_{j=1}^{k} p_j^2 \quad \text{where} \quad p_j = \frac{1}{Nn}\sum_{i=1}^{N} x_{ij} \tag{2.3}$$

- N = total number of subjects or text instances
- n = number of raters per subject
- k = number of categorical choices (e.g., Positive, Negative, Neutral)
- x_{ij} = number of raters who assigned the i-th subject to the j-th category

2.11 Theoretical Frameworks

2.11.1 Multilingual Transformers

This method of Multilingual Transformers is grounded in the methodology of Maceda et al. [37]. Their study highlights the need for multilingual models to process code-mixed data, demonstrating that traditional approaches are insufficient for analyzing text that blends languages such as English and Filipino due to inconsistencies in grammar, vocabulary, structure, and context. By implementing a fine-tuned multilingual Bidirectional

Encoder Representations from Transformers (mBERT) [15] [37], they were able to capture contextual relationships across multiple languages. The sequential process for this framework is depicted in Figure 1. This approach establishes that multilingual transformers are well-suited for handling linguistically diverse datasets, supporting their application in studies involving mixed-language inputs. Similarly, Patil et al. [49] emphasized that transformer-based approaches consistently outperform traditional feature-based models when handling noisy text. Therefore, multilingual transformers are considered an optimal modeling framework for analyzing unstructured, code-mixed consumer reviews where linguistic variability is prevalent.

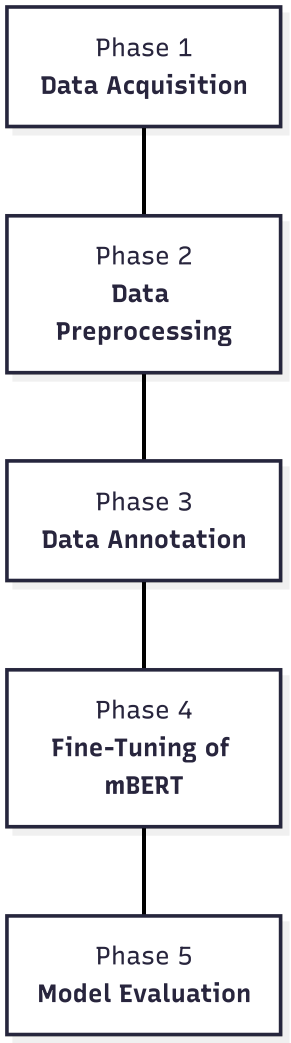


Figure 1: *Multilingual Transformers*

Phase 1: Data Acquisition The framework begins by collecting unstructured, code-mixed textual data from multiple social media platforms, such as Facebook, Twitter, and YouTube. Scraper libraries were utilized to systematically retrieve relevant posts, tweets, and comments using predefined keywords and sources [37].

Phase 2: Data Preprocessing The noisy and inconsistent nature of the code-mixed

dataset necessitated preprocessing to clean and standardize the data. Duplicate entries were removed using URL-based identification, and irrelevant content was filtered manually. Minimal preprocessing techniques were applied, including the correction of informal spelling, normalization of special characters, removal of unnecessary whitespace and punctuation, and conversion of emojis into their textual representations. This approach preserves contextual meaning while improving data consistency [37].

Phase 3: Data Annotation From the preprocessed dataset, data points were manually labeled and validated by experts. Each instance was classified into three sentiment categories: neutral, negative, and positive, encoded as 0, 1, and 2, respectively. The annotated dataset serves as the ground truth for training and evaluating the sentiment classification model, following established annotation practices [37].

Phase 4: Fine-Tuning of mBERT The framework utilizes a fine-tuned Multilingual Bidirectional Encoder Representations from Transformers (mBERT) model for sentiment classification. Using transfer learning, the model will leverage pre-trained global linguistic knowledge and adapt it to the specific variations in the mode-mixed dataset. Built on the Transformer architecture, mBERT captures contextual relationships across multiple languages, making it suitable for code-mixed data. The model processes input sequences using special tokens such as [CLS] for classification and [SEP] for segmentation, where the final hidden state of the [CLS] token is used for prediction. Fine-tuning was performed using task-specific parameters, including learning rates ($2e-5$, $3e-5$, $5e-5$), batch sizes (16 and 32), and four training epochs. A maximum sequence length of 160 tokens was applied, and class weights were used to address class imbalance [37].

Phase 5: Model Evaluation The model's performance was evaluated using accuracy and F1-score metrics to assess its effectiveness in classifying sentiment in code-mixed text. The best-performing model configuration was then used to predict sentiment labels for the remaining dataset, enabling the analysis of overall sentiment distribution. This evaluation ensures the reliability and robustness of the framework in handling multilingual and unstructured data [37].

2.11.2 Aspect-based sentiment analysis

The method of Aspect-based sentiment analysis is grounded by the methodology by Suciati and Budi [58]. Their research addresses computational complexities of processing code-mixed (Indonesian-English) restaurant reviews by proposing a dual-pipeline classification architecture, as illustrated in Figure 2. Since standard single-language models fail to capture the nuances of bilingual reviews [25], their framework compares the potency of traditional Machine Learning (ML) models against advanced Deep Learning (DL) networks in performing multi-label classification for both sentiment and emotion.

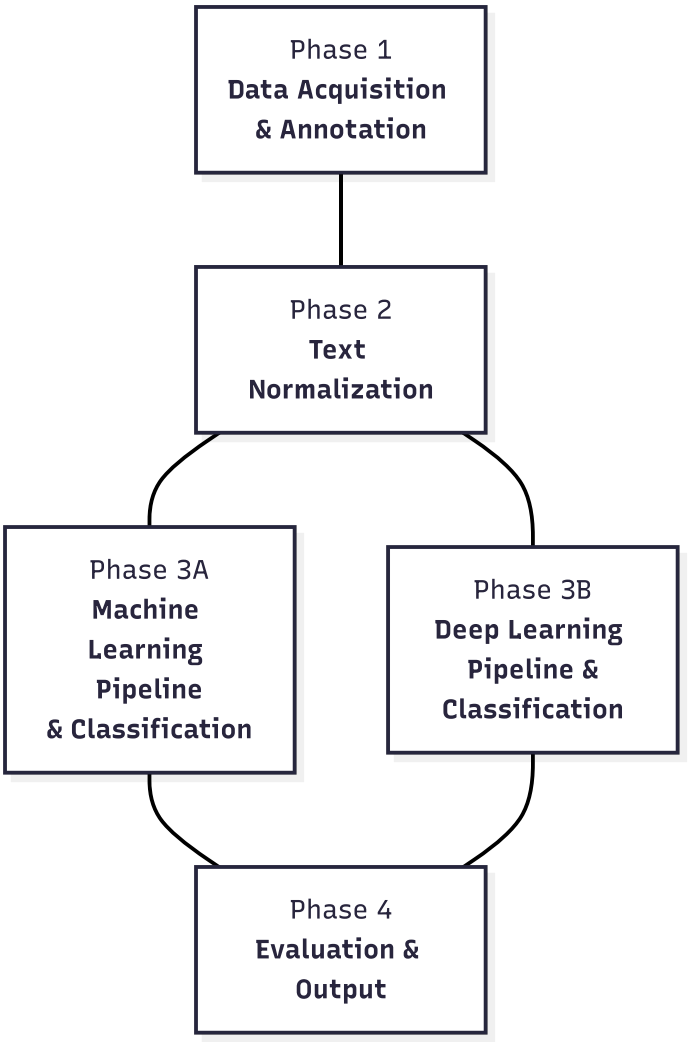


Figure 2: *Aspect-based sentiment analysis*

Phase 1: Data Acquisition and Annotation The framework is based on collecting unstructured review data that contains naturally mixed formal language and informal code-switching. The data then goes through the annotation processes using crowdsourcing. Multiple annotators label the text based on predefined guidelines, and when annotators reach a disagreement on how something should be labeled, a majority vote settles the call [58].

Phase 2: Text Normalization (Pre-processing) As code-mixed social media text is highly noisy and inconsistent, the framework implements an extensive eight-step preprocessing layer. Abbreviations are corrected using self-developed bilingual dictionaries, while word stemming is handled by two dedicated tools: the Snowball Stemmer for English and the Sastrawi Stemmer for Indonesian. Emoticons are also converted into their text equivalents to ensure that meaningful emotional cues are preserved throughout the cleaning process [58].

Phase 3: Dual Classification Pipelines

- *Machine Learning Pipeline:* The Machine Learning pipeline relies on traditional feature extraction methods through N-gram language models, such as unigrams, bigrams, and a combination of both, paired with Chi-Square for feature selection. Since aspect-based classification often involves reviews that address multiple targets at the same time, the framework applies problem transformation methods such as Binary Relevance, Classifier Chain, and Label Powerset to adapt standard algorithms including SVM, Random Forest, Decision Tree, and Extra Tree into multi-label classifiers [58].
- *Deep Learning Pipeline:* The deep learning pipeline takes a different approach by eliminating the need for manual feature engineering. Instead, it constructs custom word embeddings using the Skip-gram method to capture the semantic relationships present in code-mixed text. Deep learning architectures, specifically BiLSTM and GRU, are then deployed using a sigmoid activation function and binary cross-entropy loss to directly predict multiple aspect-sentiment pairs at the same time [58].

Phase 4: Evaluation and Output The performance of both pipelines is thoroughly tested using a 10-fold cross-validation technique. The framework evaluates each model based on its F1-Score, measuring how accurately it was able to isolate and classify sentiment and emotion across four distinct business dimensions: food, price, service, and ambience [58].

2.11.3 Aspect Extraction

This framework draws from Natural Language Processing, transfer learning, and Aspect-Based Sentiment Analysis. The core problem with traditional sentiment analysis is that it reduces an entire review to a single polarity label, and that is simply too blunt for user-generated content [60]. In the study, hotel guests routinely praise one aspect of their stay while criticizing another in the same sentence. Recovering that granularity requires Aspect Extraction. It is the process of identifying which specific service attributes a reviewer is actually referring to [60].

The study uses XLM-RoBERTa, a multilingual transformer architecture, to handle the code-mixed language that commonly appears in reviews from multilingual settings. Since no locally annotated dataset exists for this domain, zero-shot learning was adopted. The ABSA task is reformulated as a Natural Language Inference problem, which lets the models assess sentiment without needing labeled training examples from the target domain [60]. Figure 3 outlines the overall workflow and the structural phases of the study.

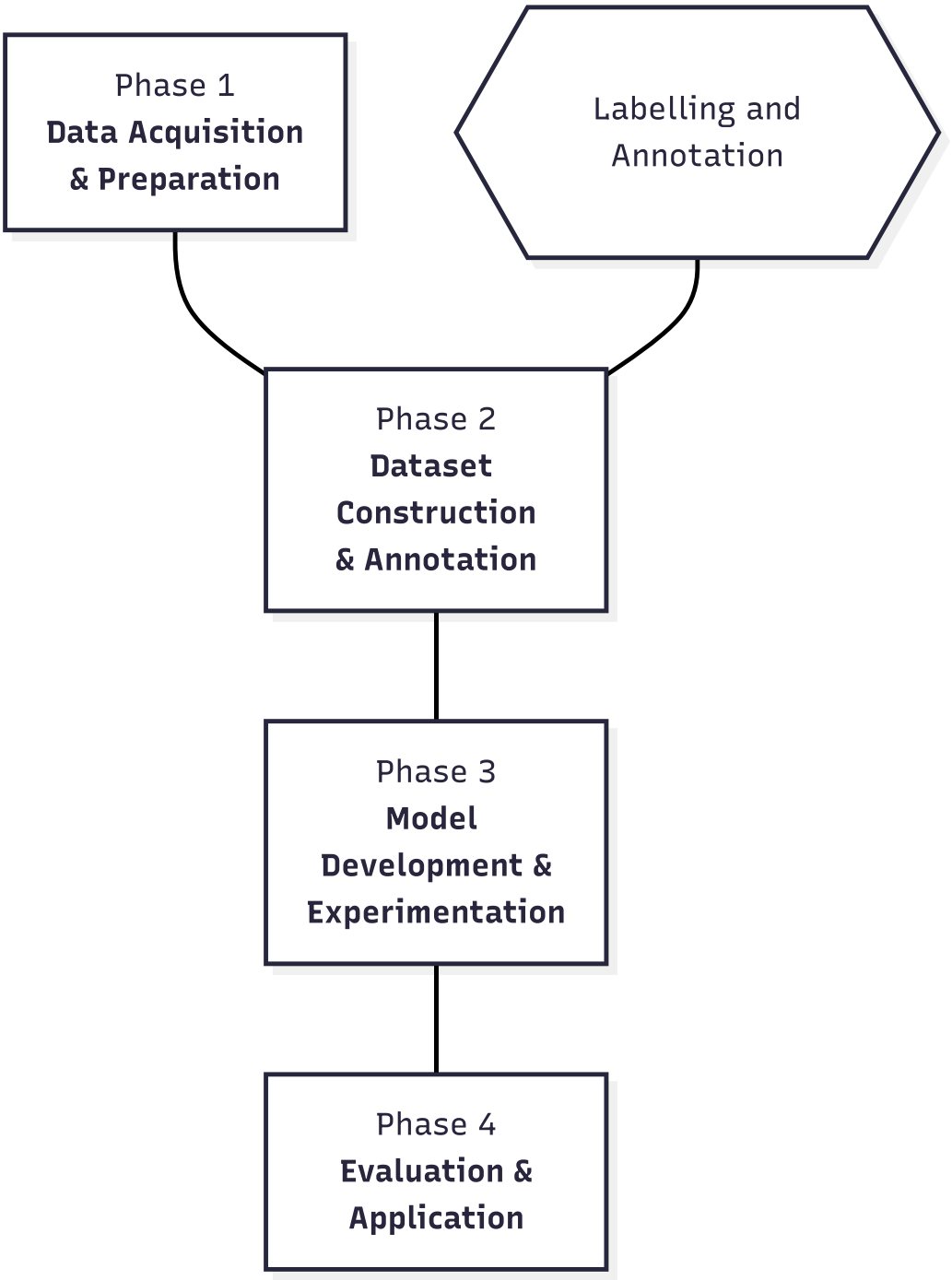


Figure 3: *Aspect Extraction*

Phase 1: Data Acquisition and Preparation Any NLP pipeline depends on the quality and representativeness of its corpus. For this study, over 100,000 reviews were collected from TripAdvisor and Google Reviews across 43 five-star hotels in Jakarta. The challenge at this stage is that user-generated content is unstructured, inconsistent, and often multilingual, with some reviewers switching between languages mid-text [60].

- *Collection and Preprocessing* Covers data collection and preprocessing. Duplicate entries and empty strings are removed to reduce model bias and prevent overfitting. A language filter then isolates reviews in Indonesian, English, and code-mixed text. This filtering decision was deliberate: mBERT and XLM-RoBERTa were pre-trained on multilingual corpora, and keeping the input data within those language boundaries ensures the models are operating within their intended distribution [60].

Phase 2: Dataset Construction and Annotation This is where Zero-Shot Learning and Aspect Extraction are present. A major hurdle in ABSA is the "cold-start" problem: the lack of massive, human-annotated datasets for specific domains (like Indonesian hotel reviews). This phase leverages the theory of Zero-Shot Transfer Learning, in which a model pre-trained on general tasks is used to classify unseen data without requiring domain-specific fine-tuning. Crucially, it relies on aspect extraction as the prerequisite mechanism to break down monolithic blocks of review text into granular, analyzable targets before any sentiment can be measured [60].

- *Zero-Shot Labeling and Aspect Extraction:* multilingual zero-shot classifier acts as a proxy annotator. The indispensable first step is aspect extraction, where the model automatically scans the raw text to identify what the user is talking about, categorizing the text into seven predefined aspects (Cleanliness, Facilities, Food, Location, Price, Room, and Service). Because a single review might discuss multiple aspects simultaneously. In the dataset, sentiments were isolates each extracted review-aspect pair into its own unique instance (generating 352,720 records). Only after these target aspects are successfully extracted and separated can a subsequent zero-shot pass accurately determine the specific sentiment (Positive, Negative, Neutral) attached to them [60].

- *Sampling and Splitting:* To optimize computational efficiency while preserving statistical significance, a representative sample of 20,000 aspect-level instances is extracted and rigorously partitioned (70% Train, 15% Validation, 15% Test) to ensure the model can be evaluated on strictly unseen data [60].

Phase 3: Model Development and Experimentation This phase is where the core computational work happens. Rather than treating ABSA as a standard classification task, the study reformulates it as a Natural Language Inference problem [60].

- *Sentence-Pair Construction and Tokenization:* Each review is restructured into a sentence pair. The review text serves as the premise, and a constructed auxiliary sentence. The model then predicts whether the premise entails the hypothesis. Framing the task this way gives the transformer explicit structural context about the aspect and sentiment orientation being evaluated. Tokenization is handled by subword algorithms appropriate to each model. WordPiece for mBERT and SentencePiece for XLM-RoBERTa, with the other being better equipped for the morphological complexity and code-switching present in the corpus, morphological variation and code-switching [60].

- *Training and Fine-Tuning:* One persistent problem with review datasets is se-

vere class imbalance, as positive reviews typically account for more than 90% of the data. A model trained on distribution can achieve deceptively high accuracy by defaulting to positive predictions while failing almost entirely on negative and neutral cases. Rather than correcting this through oversampling, which carries a real risk of overfitting, the framework applies Weighted Loss Functions under a cost-sensitive learning approach. Misclassifications of minority class instances are penalized more heavily during training, pushing the model to learn the distinguishing patterns of negative and neutral reviews rather than treating them as statistical noise [60].

Phase 4: Evaluation and application The final phase connects model outputs to practical use within a hotel management context.

- *Evaluation:* On imbalanced datasets, standard accuracy is an unreliable measure of model performance. A classifier that predicts positive sentiment for every review would exceed 90% accuracy on this corpus while failing entirely to detect negative or neutral cases. For this reason, the framework uses the F1-score as its primary evaluation metric, which computes the harmonic mean of Precision and Recall. Precision measures how many of the model's predicted positives are correct, while Recall measures how many of the model's predictions are correct. Confusion matrices and error analyses supplement these scores by identifying which linguistic structures consistently cause misclassification, providing diagnostic depth beyond what aggregate metrics alone can offer [60].
- *Application and Visualization:* Deploys the better performing architecture, XLM-RoBERTa, across the approximately 76,000 reviews that were not used in training or evaluation. The resulting sentiment predictions are mapped into visualizations including heatmaps and correlation charts. This translation step is what makes the model outputs usable in practice so that hotel management can identify which service aspects are consistently underperforming across reviews without having to interpret raw model inferences directly [60].

2.11.4 Aspect Taxonomy

This research builds on two related ideas: the Principle of Cross Lingual Semantic Resilience and the Theory of Generative Attribute Induction. Dionela et al. [16] incorporated both into their Aspect-Based Sentiment Analysis framework [16]. The claim is that generative language models can process messy, code-switched consumer feedback without relying on predefined word lists. Dionela et al. [16] points out that traditional Natural Language Processing fails in low-resource, mixed-language environments [16]. Their study of Philippine e-commerce reviews shows the problem clearly. Rule based models and small fine tuned models struggle with code-switched Taglish since they depend on keyword matching [16]. When consumers use informal language or imply an aspect without stating it directly, the system misses it. For instance, "It took a week to arrive" signals a delivery problem, but the keywords might not be obvious [16]. Refer to Figure 4.

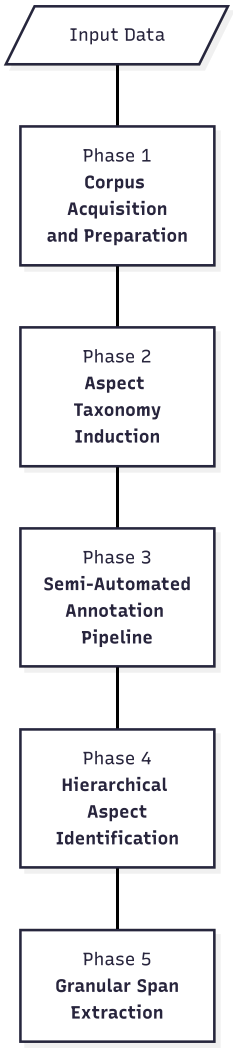


Figure 4: *Aspect Taxonomy*

Phase 1: Corpus Acquisition and Preparation. Raw, unstructured code-switched consumer reviews are processed to create a baseline corpus [16]. The dataset contains 10,510 Taglish reviews from Google Maps and Shopee Philippines. Human annotators labeled both general and specific aspect categories [16].

Phase 2: Aspect Taxonomy Induction Rather than starting with predefined categories, this phase builds a Hierarchical Aspect Framework by combining four approaches: Open Coding, Latent Dirichlet Allocation for topic modeling, BERTopic for multilingual embeddings, and Generative LLMs to integrate the results [16]. The outcome was a taxonomy with 4 General Categories (Product, Delivery, Price, Service) and 21 Specific Aspects derived from the actual corpus [16].

Phase 3: Semi-Automated Annotation Pipeline To annotate implicit aspects at scale, human researchers first labeled 67 reviews as a reference set, achieving a Fleiss Kappa of 0.691. This reference set was then used as few-shot examples for Gemini 3.0, which automatically annotated the remaining reviews to create a train-

ing dataset [16].

Phase 4: Hierarchical Aspect Identification The framework uses conditional classification: broad domains like Product or Service are identified first, and these results determine which specific aspect classifiers run next. It was then compared to four approaches. A Rule Based system, Gemini 3.0, and two versions of Fine Tuned Gemma 3 (1B) [16]. The Generative LLM achieved a Macro F1 of 0.91, outperforming the others. The smaller fine tuned models had difficulty with noisy syntax and imbalanced classes in code-switched text [16].

Phase 5: Granular Span Extraction The system extracts exact text spans for the identified aspects, mapping code-switched vocabulary to the taxonomy. LLMs extracted numerical concepts like price accurately, but struggled with Over Extraction and Partial Matches when faced with variations in Taglish Product and Service terminology [16].

2.11.5 Active Learning

This method of Active Learning is grounded in the methodology established by Muralidhar and Hulipalled [44]. Their research addresses the high cost and time needed to manually label large social media datasets by proposing a Deep Active Learning pipeline as depicted in Figure 5 To reduce misclassification of objective and neutral statements, their framework combines lexicon-based techniques with human judgment to distinguish positive from negative sentiments. Text normalization, aspect categorization, and a Long Short-Term Memory (LSTM) network form the core of this pipeline. The model queries human experts only when classification confidence falls below a set threshold, resulting in efficient, aspect-specific sentiment predictions for mobile application reviews [44].

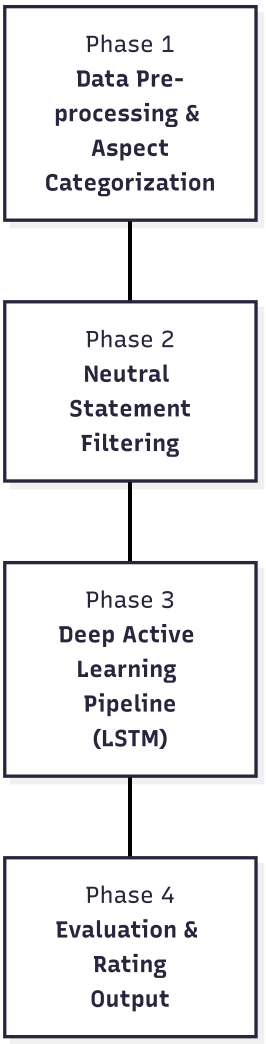


Figure 5: *Active Learning*

Phase 1: Data Pre-processing and Aspect Categorization Raw social media text presents several preprocessing challenges, such as misspellings, informal jargon, and hashtags that merge multiple words together. The framework addresses these using Ekphrasis, a tool that segments compound tokens, corrects spelling errors, and applies lemmatization to reduce words to their root forms without losing contextual meaning. Emojis and emoticons are converted to their text equivalents before the cleaned text is split into smaller phrases along punctuation and conjunction boundaries. Each phrase is then assigned to a product aspect such as "battery" or "camera" using predefined keyword dictionaries [44].

Phase 2: Neutral Statement Filtering Neutral, non-opinionated statements may distort classifier training if left in the dataset. To filter these out, the framework scores each text entry using a composite lexicon built from VADER, AFINN, and SentiWordNet. Since domain-specific language can complicate automated scoring "heavy" carries negative sentiment in a phone review but registers as neutral in a general lexicon. Then, a human annotator reviews the 100 most frequent words

flagged as neutral. Words identified as false neutrals are reassigned sentiment polarities and reintegrated into the lexicon. The dataset is then recorded, and remaining neutral entries are discarded, leaving a training set composed entirely of opinionated samples [44]. While Muralidhar and Hulipalled [44] utilized standard lexicons for this filtering phase, this study's conceptual framework adapts this approach to accommodate localized code-switching.

Phase 3: Deep Active Learning Pipeline The framework of the specified study uses an LSTM-based Active Learning strategy that selectively involves human input. A human annotator manually labels an initial batch of 5,000 reviews to train the model. The LSTM then processes the next 5,000 reviews: predictions made with at least 75% confidence are accepted automatically, while those falling below this threshold are routed back to the annotator for manual labeling. Each newly labeled sample is used to retrain the model, progressively improving its classification boundaries while keeping total annotation effort low [44].

Phase 4: Evaluation and Rating Output Model performance is evaluated against standard baselines, including SVM, using precision, recall, and F-measure. Beyond classification accuracy, the framework converts binary sentiment predictions into a 1-to-10 aspect-level rating scale, which is calculated by dividing the count of positive reviews by the total number of reviews for each aspect, thus producing a metric directly usable in commercial product evaluation [44].

2.11.6 Manual Labeling in Sentiment Analysis

This method of optimizing data selection for Aspect Based Sentiment Analysis (ABSA) draws from the methodology of Alturayeif et al. [6]. Their study identifies manual annotation as a serious bottleneck when fine tuning pretrained language models for granular sentiment tasks. To address this, they proposed the Enhanced Active learning framework for Aspect-based Sentiment Analysis (EASE), demonstrating that filtering redundant data before it reaches human annotators can significantly reduce labeling costs without harming model accuracy [6]. The framework works by combining uncertainty sampling, sample diversity, and targeted data augmentation so that every manually labeled instance actually contributes something useful. The full sequential process is illustrated in Figure 6.

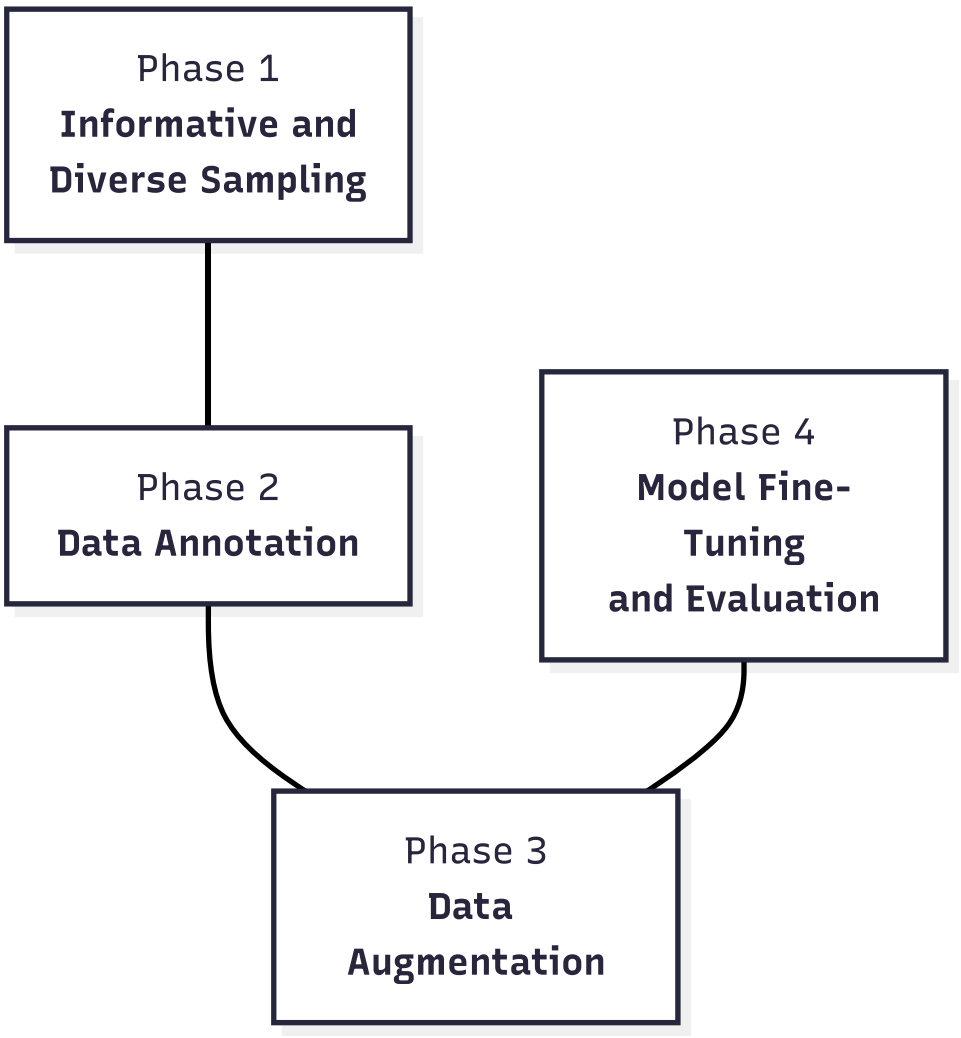


Figure 6: *Manual Labeling in Sentiment Analysis*

Phase 1: Informative and Diverse Sampling The framework starts by evaluating a large pool of unlabeled text. Rather than picking instances at random for human review, the algorithm computes two specific metrics to find the most valuable data points. The first is model uncertainty, which surfaces sentences the model finds ambiguous. The second is a contextual similarity score computed using Sentence-BERT, which checks whether a candidate sample is meaningfully different from the selected sample. Averaging these two scores together, the framework flags only the instances that are both uncertain and linguistically distinct from one another [6].

Phase 2: Data Annotation The informative and diverse instances identified in Phase 1 are then sent to human annotators for manual labeling. Since the algorithm removed semantically similar reviews, annotators spend their time on genuinely unique and complex patterns rather than variations of the same thing. This targeted approach reduces the total number of samples that need manual labeling by up to 94% compared to random sampling [6].

Phase 3: Data Augmentation To get more out of the newly annotated data and to handle potential class imbalances, the labeled instances go through Easy Data Augmentation (EDA). This step generates additional training samples from the seed data, which expands the overall training corpus while preserving the contextual integrity of the edge case reviews that were originally selected [6].

Phase 4: Model Fine-Tuning and Evaluation The final dataset which consists of a diverse manually annotated core supplemented by the augmented synthetic samples, is used to fine-tune the pretrained language model for ABSA. The model is then tested on benchmark datasets to confirm that classification accuracy holds up even with a drastically reduced annotation effort [6].

2.11.7 Synthetic Data Generation

To address data scarcity and class imbalance, the introduced study adopts the Self-Instruct framework developed by Wang et al [67]. The procedural steps of which are shown in Figure 7. The approach uses a pre-trained Large Language Model to generate training data from a small set of human-authored seed examples, expanding the dataset substantially without additional manual annotation.

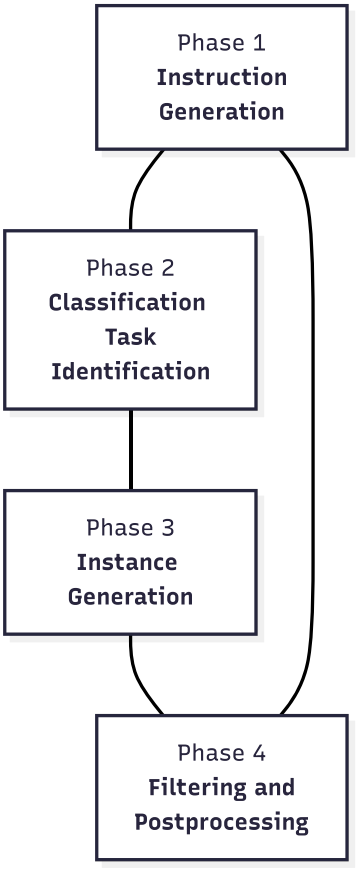


Figure 7: *Synthetic Data Generation*

Phase 1: Instruction Generation The process starts with a Task Pool containing a

small set of human-written seed examples. The LLM draws from these to generate new task instructions recursively, using existing examples as context for each subsequent generation [67].

Phase 2: Classification Task Identification The model determines whether it is a classification task, such as assigning sentiment polarity to a review, or a non-classification task. This distinction controls how the input and output are structured in the next phase [67].

Phase 3: Instance Generation Using the instruction and its identified task type, the model generates a synthetic input-output pair. For classification tasks, the input is a synthetic text review, and the output is a set of aspect and sentiment labels. The structure differs for non-classification tasks based on what the instruction specifies [67].

Phase 4: Filtering and Postprocessing Before entering the Task Pool, generated instructions are checked for redundancy and validity. Instructions with a ROUGE-L overlap of 0.7 or above against any existing pool entry are discarded, including entries the system cannot parse. The remaining instances are added to the pool and become available as in-context examples for future rounds, growing the dataset over successive iterations [67].

2.11.8 Constraint of Synthetic Data: Model Collapse and Stabilization Framework

While the Self-Instruct framework by Wang et al. [67] establishes the methodology for generating synthetic data, scaling the dataset effectively requires managing the inherent properties of AI-generated text [22]. Preserving the code-switched syntax of localized dialect reviews shapes how the augmentation strategy must be designed. Figure 8 visualizes the framework for mixing synthetic and authentic data to mitigate these constraints. Thus, the theoretical framework of Guo et al. [22] demonstrates that training models on synthetic outputs causes a persistent decrease in linguistic diversity, a phenomenon that cannot be fully cured even when mixed with authentic human data. To resolve this, He et al. [23] introduces a Weighting-Based Stabilization Framework to calculate exact mathematical limits for synthetic integration.

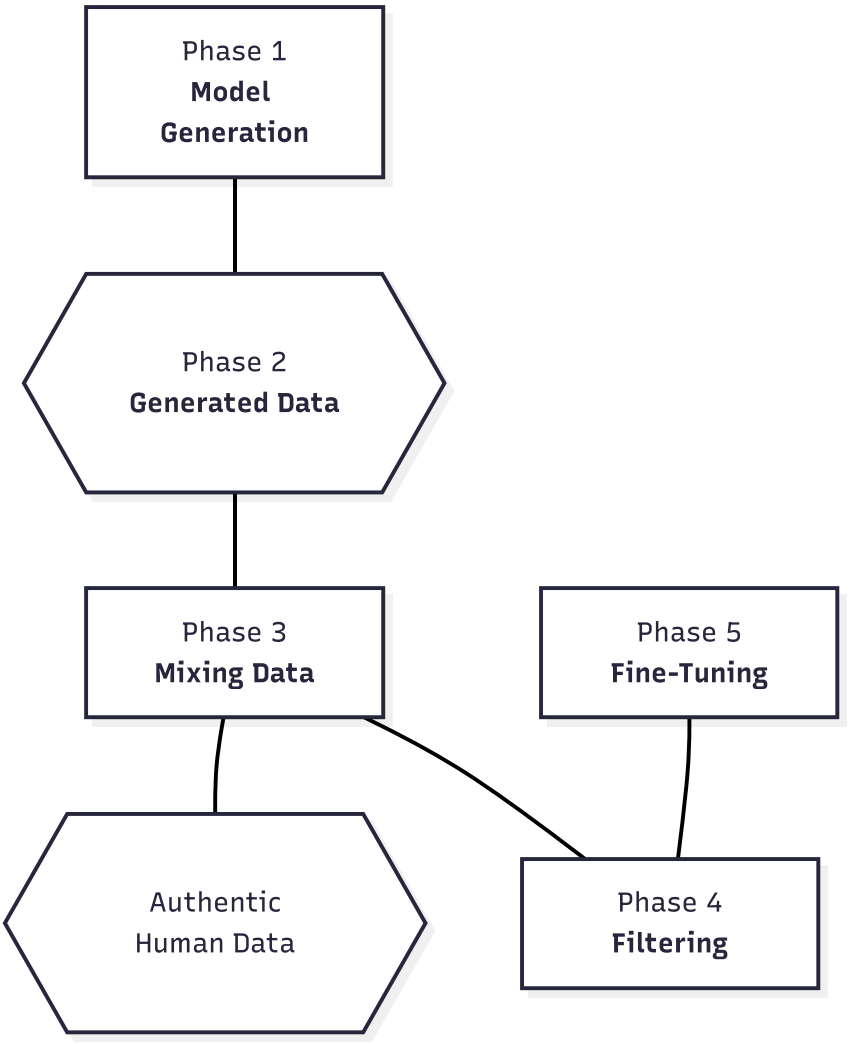


Figure 8: *Synthetic Data Causing Model Collapse*

Phases 1 and 2: Generation and Diversity Loss The cycle begins with a language model generating a batch of synthetic text. Guo et al. [22] find that synthetic data consistently underrepresents the lexical, syntactic, and semantic range of human-authored text. This deficit compounds across generation rounds, producing diversity loss even in tightly constrained tasks.

Phase 3: Data Mixing To reflect a realistic training scenario, the framework introduces a mixing phase in which synthetic outputs are combined with a fixed proportion of human-authored data [22].

Phases 4 and 5: Persistent Degradation After filtering and fine-tuning the subsequent model, Guo et al. [22] show that the added human data produces only marginal gains in diversity. The core finding is that model collapse is not a problem mixing alone can solve. Diversity loss accumulates across iterations. Hence, reliance on synthetic data still causes a persistent, cumulative decline in linguistic diversity, eventually leading to *"model collapse"* [22].

Guo et al. [22] showed that while mixing human and synthetic data mitigates diversity loss, it does not completely eliminate it. Addressing this degradation, He et al. [23] developed a framework for recursive learning stabilization. They determined the optimal weight for real data sits at the reciprocal of the golden ratio (roughly 61.8%). This creates a hard ceiling: synthetic data must stay below 38.2% to avoid performance drops.

These mathematical limits directly shape the study's augmentation strategy. Local coffee and matcha shop reviews in Davao City rely on code-switched Bislish and Taglish linguistic nuances that are notoriously difficult to replicate. Flooding the corpus with synthetic data would wash out the exact syntactic patterns the model needs to learn. Instead, synthetic generation via Wang et al.'s [67] Self-Instruct framework acts purely as a safety net. The pipeline only triggers when a specific aspect-sentiment class lacks enough real samples to be usable. Capping synthetic data at 30% keeps the project well within the 38.2% safety limit, balancing the training distribution without risking generative collapse.

To effectively analyze code-switched consumer reviews, this research integrates the Multilingual Transformers framework with the selective querying principles of Deep Active Learning, the unsupervised discovery of an Aspect Taxonomy, and the structural granularity of Aspect Extraction. While the primary objective focuses on comparing the proposed model against lexicon-based approaches, traditional feature-based pipelines relying on N-grams and algorithms like SVM are retained as supplementary baselines. Literature consistently demonstrates that transformer-based architectures outperform these traditional models on noisy text [49, 37, 68]; however, including SVM provides a rigorous empirical benchmark to validate the necessity of contextual embeddings for Taglish and Bislish. Furthermore, while the foundational Aspect-Based Sentiment Analysis (ABSA) framework established by Maceda et al. [37] utilized mBERT, this study implements XLM-RoBERTa as the primary classification engine, given its superior capacity to capture complex contextual relationships and resolve ambiguities in highly code-switched Southeast Asian languages [55].

Crucially, since traditional sentiment analysis reduces an entire review to a single polarity label, it is often too blunt for complex user-generated content where consumers frequently praise one attribute while criticizing another in the same sentence [60]. To address this, the framework first incorporates generative Aspect Taxonomy Induction [16] to discover localized operational dimensions from the raw text. This serves as a prerequisite phase for Aspect Extraction. Thus, ensuring the review text is broken down into granular, analyzable targets governed by a data-driven master dictionary rather than assumed categories. By identifying these specific service attributes and reformulating the classification task as a Natural Language Inference problem [60], the model can isolate and evaluate individual review-aspect pairs efficiently.

However, fine-tuning the classification engine requires a large volume of ground truth labels, and getting those labels manually is where the process tends to slow down considerably. To work around this, the methodology draws on the principles of the Enhanced Active learning framework for Aspect based Sentiment Analysis [6]. Instead of labeling everything or pulling samples at random, the pipeline uses uncertainty and sample

diversity metrics to identify only the reviews that are genuinely ambiguous and linguistically distinct from one another. By narrowing the annotation queue down to these complex, non redundant cases, human effort gets directed toward the instances that actually matter rather than being spread thin across data that would not meaningfully improve the model [6].

The framework uses the Self-Instruct methodology [67] to address data scarcity without triggering model collapse. Since mixing authentic data does not completely reverse the drop in linguistic diversity caused by relying heavily on AI text [22], the pipeline avoids bulk synthetic generation entirely. Instead, it applies He et al.'s [23] Weighting-Based Stabilization Framework. Synthetic data acts strictly as a targeted fallback capped at 30% of the training volume. This hard limit keeps the system safely below the 38.2% threshold of generative degradation.

Taken together, these approaches form a comprehensive theoretical framework that integrates XLM-RoBERTa's robust classification capabilities, generative Aspect Taxonomy discovery, Active Learning driven by Fleiss' Kappa validation, and mathematically constrained synthetic augmentation. The literature supports this specific combination as a highly effective architecture to balance class distribution, minimize human annotation costs, and mitigate generative collapse, all while preserving the complex, code-switched syntax that makes localized Bislish and Taglish reviews worth studying.

Chapter 3

Methodology

3.1 Conceptual Framework

The study's conceptual framework will be described using the Input-Process-Output (IPO) model, as illustrated in Figure 9, which maps the transformation of unstructured, multilingual consumer reviews into actionable market intelligence. The IPO model illustrated the flow of data from raw collection to final classification. Emphasizing the combined application of web scraping, active learning, synthetic data augmentation, and transformer-based sentiment analysis, thus, effectively addressing the limitations of traditional review evaluation in localized, code-switched domains.

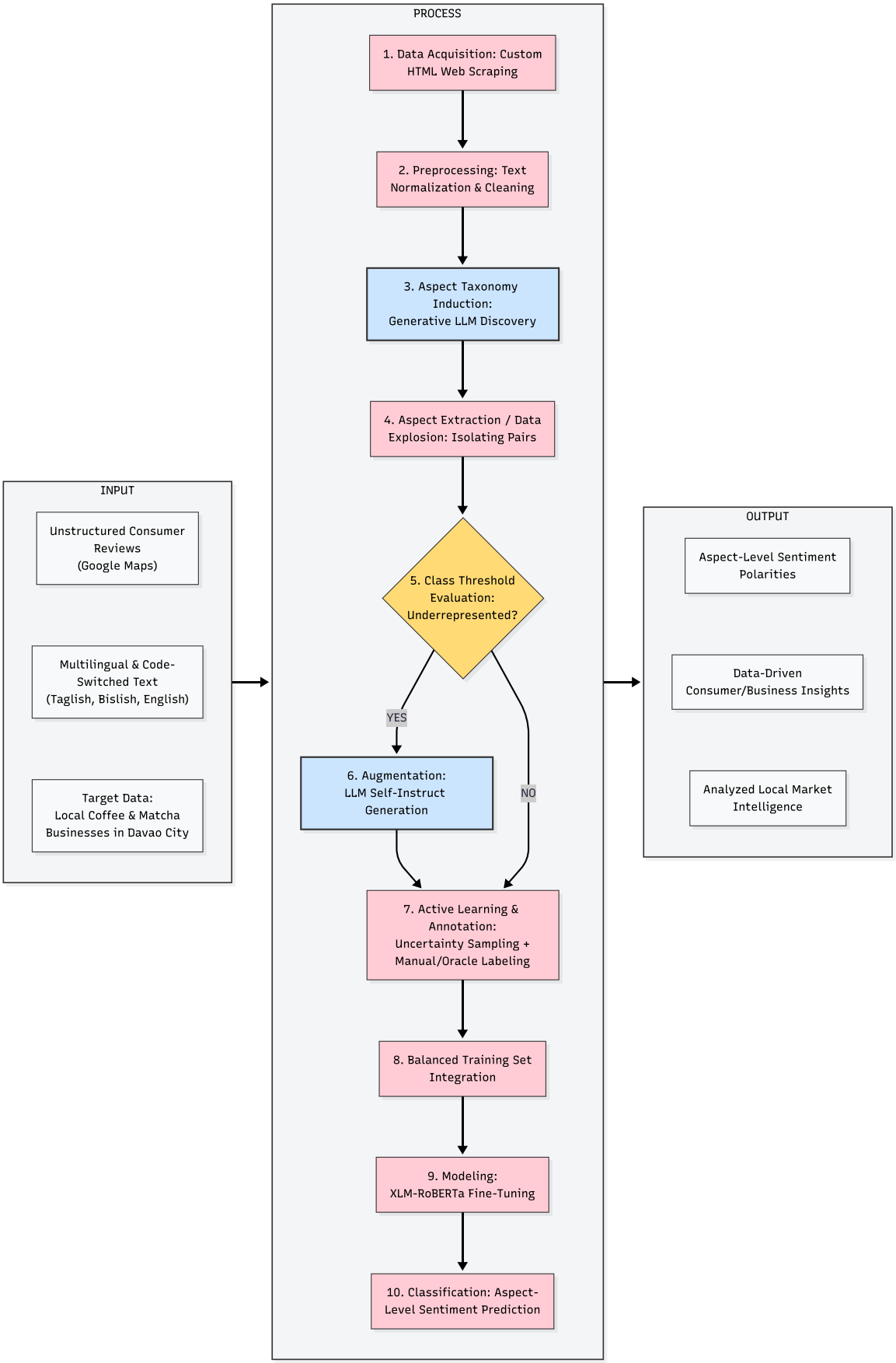


Figure 9: *IPO Chart (Input-Process-Output)*

3.2 Input Phase

This defines the foundation for the data requirement. The primary data source consists of unstructured consumer reviews of local coffee and matcha businesses in Davao City. These reviews are gathered through scraping from Google Maps reviews. The collected raw reviews are then stored in segregated CSV files for coffee and matcha shops, prior to entering the processing pipeline. Its linguistic complexity is one of its unique characteristics. It is multilingual and features heavy code-switching among English, Tagalog (Taglish), and Bisaya (Bislish) [46].

3.3 Process Phase

This outlines the machine learning and data engineering pipeline for handling the input data.

1. **Custom HTML Scraping:** A sufficient volume of reviews is needed, and this requires going beyond the strict extraction limit of only 5 instances imposed by the standard Google Places API [21]. Data is collected from Google Maps through a semi-automated DOM scraping workflow using a custom JavaScript-based scraping utility [65]. Researchers manually navigate to each target establishment, load and scroll through the review section, and initiate or terminate the scraping process as needed. Once the scraper is running, the script automates the repetitive detection and extraction of review elements from the loaded DOM, including star ratings and review text, while also performing basic text cleaning and duplicate detection. The extracted records are exported into CSV files, with the output split into separate files when necessary to facilitate data handling and subsequent preprocessing.

2. **Normalization and Cleaning:** Noisy, unstructured text undergoes preprocessing to correct informal spelling, normalize special characters, and convert emojis to textual representations. Thus, ensuring clean and consistent data for model ingestion [58].

3. **Generative Aspect Taxonomy Induction:** Rather than relying on assumed, static industry categories, generative Large Language Models (LLMs) are used to discover recurring themes and implicit operational dimensions specific to the regional Davao City consumer context (e.g., [Wi-Fi Reliability], [Parking Availability]). This establishes a data-driven master dictionary [16].

4. **Aspect Extraction:** Because reviews frequently contain mixed sentiments directed at different attributes, traditional document-level classification is insufficient. In this stage, the model isolates specific areview-aspect pairs (e.g., Taste, Price, Service, Ambiance) from the unstructured text. This serves as a prerequisite mechanism to break down complex sentences into granular, analyzable targets before any sentiment is evaluated [60].

5. **LLM Synthetic Augmentation:** To catch class imbalances early in the pipeline, a class deficit evaluation checks the distribution across the isolated aspect-sentiment dimensions following aspect extraction. Rather than unconstrained oversampling,

which risks introducing synthetic noise and inducing generative collapse [22], augmentation is deployed strictly as a targeted corrective measure for severely underrepresented classes. Synthetic generation triggers only for minority complaint categories that fail to reach a viable sample-size floor, supplementing them up to a functional operational volume [44] while remaining bounded by the 30% mathematical stabilization ceiling derived from recursive learning theory [23].

To avoid model collapse [22], the threshold acts as a dynamic ceiling instead of a hard cutoff. When a minority category drops below its baseline, the pipeline triggers Gemini 3 [61] to synthesize more samples. Following the stabilization framework from He et al. [23], the system caps this generation at 30% of the total training volume. The process stops the moment the class hits the empirical threshold, matches the median frequency of natural classes [44], or reaches the 30% limit. Whichever condition happens first stops the script. This setup prevents runaway augmentation and mathematically blocks generative degradation.

6. **Active Learning and Manual Labeling:** To overcome the bottleneck of massive manual annotation and verify the quality of the dataset, the pipeline deploys the EASE framework [6]. Instead of randomly labeling data, the system uses informative and diverse sampling to selectively flag highly uncertain and non-redundant textual edge cases. These filtered samples are then routed to human annotators who perform manual labeling, acting as the "oracle" to provide definitive ground-truth sentiment labels for the most ambiguous code-switched text while drastically reducing overall human effort.

7. **Multilingual Transformer Fine-Tuning:** A pre-trained Cross-lingual Language Model (XLM-RoBERTa) serves as the core linguistic engine. In this phase, the model is actively fine-tuned using the study's constructed dataset to accurately interpret the contextual relationships and complex syntax of the localized, code-switched text. Building on the framework established by Maceda et al. [37], this study introduces several architectural modifications. Where the baseline study used a static, manually annotated dataset for general ternary sentiment classification, this pipeline works with a dynamic, hybrid dataset that combines human-annotated edge cases and strictly regulated LLM-generated synthetic data.

8. **Classification (Aspect-Level Sentiment Prediction):** Once the target aspects have been extracted and the model fine-tuned, the pipeline evaluates the emotional polarity attached to each individual feature [60]. The model classifies the sentiment specific to that isolated review-aspect pair, enabling a highly nuanced and granular understanding of consumer feedback [58][7].

3.4 Output Phase

The output phase represents the final deliverables after the process. This will present Aspect-level polarities, Consumer insights, and Local Market Intelligence, transforming raw, unstructured feedback into metrics. Relying on an ambiguous average star rating often fails to provide specific operational insights. Extraction of granular aspect-level po-

larities allows consumers to compare establishments [29]. These extracted insights provide businesses with market intelligence enabling them to pinpoint operational strengths and address targeted areas for improvement [40] [58].

The conceptual framework demonstrated how challenges of processing large volumes of unstructured, code-switched local review can be effectively managed. Routing raw text through a robust and detailed pipeline of active learning, augmentation, and transformer-based aspect classification. It bridges the gap between online reviews and structured, actionable intelligence. Further discussions of technical configurations, procedures, and executions will be detailed in the succeeding methodology.

3.5 Research Design

The research follows a multi-stage computational pipeline designed to perform Aspect-Based Sentiment Analysis (ABSA) on code-switched consumer feedback. The architecture integrates transformer-based deep learning, generative artificial intelligence for taxonomy discovery, and human-in-the-loop active learning to provide granular market intelligence.

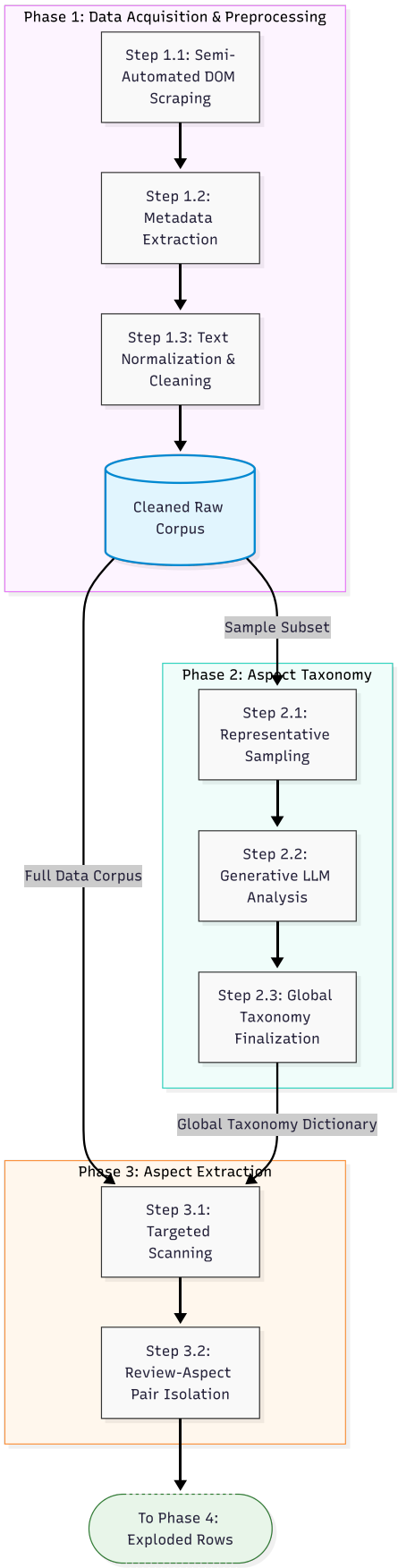


Figure 10: *Data Acquisition and Aspect Extraction*

3.5.1 Phase 1: Data Acquisition and Textual Preprocessing

The initial phase, as shown in Figure 10, is the automated harvesting and textual normalization of the primary research corpus.

1. **Step 1.1: Semi-Automated DOM Scraping:** Bypassed Google Places API restrictions using a custom JavaScript scraper executed directly in the browser console. Researchers manually navigate to a venue's review tab and run the script while scrolling to trigger lazy loading. As new content renders, the script continuously extracts the coffee shop's name, geographical coordinates, star ratings, and raw review text. It handles data persistence in the background, caching the extracted nodes in memory before the virtual DOM unmounts them. The researcher monitors the feed and halts execution when the reviews run out or the page layout breaks. The script then compiles the persisted data and exports it directly into a CSV file for that specific establishment.

2. **Step 1.2: Data Extraction:** For each retrieved entry, the system extracts the raw review text including numerical star ratings, geographic coordinates, and establishment identifiers.

3. **Step 1.3: Textual Normalization:** Utilizing the *Pandas* library and Regular Expressions (Regex), the corpus undergoes rigorous cleaning. This process involves the removal of HTML artifacts, normalization of idiosyncratic characters, and the conversion of emoticons into descriptive text to preserve semantic sentiment cues during model training.

3.5.2 Phase 2: Aspect Taxonomy

Rather than adopting a static industry standard taxonomy or relying on traditional unsupervised topic modeling like Latent Dirichlet Allocation (LDA), which already tends to struggle with the noisy, informal syntax of code-switched regional text, this research uses a direct generative approach to identify localized operational dimensions instead. Recent work by Boughanmi et al. [9] shows that large language models can directly produce semantic taxonomies that match or even outperform human coding, without needing an intermediate LDA step.

1. **Step 2.1: Representative Stratified Sampling:** A subset of normalized reviews is randomly sampled to serve as a representative seed for exploratory analysis.

2. **Step 2.2: Generative Semantic Analysis:** The sample is processed via a Generative Large Language Model (Gemini 3) [61]. The prompt design explicitly instructs the model to act as a qualitative researcher, tasked with identifying recurring linguistic patterns, frequent noun phrases, and implicit themes specific to the regional Davao City consumer context (e.g., isolating Bislish terms for taste or ambiance).

3. **Step 2.3: Global Taxonomy Formalization and Validation:** A master dictionary of aspects is generated based on the LLM output. Before deployment, this dictionary undergoes a human-in-the-loop validation check by the researchers to

ensure context-specific dimensions, such as [Parking Availability] and [Wi-Fi Reliability], are accurately captured alongside traditional metrics.

3.5.3 Phase 3: Granular Aspect Extraction and Data Decomposition

To achieve high-resolution sentiment mapping, unstructured feedback is decomposed into granular units. Refer to Figure 10.

1. **Step 3.1: Targeted Scanning and Sentence-Pair Formatting:** An extraction algorithm utilizes the formal taxonomy dictionary to scan the comprehensive review dataset. Using the Python *re* module and *nltk.sent_tokenize*, reviews are split into individual sentences. Strict regular expressions (with word boundary anchors) mapped to the LLM-discovered dimensions isolate explicit and implicit aspect mentions. Adapting the input strategy by Suryabrata et al. [60], the extracted text is then explicitly formatted into sentence-pairs (e.g., *[Review Text] + [Target Aspect]*) to provide the downstream transformer with rigorous structural context before ternary sentiment classification.
2. **Step 3.2: Review-Aspect Pair Isolation:** This process, referred to as the "Data Explosion," partitions multi-aspect reviews into structured rows using *Pandas* (*df.explode*). Each row maps a specific textual segment to a single target aspect. This prevents "sentiment cancellation," in which a single review shows conflicting polarities across different operational dimensions.

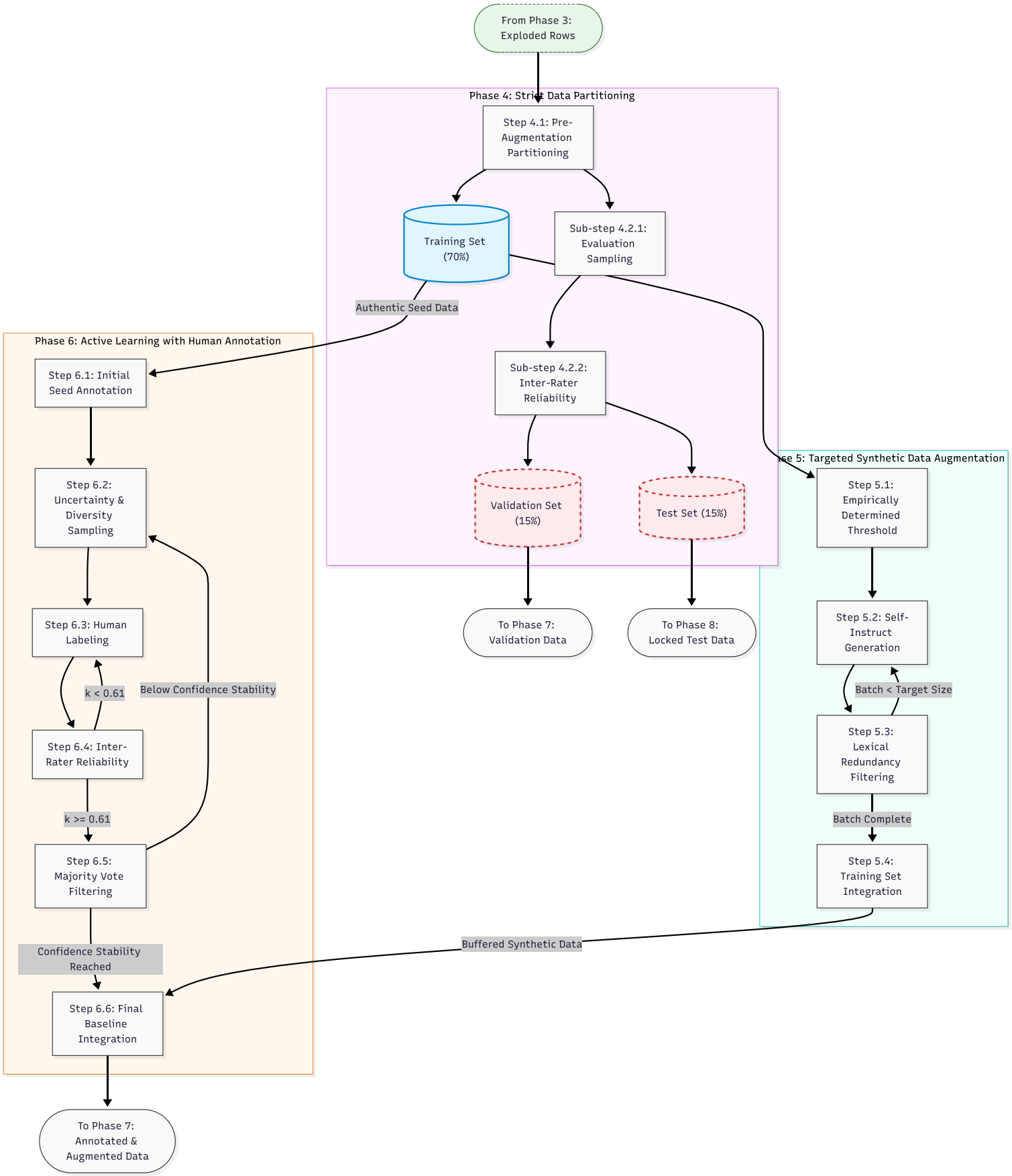


Figure 11: *P*artitioning, Augmentation and Annotation

3.5.4 Phase 4: Strict Data Partitioning

To maintain experimental integrity and preclude data leakage, strict partitioning is enforced, as shown in Figure 11.

1. **Step 4.1: Pre-Augmentation Partitioning:** The authentic dataset is partitioned into Training (70%), Validation (15%), and Test (15%) sets. This was done to prevent data leakage and maintain experimental integrity. Additionally, a composite variable combining the target aspect and the sentiment polarity (e.g., "Customer Service 1-Star") is generated before splitting to ensure class imbalances are proportionally distributed across all sets, preventing a scenario where the Test set entirely lacks minority negative sentiments.

2. **Step 4.2: Evaluation Firewall:** The Validation and Test sets are isolated from all synthetic data and training iterations. This ensures that final performance metrics represent the model's genuine generalizability to authentic human language.

   (a) **Sub-step 4.2.1: Evaluation Sampling:** Raw star ratings fail as an exact proxy for aspect-level sentiment. To fix this, the researchers built a human-annotated gold standard. To offset the skew of raw ratings [26], the researchers pulled a stratified sample of 400 exploded review-aspect pairs from both the Validation and Test partitions for an 800-instance evaluation corpus. To ensure sufficient representation of negative and neutral cases across all operational dimensions, the sampling was stratified jointly by target aspect and proxy star rating (1 to 2 stars, 3 stars, and 4 to 5 stars). Real-world annotation capacity, not a theoretical statistical power target, dictated this sample size. Three independent annotators evaluated every instance, aligning with recent verification practices in ABSA benchmark construction [57]. Thus, a 400-instance partition requires 1,200 individual annotations, which represents the largest workload the team could complete without sacrificing label quality.

   (b) **Sub-step 4.2.2: Inter-Rater Reliability Validation:** Agreement across the three annotators is statistically verified using Fleiss' Kappa [19]. Discrepancies are resolved through majority consensus to establish the definitive answer keys for hyperparameter validation and final benchmarking.

3.5.5 Phase 5: Targeted Synthetic Data Augmentation

To calculate sufficient data thresholds. The dataset in phase 5, which originally has 1-to-5 star ratings are converted into proxy labels (e.g 1-2 stars mapped "Negative'). These proxy labels are used exclusively to train a TF-IDF and Logistic Regression baseline to plot learning curves. To avoid class imbalances found in niche operational categories, synthethic data augmentation is strictly utilized as a fallback mechanism, shown in Figure 11.

1. **Step 5.1: Minority Class Deficit and Threshold Evaluation:** Rather than arbitrary unconstrained oversampling, which risks introducing synthetic noise and inducing generative collapse [22], data sufficiency across the review-aspect partitions

was evaluated by establishing a minimum functional operational floor for underrepresented categories [44]. While dominant dimensions naturally satisfied training volume requirements, minority complaint categories (specifically Negative Ambiance, Facilities, Price, and Store Operations) exhibited severe data deficits. Rather than artificially forcing parity with majority classes, target augmentation volumes were calibrated to supply sufficient gradient signals to the minority loss space [60, 37] while strictly respecting the 30% mathematical stabilization ceiling derived from recursive learning theory [23].

2. **Step 5.2: Self-Instruct Generation:** If a specific category and star rating remain underrepresented and cannot naturally meet the empirically determined threshold, a Large Language Model (Gemini 3) [61] will be utilized. It will use the Self-Instruct framework to synthesize additional training samples. Authentic reviews serve as "seed context" to ensure that the synthetic output accurately replicates the regional Bislish and Taglish syntactic nuances.

3. **Step 5.3: Lexical Redundancy Filtering (TF-IDF):** Synthetic data generation often produces repetitive phrasing that dilutes the dataset. To enforce true lexical variance, candidate synthetic rows are transformed into sparse feature vectors using the TF-IDF vectorizer from Step 5.1. Pairwise cosine similarity is then computed both across the new synthetic batch to isolate internal duplicates and against the authentic baseline data. If a synthetic instance reaches a similarity score of ≥ 0.85, it is discarded. Because this pruning shrinks the batch below target dimensions, the generation pipeline operates as a continuous loop. Exact class deficits are recalculated, and subsequent generation prompts are updated with negative constraints, explicitly banning the n-grams of the rejected text. This cycle repeats until the precise volume of structurally distinct Bislish and Taglish reviews required for the final dataset is achieved.

4. **Step 5.4: Synthetic Data Buffering:** Rather than immediately merging the synthetic data into the training corpus, the generated instances are temporarily isolated. This strict separation guarantees that the human peer annotators in Phase 6 evaluate only authentic consumer text, preventing wasted manual effort on AI-generated reviews.

3.5.6 Phase 6: Active Learning with Human Annotation

Annotation efficiency is maximized through an iterative labeling process (see Figure 11).

1. **Step 6.1: Initial Seed Annotation:** Since distant supervision proxy labels utilized in Phase 5 contain inherent noise (e.g., a five-star review expressing a specific operational complaint), manual intervention is strictly required to establish definitive ground truth. Four (4) Domain-Competent Peer Annotators, comprising undergraduate students from Ateneo de Davao University who are native speakers of regional Bislish and Taglish dialects, will review a seed batch of the authentic training data. Rather than labeling from a blank slate, their task is to validate or correct the weak proxy labels. Their localized linguistic fluency qualifies them to accurately

interpret highly contextual code-switching, thereby establishing the clean baseline required to initialize the XLM-RoBERTa active learning querying engine.

2. **Step 6.2: Uncertainty and Diversity Sampling:** A preliminary baseline instantiation of the XLM-RoBERTa model is trained on the seed data to act as the active learning querying engine. This in-loop model flags "low-confidence" predictions (where the probability margin between the top predicted classes is minimal) and non-redundant samples [44]. These instances are prioritized for human intervention, ensuring that manual labeling efforts are focused strictly on the most ambiguous segments rather than redundant data [6].

3. **Step 6.3: Human Labeling:** The flagged edge cases are routed back to the Domain-Competent Peer Annotators for a targeted round of manual annotation.

4. **Step 6.4: Inter-Rater Reliability (Fleiss' Kappa):** The overall consistency of the annotations is mathematically evaluated using the Fleiss' Kappa statistic [19]. To ensure high-quality ground truth, the study aims for a rigorous standard. If the batch achieves a score of $k \geq 0.61$ (Substantial Agreement) [33][32], it is validated and proceeds to the next step. If a batch falls below this threshold, it triggers a mandatory recalibration phase where annotators must review discordant instances, refine guidelines, and re-annotate until the substantial reliability threshold is met.

5. **Step 6.5: Majority Vote Filtering:** For the mathematically validated batch, individual ground-truth labels are finalized using a strict majority-voting mechanism. In the event of an exact 2-2 tie among the four peer annotators, a fifth tie-breaker (the lead researcher) manually adjudicates the instance. This guarantees that all 500 budgeted rows are safely preserved and passed to the fine-tuning phase without discarding expensive human-labeled data.

6. **Step 6.6: Final Baseline Integration for Phase 7:** Upon meeting the active learning stopping criterion, the final human-annotated authentic dataset (500 instances) is officially merged with the targeted synthetic instances held in the Phase 5 buffer (214 instances). This final integration strictly caps the synthetic influence at exactly 29.97% of the overall training volume. This step acts as a final structural safeguard, resolving any remaining minority class deficits without overriding the authentic linguistic feature space [23], thereby establishing a balanced 714-row baseline for Phase 7 model training.

Since active learning runs as an iterative cycle, a stopping criterion is enforced to keep the process from querying indefinitely. The annotation loop repeats until the model reaches Confidence Stability, which is the point where the remaining unlabeled pool no longer produces instances that fall below the uncertainty threshold. As a logistical failsafe, the loop will also stop if it hits a predefined maximum annotation budget, essentially a hard cap on the total number of instances based on the time and availability of the volunteer peer annotators. Once either condition is met, the active learning phase ends and the dataset moves to Phase 7.

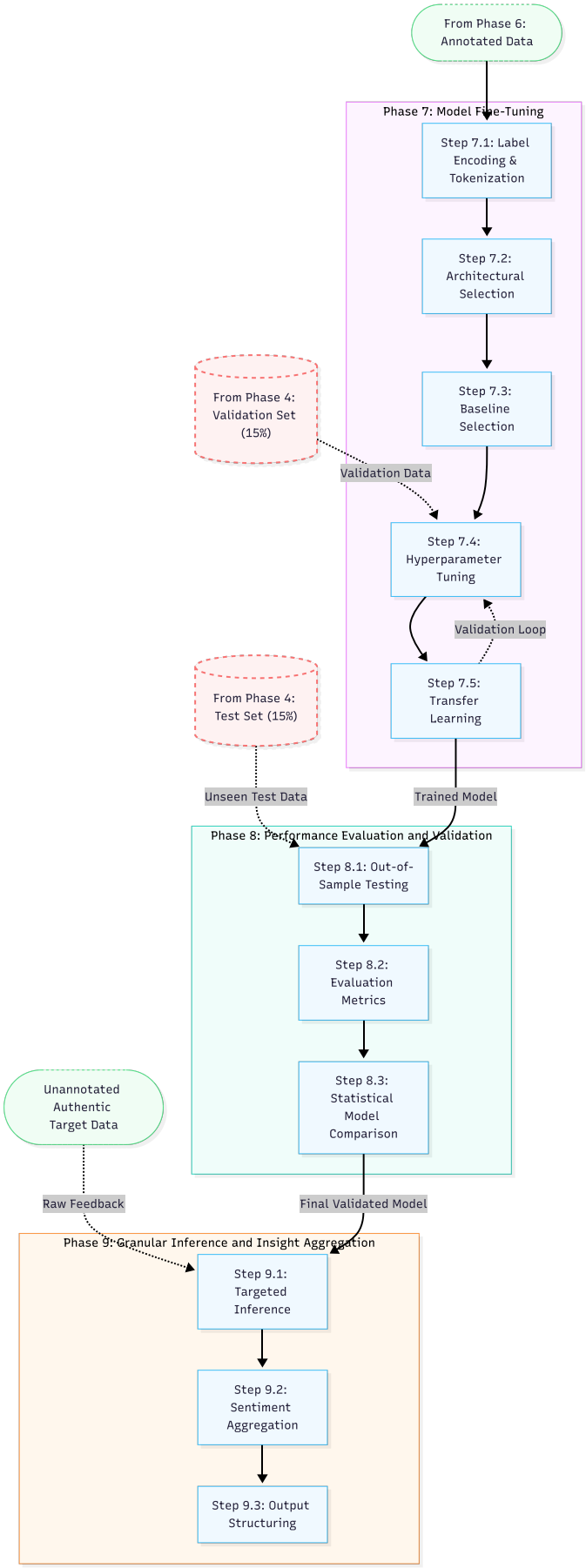


Figure 12: *Modeling and Evaluation*

3.5.7 Experimental Setup and Reproducibility

Hardware and Software Environment: All computational experiments, data preprocessing, and model fine-tuning are executed on a custom PC machine equipped with a Ryzen 7 5800x processor, 16GB of RAM, and an NVIDIA RTX 3050 (8GB VRAM) GPU. The software environment operates on Windows 11 using Python 3.10. The core machine learning pipeline relies on PyTorch (v2.5.1+cu121) [48] as the primary tensor backend and the Hugging Face Transformers library (v5.17.0) [69] for model architecture and tokenizer implementations. Data manipulation and numerical operations are handled via Pandas (v3.0.5) and NumPy (v2.5.1), while Scikit-Learn (v1.9.0) [50] is utilized for baseline traditional machine learning models and metric evaluations.

Version Control: To ensure full transparency and reproducibility, the source code, dataset samples, and preprocessing scripts are version-controlled and publicly accessible via a GitHub repository (`https://github.com/z-sulit/Thesis-Project-Aspect-Based-Sentiment-Analysis`). The repository includes a comprehensive `README.md` detailing dependency installation via `requirements.txt` and execution instructions for running a minimal working example of the fine-tuning pipeline.

3.5.8 Phase 7: Model Fine-Tuning

The core sentiment engine is optimized for the code-switched linguistic landscape of the Philippines (see Figure 12).

1. **Step 7.1: Label Encoding and Tokenization:** The pipeline formats the training corpus for PyTorch execution by mapping categorical sentiment targets directly to zero-indexed integers: Positive as 0, Neutral as 1, and Negative as 2. This categorical indexing is strictly required to formulate the multi-class training loss objective. Concurrently, the pre-trained XLM-RoBERTa SentencePiece tokenizer processes the paired review-aspect sequences, generating attention masks and padded input ID tensors bounded by the task sequence constraints [48].

2. **Step 7.2: Architectural Selection:** The research implements the xlm-roberta-base architecture (270M parameters) via the Hugging Face Transformers library [69]. XLM-RoBERTa was selected over monolingual baselines due to its larger shared SentencePiece vocabulary (250,000 tokens) and extensive pre-training on a 2.5TB Common Crawl corpus spanning 100 languages, which explicitly includes low-resource representation for Tagalog and Cebuano (Bisaya) [55]. This extensive vocabulary coverage mitigates out-of-vocabulary subword fragmentation when parsing morphologically rich Taglish and Bislish text.

3. **Step 7.3: Baseline Selection:** This step establishes the comparative benchmarks for Phase 8 evaluation across identical data partitions and matching human-annotated rows, ensuring valid pairwise statistical comparisons.

   (a) **VADER Lexicon:** Deployed as the rule-based lexical benchmark [28]. As a static, English-oriented lexicon, it inherently lacks representation for regional sentiment roots (e.g., the Bisaya adjective *lami*) and unstandardized code-switched morphology [8]. Because VADER evaluates global sentence polarity

without native aspect conditioning, its compound polarity scores were mapped to discrete classes using the canonical thresholds (≥ 0.05 for Positive, ≤ -0.05 for Negative, and intermediate scores for Neutral) established by Hutto and Gilbert [28].

(b) **SVM with TF-IDF:** This setup serves as the traditional machine learning baseline following Suciati and Budi [58]. The script inserts the target aspect directly into the review segment to force strict aspect conditioning. The architecture demands a fair benchmark against the fine-tuned transformer. To guarantee this, a 5-fold cross-validated grid search runs directly on the training split. This search locks in the optimal n-gram range, maximum feature vocabulary, regularization penalty C, and cost-sensitive class balancing. This baseline directly tests whether sparse, rigid n-gram representations can actually separate aspect-level sentiment boundaries when the text suffers from massive lexical overlap.

4. **Step 7.4: Hyperparameter Optimization and Model Selection:** Hyperparameter tuning was conducted to balance convergence stability against collapse on small domain batches [13, 38].

(a) **Search Space and Strategy:** The pipeline executes a Grid Search to test hyperparameters against established transformer benchmarks [38]. The setup locks a random seed of 42 across the PyTorch, NumPy, and Python environments. This hardcoded seed guarantees deterministic DataLoader shuffling and identical dropout initialization during every run. The search tests learning rates of 2e-05, 3e-05, and 5e-05 against batch sizes of 16 and 32. The 8GB VRAM limit of the NVIDIA RTX 3050 GPU physically caps the maximum batch size at 32.

(b) **Regularization and Capacity Control:** The AdamW optimizer applies a decoupled weight decay of 0.1 to penalize extreme parameter drift and control gradient spikes [36]. All active transformer layers retain the standard 0.1 dropout rate to stop features from co-adapting. A massive 270M-parameter architecture will instantly memorize a tiny 714-row training set if left unchecked. Thus, to physically prevent this, freezing the embedding layer and the bottom six encoder layers (layers 0 through 5) is necessary. This locks in XLM-RoBERTa's pre-trained cross-lingual representations while severely restricting the trainable parameter space.

(c) **Loss Formulation and Learning Rate Schedule:** The pipeline implements Inverse Class-Weighted Cross-Entropy Loss. This configuration directly addresses severe minority class starvation without distorting decision boundaries. The system computes class weights using balanced inverse frequencies:

$$w_c = \frac{N}{K \cdot N_c}$$

It assigns a penalty weight to the starving class. The optimization schedule runs a 10% linear warmup before dropping into a standard linear decay. This

ramp-up is strictly necessary. It prevents the massive minority loss gradients from instantly destabilizing the unfrozen upper encoder layers during early iterations. This cost-sensitive architecture mirrors the exact approaches validated by Suryabrata et al. [60] and Maceda et al. [37]. Their work proved that weighted loss functions penalize minority errors. This approach entirely bypasses the severe overfitting risks caused by naively oversampling messy, code-switched text.

(d) **Epoch Allocation and Final Checkpoint Selection:** Training operates under a strict 5-epoch ceiling. The system uses dynamic checkpointing to aggressively cut off overfitting. Checkpoint selection relies purely on peak Validation Macro-F1 to prioritize actual minority class boundary separation over basic loss fluctuations. The script evaluates the model weights at the end of every epoch. It then automatically restores and saves the state dictionary that hit the highest validation Macro-F1 score as the final optimal model.

5. **Step 7.5: Transfer Learning and Task Adaptation:** The model parameters are updated across the balanced 714-row training partition, comprising 500 authentic human-annotated rows integrated with 214 collision-filtered synthetic instances capped at 30%. This fine-tuning adapts the transformer's upper representational layers to the domain-specific semantics of localized coffee and matcha reviews, preparing the network for out-of-sample evaluation on the 100% authentic test firewall in Phase 8.

3.5.9 Phase 8: Performance Evaluation and Validation

The pipeline's effectiveness is validated against the actual data (see Figure 12).

1. **Step 8.1: Out-of-Sample Testing:** The fine-tuned XLM-RoBERTa model is evaluated using the 400-instance human-annotated Test sample established in Step 4.2.1. This ensures that performance metrics are calculated strictly against verified, authentic human feedback rather than noisy proxy ratings.

2. **Step 8.2: Evaluation Metrics:** Model performance is evaluated using Precision, Recall, and the F1-Score [71]. Given the inherent class imbalance of consumer review data, the Macro-F1 score is the primary evaluation metric. Macro-averaging computes the metric independently for each class and then takes the unweighted mean, ensuring the model is penalized if it defaults to predicting the dominant positive sentiment class while failing on the minority negative classes [59].

3. **Step 8.3: Statistical Model Comparison:** To determine if the performance difference between the fine-tuned XLM-RoBERTa model and the lexicon-based VADER baseline is statistically significant, the study employs McNemar's Test [41]. This non-parametric test is specifically designed for comparing two classifiers evaluated on the exact same 400-instance nominal test set. It isolates the discordant predictions (cases where one model predicts correctly and the other fails) to calculate the test statistic. The equation is defined as:

$$\chi^2 = \frac{(b-c)^2}{b+c}$$

Where b represents the number of test instances where XLM-RoBERTa was correct and VADER was incorrect, and c represents the number of instances where VADER was correct and XLM-RoBERTa was incorrect. A significance level of $\alpha = 0.05$ will be used to reject the null hypothesis, formally satisfying the objective to compare transformer performance against traditional lexicon methods.

3.5.10 Phase 9: Granular Inference and Insight Aggregation

The final phase translates the computational outputs into structured, actionable insights designed to support decision-making for both consumers and business owners (see Figure 12).

1. **Step 9.1: Targeted Inference:** The validated model processes authentic feedback for individual target establishments to generate aspect-level sentiment predictions.
2. **Step 9.2: Sentiment Aggregation:** The predicted polarities are compiled to provide a comprehensive, quantitative overview of consumer satisfaction. The system aggregates the counts of positive, negative, and neutral sentiments for each discovered operational dimension and calculates their prevailing sentiment percentages (e.g., determining that 85% of reviews mentioning "Wi-Fi Reliability" are positive).
3. **Step 9.3: Output Structuring (Granular System Interface):** Rather than returning raw computational output, the final vision for system integration is to present these aggregated metrics in a structured, consumer facing format. The system will display a directory-style listing of local coffee and matcha establishments, replacing vague average star ratings (4.2 stars, 4.0 stars) with granular, aspect-specific scores. A specific shop might show a 75% positive sentiment for Taste but a 60% negative sentiment for Parking Availability. Following the principle of translating model outputs into something practically useful and easy to read, this structured output works as a proof of concept.

3.6 Data Collection

The primary dataset for this study consists of publicly available consumer reviews of coffee and matcha establishments in Davao City, collected from Google Maps. Although the Google Places API offers a structured way to retrieve this kind of data, it limits results to roughly five reviews per establishment. Chen et al. (2024) [12] and Trezza (2023) [63] both noted that this restriction makes APIs impractical for training machine learning models, and this study encountered the same limitation. A custom web scraper was developed to collect sufficient review data to fine-tune the XLM-RoBERTa model (Figure 13).

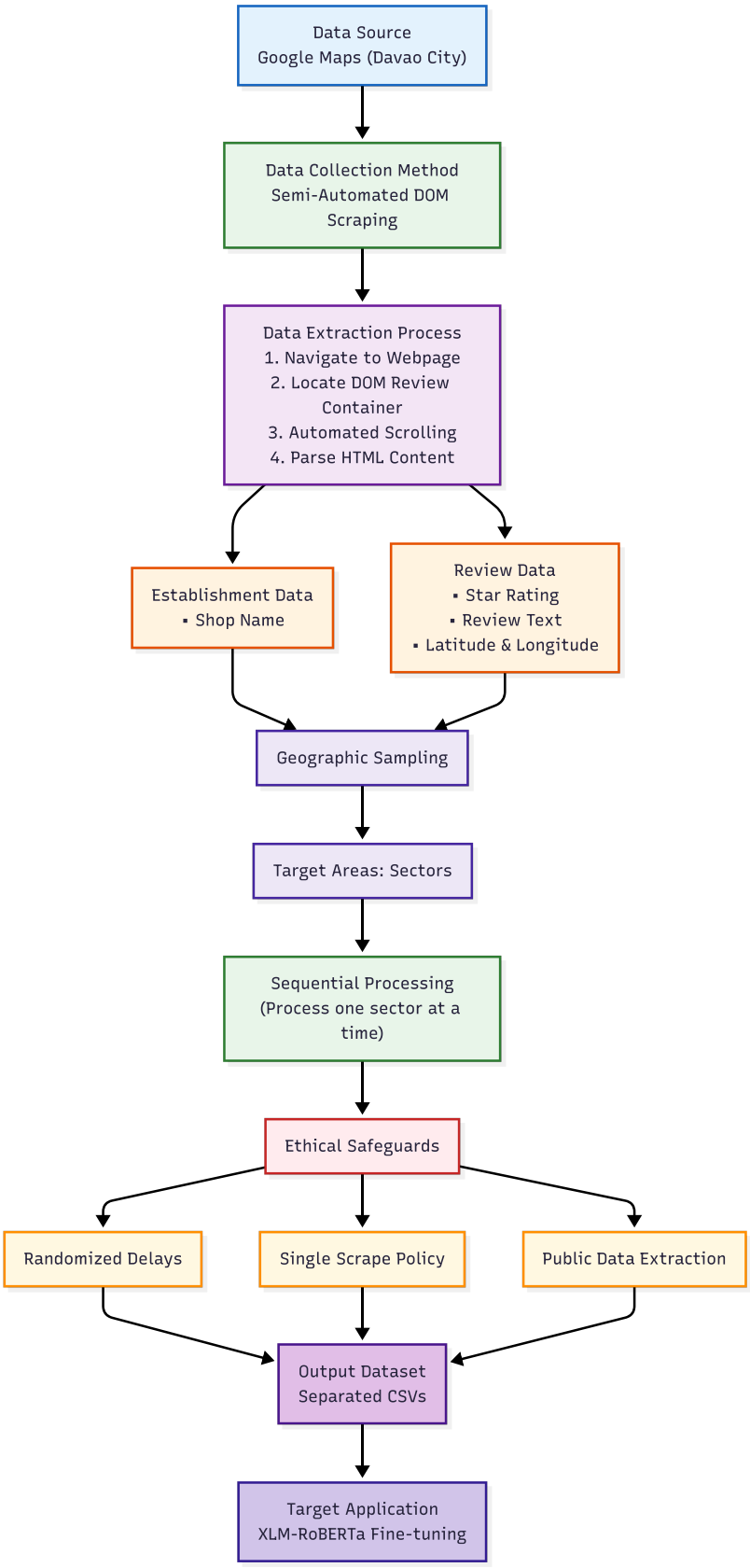


Figure 13: *Data Collection Process*

3.6.1 Extraction Architecture and Logic

Data collection was carried out using custom DOM JavaScript and Python web scraping scripts. Google Maps loads its content dynamically through JavaScript and utilizes DOM virtualization. To bypass this, a semi-automated DOM scraping approach was implemented. The review pane was manually scrolled to trigger the network requests and force the lazy-loaded content to render fully within the browser. Once the DOM was populated, custom JavaScript routines extracted the target shop's name, overall rating, and granular data for each entry, including the reviewer's name, star rating, full review text, and publication date. The extracted payload was then passed to a Python script for structuring. To maintain dataset organization, the resulting structured data was partitioned and exported into multiple CSV files, strictly categorized by business type (e.g., separating Coffee Shops and Matcha Shops into distinct files).

3.6.2 Systematic Geographic Sampling

To cover Davao City without making the data collection process unmanageable, the study divided the city into key commercial sectors such as Ecoland, Matina Crossing, Poblacion District, and others. The extraction process targeted all identified coffee and matcha shops within one sector at a time before moving on to the next, which kept the process organized and reduced the risk of geographic gaps in coverage.

3.6.3 Ethical Considerations and Technical Safeguards

Automated scraping can place unintended loads on servers and trigger IP blocks if executed recklessly. Therefore, the data collection followed polite scraping principles. The semi-automated approach naturally introduced human-like time delays during the manual scrolling phase, avoiding the rapid, predictable request patterns that trigger anti-bot protections. Each establishment was scraped only once rather than on a recurring schedule to avoid straining the host servers. All collected data was publicly visible on Google Maps and was utilized strictly for this academic study with no commercial intent.

3.7 Data Description

3.7.1 Raw Data

1. **Volume and Attrition:** The master dataset contains 19,586 total reviews across 400 unique establishments, split between 17,454 coffee shop reviews across 349 locations and 2,132 matcha shop reviews across 51 locations. A significant attrition rate is present, as 7,015 instances (6,350 coffee; 665 matcha) are rating-only entries entirely lacking text. These empty entries are systematically dropped during preprocessing, yielding a final usable textual corpus of 12,571 reviews. Furthermore, the raw corpus extracted via the custom Semi-Automated DOM Scraping is inherently susceptible to sampling gaps and selection bias.

2. **Baseline Distribution Skew:** Response bias was observed in the dataset. The

frequency of 5-star ratings vastly outnumbers all other categories combined, while 1-star, 2-star, and 3-star ratings are severely underrepresented. This massive deficit in negative baseline data establishes the empirical need for the targeted LLM synthetic augmentation phase later in the pipeline to prevent model bias.

3. **Vocabulary, Bigrams, and Linguistic Metrics:** Code-switching is a consistent feature, maintaining a 5.04% overall rate across the combined corpus (5.05% in coffee; 4.97% in matcha). The top extracted bigrams reflect broad, generic sentiment markers rather than specific operational feedback, dominated by phrases such as "good food" (462 occurrences), "nice place" (393 occurrences), and "best matcha" (58 occurrences).

   The corpus contains a high frequency of sentiment-critical emojis, led by the red heart (378 occurrences), star (227 occurrences), and thumbs-up (219 occurrences). These elements undergo systematic demojization during the normalization phase, converting symbols into standardized text tokens to preserve semantic intent before transformer fine-tuning.

Table 1: Dataset Overview Summary

Metric Description	Coffee Shops	Matcha Shops	Combined Corpus
Total Master Reviews	17,454	2,132	19,586
Reviews with Text	11,104	1,467	12,571
Rating-Only Reviews (Empty Text)	6,350	665	7,015
Unique Establishments	349	51	400

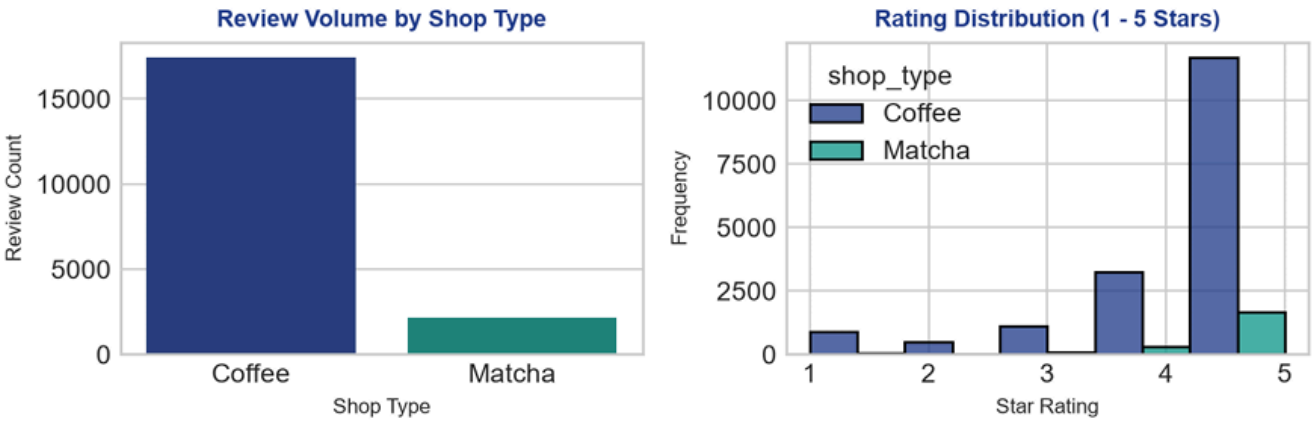


Figure 14: *Dataset Volume and Rating Distribution Skew*

Table 2: Vocabulary, Bigrams & Linguistic Metrics

Category	**Top Bigrams (Count)**	**Type-Token Ratio (TTR)**	**Code-Switching %**
Coffee	good food (462), coffee shop (430), nice place (393), food good (345), good place (273)	0.0823	5.05%
Matcha	matcha latte (73), best matcha (58), love matcha (54), good matcha (50), matcha good (39)	0.1718	4.97%
Overall	Combined corpus bigram profiling	0.0792	5.04%

Table 3: Top Frequency Emoji Analysis

Rank	**Demojized Name**	**Occurrences**
#1	:red_heart:	378
#2	:star:	227
#3	:thumbs_up:	219
#4	:smiling_face_with_smiling_eyes:	169
#5	:smiling_face_with_heart-eyes:	165
#6	:hot_beverage:	156
#7	:hundred_points:	125
#8	:face_savoring_food:	124
#9	:smiling_face_with_hearts:	120
#10	:sparkles:	108

3.7.2 Review Aspect Pairs Data

1. **Class Imbalance:** As detailed in Table 4, the dataset exhibits severe class imbalance, with Product Quality dominating at 47.88%, while Price and Value (5.18%) and Store Operations (5.24%) are severely underrepresented. Training a model on this raw distribution will cause it to overpredict the majority class and fail on the minority categories. This validates the necessity of executing Phase 5 (Targeted Synthetic Data Augmentation) to generate targeted text for these specific deficits before moving into the active learning loop.

2. **Validation of the De-composition Strategy:** Because over 54% of the scraped reviews mention two or more distinct operational aspects (Table 5), standard document-level sentiment analysis would have failed on this corpus. This high aspect density proves that the Phase 3 decision to explode the reviews into granular, sentence-level chunks was necessary for isolating targeted sentiment.

3. **Volatility in Customer Service:** While "Ambiance and Atmosphere" holds the highest average rating (4.46), Table 6 demonstrates that "Customer Service" is highly polarizing. It has the lowest mean rating (4.00) alongside the highest standard deviation (1.444), indicating that customers primarily mention service when it fails.

Table 4: Aspect Class Imbalance Summary

Target Aspect	**Count**	**Percentage (%)**
Product Quality	11,040	47.88%
Customer Service	4,611	20.00%
Ambiance and Atmosphere	3,085	13.38%
Facilities and Amenities	1,919	8.32%
Store Operations and Accessibility	1,208	5.24%
Price and Value	1,195	5.18%

Table 5: Aspects by Shop Type (Frequencies and Percentages)

Target Aspect	**Coffee (Count)**	**Matcha (Count)**	**Coffee (%)**	**Matcha (%)**
Product Quality	9,704	1,336	46.64%	59.30%
Customer Service	4,206	405	20.22%	17.98%
Ambiance and Atmosphere	2,799	286	13.45%	12.69%
Facilities and Amenities	1,820	99	8.75%	4.39%
Store Operations and Accessibility	1,155	53	5.55%	2.35%
Price and Value	1,121	74	5.39%	3.28%

Table 6: Overall Rating vs. Target Aspect

Target Aspect	**Count**	**Mean**	**Median**	**Std. Dev.**
Ambiance and Atmosphere	3,085	4.463	5.0	0.911
Product Quality	11,040	4.286	5.0	1.171
Facilities and Amenities	1,919	4.138	5.0	1.196
Price and Value	1,195	4.035	5.0	1.317
Store Operations and Accessibility	1,208	4.028	4.0	1.238
Customer Service	4,611	4.005	5.0	1.444

Table 7: Synthetic Ratio Validation

Target Aspect	**Synthetic Data %**
Ambiance and Atmosphere	6.01%
Customer Service	0.0%
Facilities and Amenities	6.47%
Price and Value	13.1%
Product Quality	0.0%
Store Operations and Accessibility	13.94%

Table 8: Cross-Tabulation: Target Aspect by Sentiment (Added Synthetic Data)

Target Aspect	**Negative**	**Neutral**	**Positive**	**Total**
Ambiance and Atmosphere	63	4	187	254
Customer Service	14	6	34	54
Facilities and Amenities	57	2	57	116
Price and Value	56	1	10	67
Product Quality	21	24	102	147
Store Operations and Accessibility	55	2	19	76
Total	**266**	**39**	**409**	**714**

3.7.3 Synthetic Data Augmentation and Ratio Validation

Table 7 confirms that the targeted augmentation protocol prevented distribution shift and generative model collapse. In Phase 5, the percentages in Table 7 measure the generated synthetic rows against the Phase 5 subsampled training pool for each respective target aspect prior to active learning filtering. By measuring against the pre-annotation category volume, the pipeline verified that synthetic text never washed out the natural category distribution. At most, synthetic data accounts for 13.94% of the Store Operations and Accessibility pool and 13.10% of Price and Value, while bypassing Customer Service and Product Quality entirely (0.0%) since their authentic data already satisfied the empirical threshold.

In contrast, Table 8 in the following section details the final 714-row active learning training set constructed in Phase 6. Within this condensed training partition, the 500 human-annotated authentic instances are merged with the 214 buffered synthetic rows (roughly 53 to 55 targeted negative instances across each of the four minority dimensions), bringing the aggregate synthetic share to 29.97% of the 714 total instances. This two-stage validation ensures that minority complaint classes receive the concentrated statistical weight needed during XLM-RoBERTa fine-tuning while remaining strictly capped below the 30% mathematical safety ceiling.

3.7.4 Active Learning and Annotations

Table 8 shows the balanced structure produced by the Phase 6 baseline integration. By injecting 30% synthetic data (214 instances), the pipeline boosted the critical minority classes, such as Negative Ambiance, Facilities, Price, and Store Operations, to a working range of 55 to 63 instances each. This targeted augmentation gives the XLM-RoBERTa model enough statistical weight to map decision boundaries for rare complaints without overfitting or triggering generative collapse.

The broader dataset distribution confirms the project avoided generative collapse. Positive sentiment still dominates with 409 total instances, keeping the model grounded in real Bislish and Taglish. Since they did not need the boost, Negative Customer Service and Product Quality were left untouched to minimize synthetic noise, holding their original counts of 14 and 21 instances. The Neutral class was also kept intentionally sparse, steering the model toward actionable business insights rather than flat observations.

3.8 Bias Mitigation and Data Integrity Controls

To ensure the authenticity of the XLM-RoBERTa model and prevent algorithmic bias during training, the dataset underwent strict control mechanisms before fine-tuning.

1. **Strict Stratified Partitioning:** The use of the 70/15/15 split implementation as an Evaluation Firewall. By locking the validation and test sets before any data processing, we can be assured that our performance metrics measure generalization to the original Taglish and Bislish text, free from the influence of training contamination.

2. **Deduplication and Noise Filtering:** Raw text corpora contain structural noise that skews transformer-based classification. To address this, the first preprocessing step dropped 7,015 rating-only entries. Since ABSA depends entirely on extracting operational dimensions from written text, empty strings offer no value. Leaving them forces the XLM-RoBERTa tokenizer to process zero-information vectors, thus wasting compute and distorting gradient updates during fine-tuning.

3. **Empirical Class Thresholding:** The use of progressive sampling to determine the threshold for the F1-score metric. It minimizes the potential problems caused by the introduction of synthetic noise that causes the issues encountered in overusing sampling of minority data classes to balance large majority classes. The values of 500 and 1,000 instances mentioned are actual empirical values obtained from results extracted from learning curves.

4. **Validation Set Isolation:** Synthetic data was injected exclusively into the Training set. The Validation and Test sets were strictly isolated to contain 100% authentic human text, completely preventing data leakage and ensuring untainted downstream evaluation metrics.

5. **Synthetic Ratio Capping and Imbalance Resolution:** Synthetic data injection was capped at 30% of the training baseline to prevent generative collapse and keep

authentic human text as the primary learning signal. Cross-tabulating target aspects by sentiment verified the augmentation, confirming the minority classes hit their necessary data thresholds without artificially inflating the dominant positive classes.

6. **Cosine Similarity Collision Filtering:** A TF-IDF Cosine Similarity filter with a drop threshold of ≥ 0.85 was applied during data augmentation. This efficiently detected and discarded synthetic outputs that regurgitated or heavily plagiarized existing authentic reviews, ensuring dataset diversity.

7. **Minority Class Deficit Targeting:** To minimize the introduction of synthetic noise and prevent generative collapse caused by unconstrained oversampling, data sufficiency was evaluated based on the baseline class frequency distribution. Augmentation was triggered exclusively for severe minority complaint classes, buffering them to a functional operational volume while strictly respecting the 30% synthetic stabilization cap.

8. **Inter-Rater Reliability Adjudication:** Four independent annotators evaluated the dataset to establish an unbiased ground truth for the authentic text. Before finalizing the labels, Fleiss' Kappa was calculated to verify inter-rater reliability. A majority-vote algorithm resolved standard disagreements. In the event of a direct tie, the lead researcher manually adjudicated the instance to keep arbitrary machine bias out of the training set.

9. **Overfitting and Optimization Safeguards** To keep the model from exploiting minority-class heuristics during fine-tuning, the pipeline applies decoupled weight decay (0.1) and standard dropout ($p = 0.1$) directly through the AdamW optimizer. Selection strictly monitors validation macro-F1 across the 5 epoch.

Chapter 4

Results and Discussion

4.1 Annotation Reliability

Table 9: Inter-Annotator Agreement across Data Sampling and Annotation Phases (interpreted via Landis and Koch [33])

Phase	Annotation Task	Sample Size	Annotators	Fleiss' Kappa (κ)	Interpretation
Phase 6	Seed Batch (Random Sample)	100	4	0.6550	Substantial Agreement
Phase 6	Active Learning Batch 2 (Uncertainty)	400	4	0.7428	Substantial Agreement
Phase 4	Validation Set Firewall (Reconciled)	400	3	1.0000	Almost Perfect Agreement
Phase 4	Test Set Firewall (Reconciled)	400	3	1.0000	Almost Perfect Agreement

Before fine-tuning the XLM-RoBERTa architecture, the human-annotated ground truth was verified using Fleiss' Kappa (κ). Phase 6 of the methodology required a baseline threshold of $\kappa \geq 0.61$ (Substantial Agreement) for dataset validation [33].

The 100-instance seed batch achieved $\kappa = 0.6550$. The active learning engine surfaced highly ambiguous edge cases in Batch 2 (400 instances), yet annotator consistency improved to $\kappa = 0.7428$. The human-in-the-loop guidelines effectively stabilized judgment against complex Taglish and Bislish syntax.

For the Phase 4 Validation and Test sets, the metric represents a 100% Post-Adjudication Consensus instead of an independent Fleiss' Kappa score. The three raters resolved initial discrepancies to establish a unified ground truth for out-of-sample testing.

4.2 Model Optimization

Table 10: Calculated Balanced Class Weights for Cross-Entropy Loss

Class Label	Target Index	Penalty Weight (w_c)
POSITIVE	0	0.5819
NEUTRAL	1	6.1026
NEGATIVE	2	0.8947

Table 11: Phase 7: XLM-RoBERTa Hyperparameter Optimization Grid Search Summary

Batch Size	Learning Rate	Epochs	Peak Val Macro-F1	Status
16	2×10^{-5}	5	0.4736	Optimal
32	3×10^{-5}	5	0.4551	Suboptimal
16	5×10^{-5}	5	0.4547	Suboptimal
16	3×10^{-5}	5	0.4536	Suboptimal
32	5×10^{-5}	5	0.4201	Suboptimal
32	2×10^{-5}	5	0.4030	Suboptimal

Table 11 documents the hyperparameter grid search that establishes the final model state. Across the six evaluated configurations, the optimal checkpoint converged using a batch size of 16 and a learning rate of 2e-5 at 3-epoch (within the 5-epoch ceiling).

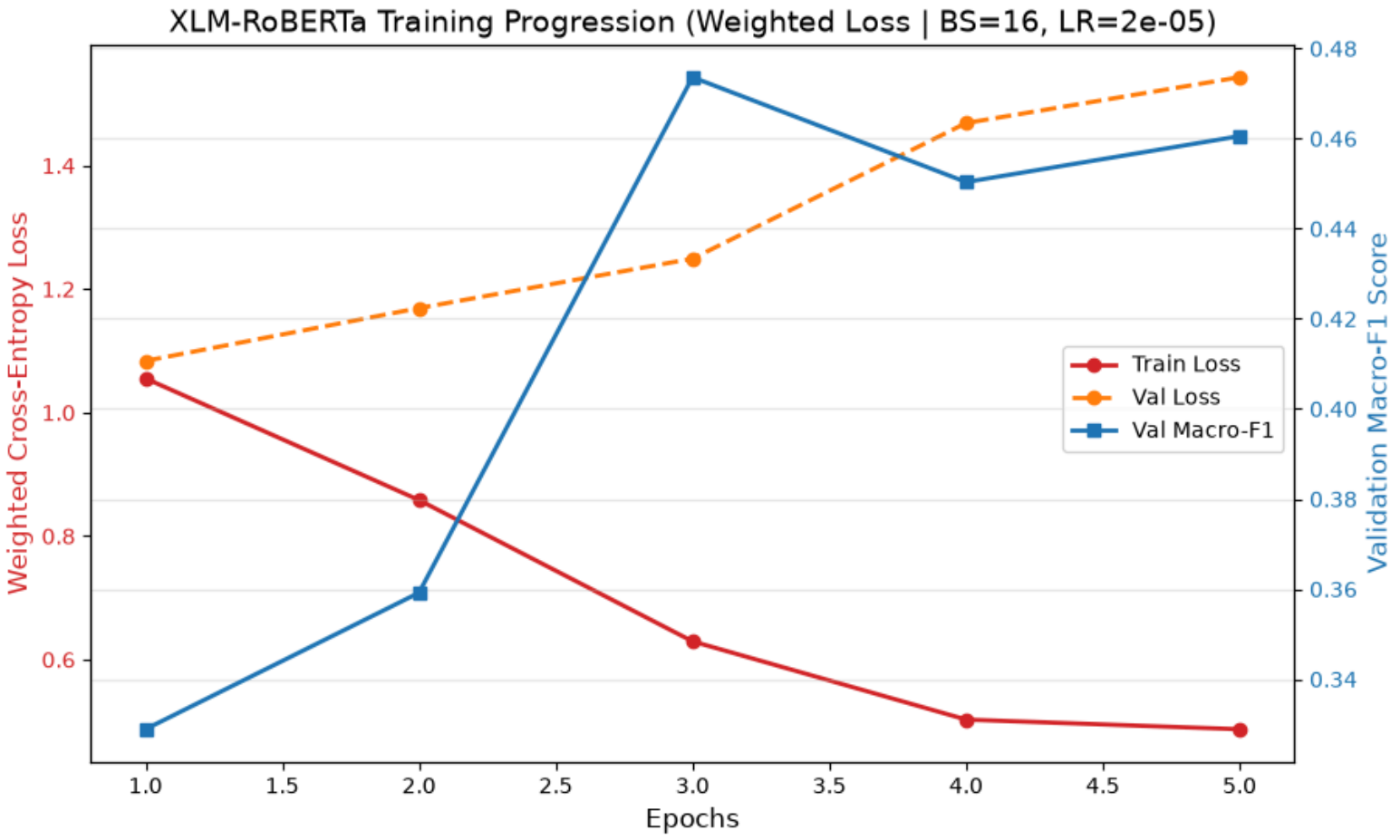


Figure 15: *Cross-Entropy Loss*

Table 12: Phase 7: XLM-RoBERTa Optimal Validation Classification Performance

Class	**Precision**	**Recall**	**F1-Score**	**Support**
POSITIVE (0)	0.7951	0.8094	0.8021	278
NEUTRAL (1)	0.1667	0.3182	0.2188	22
NEGATIVE (2)	0.4667	0.3500	0.4000	100
Accuracy			0.6675	400
Macro Avg	0.4761	0.4925	0.4736	400
Weighted Avg	0.6784	0.6675	0.6695	400

The training metrics in Figure 15 demonstrate the intentional trade-off of applying an inverse class-weighted loss function to a highly imbalanced dataset. Because the Neutral class constitutes a small fraction of the training data, the weighted Cross-Entropy Loss applies a massive penalty multiplier for minority misclassifications. Consequently, the validation loss rises continuously from Epoch 1 through Epoch 5 as the model sacrifices raw probabilistic confidence to force decision boundaries around sparse outliers.

Despite this rising validation loss, the Validation Macro-F1—the definitive metric for discrete sentiment classification—peaks distinctly at Epoch 3. The subsequent drop and fluctuation at Epochs 4 and 5 reflect the optimizer micro-adjusting and slightly overcorrecting its boundaries on the sparse validation set. This volatility actively validates the dynamic checkpointing strategy. By monitoring Macro-F1 rather than loss, the system captured the optimal decision boundary at Epoch 3 before amplified minority gradients destabilized the upper encoder layers.

By extracting the Epoch 3 checkpoint, the pipeline successfully mapped the complex Neutral and Negative boundaries, achieving a peak Validation Macro-F1 of 0.4736. Forcing a 270-million-parameter architecture to learn from a 714-row dataset naturally pushes the network toward memorization, but layer freezing, decoupled weight decay, and the Epoch 3 early cutoff successfully prevented representation collapse while optimizing operational classification power.

4.3 Performance Comparison

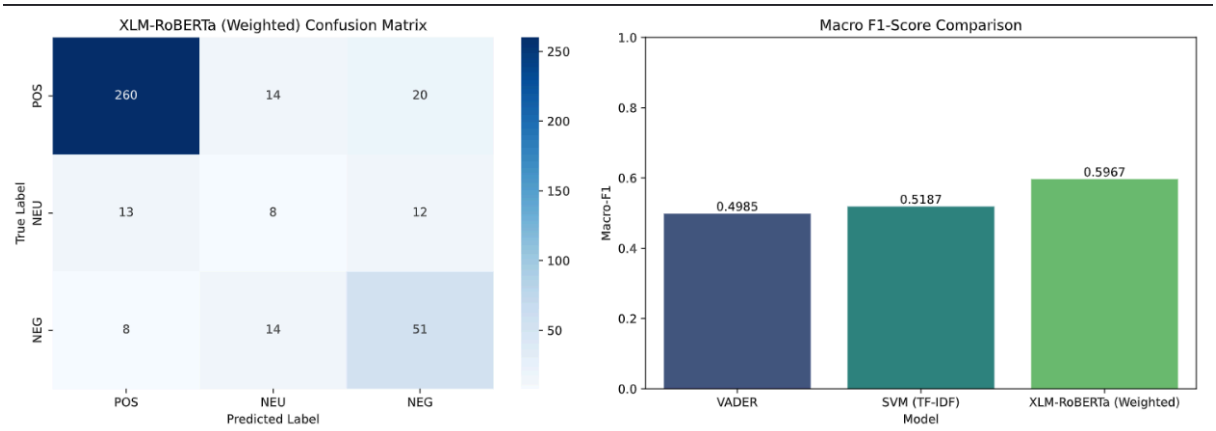


Figure 16: Confusion Matrix and Macro-f1 Score Comparison

The confusion matrix maps the decision boundaries on the out-of-sample test set ($N = 400$). XLM-RoBERTa dominated the dominant categories. It accurately caught 260 positive and 51 negative instances. The matrix also exposes a severe structural struggle with the Neutral class with the model only managing to score 8 true-positive predictions out of 33 actual neutral instances, leading to a Neutral F1-score of 0.2319.

This breakdown stems directly from the reality of consumer review datasets. Real customers rarely leave neutral feedback. They either drop polarized positive endorsements or explicit complaints. This inherent behavioral skew leaves neutral expressions extremely sparse. That lack of data actively destroys the model's ability to map a decision boundary for anything other than explicit sentiment.

Table 13: Phase 8: Final Benchmark Metrics

Model	**Accuracy**	**Precision (Macro)**	**Recall (Macro)**	**Macro-F1**	**F1 (POS)**	**F1 (NEU)**	**F1 (NEG)**
VADER	0.6875	0.5685	0.5052	0.4985	0.8293	0.2034	0.4630
SVM (TF-IDF)	0.7300	0.5123	0.5290	0.5187	0.8502	0.1875	0.5185
XLM-RoBERTa (Weighted)	0.7975	0.5873	0.6085	0.5967	0.9043	0.2319	0.6538

As shown in Table 13, XLM-RoBERTa (Weighted) hit a peak Macro-F1 of 0.5967. This cleanly beats out the rule-based VADER lexicon (0.4985) and the tuned SVM baseline (0.5187). This proves the core hypothesis. Parsing the messy code-switched Taglish and Bislish text actively demands the deep representational power of a cross-lingual transformer. Static English dictionaries and rigid n-gram filters completely fail to map this complexity.

By integrating the inverse class-weighted loss function, XLM-RoBERTa successfully prevented the representation collapse of the minority classes. It reached a 0.7975 overall accuracy, scoring a 0.9043 F1 on Positive instances and a 0.6538 F1 on Negative complaints. While VADER and the tuned SVM managed to map baseline Neutral F1 scores of 0.2034 and 0.1875, XLM-RoBERTa outperformed them both by achieving a 0.2319 F1 on the highly sparse Neutral class.

Table 14: McNemar's Statistical Test ($\alpha = 0.05$): XLM-RoBERTa vs. Baselines

Baseline	**XLM-R Correct (b)**	**Base Correct (c)**	**Chi-Square (χ^2)**	**p-value**
VADER Lexicon	76	32	17.9259	2.2967×10^{-5}
SVM (TF-IDF)	48	21	10.5652	1.1524×10^{-3}

Table 14 proves the performance gap. The transformer caught 76 instances that VADER completely misclassified ($p = 2.2967 \times 10^{-5}$). Compared against the tuned SVM baseline, XLM-RoBERTa successfully scored 48 correct predictions the traditional model missed ($p = 1.1524 \times 10^{-3}$). Both scores capped the strict 0.05 significance threshold [41]. These results mathematically proved the architectural superiority of the cross-lingual transformer over both baselines.

4.4 Error Analysis

Table 15: Qualitative Error Analysis: Correct Predictions by XLM-RoBERTa Across Discrepant Cases (Part 1: Rows 1–11)

Target Aspect	Review Text	True	VADER	SVM	XLM-R
Product Quality	Decent coffee.	POS	NEU	NEU	POS
Product Quality	REVEL by Confex is a cozy, café-style coffee shop tucked inside DBP Village along Don Julian Rodriguez Avenue in Talomo, Davao City.	POS	NEU	NEG	POS
Product Quality	Waited two hours (ordered at 8:54pm, got my drinks at 10:37!!!)	NEG	NEU	POS	NEG
Product Quality	If you're in Davao, you've got to check out Hid'n Coffee!	POS	NEU	NEU	POS
Customer Service	Food is really Good 10\10 but the staff looks like they are having holly week.	NEG	POS	POS	NEG
Product Quality	Though we understand that they do not allow outside food and we respect that.	NEU	POS	NEG	NEU
Customer Service	Very spacious and clean place you can see a big board just on the mane entrance door about wifi but no wifi , every time you go in this place is the same storie ... My tea was serve quickly but without tissue , no sugar and no spoon service is poor here, but the price is big If you want a good place with strong wifi and good service , i can recommand you coffe cat 300 m before gaisano	NEG	POS	NEU	NEG
Customer Service	very accommodating staff.	POS	NEU	NEG	POS
Product Quality	If you play Sims 4, the building design looks similar to the coffee shops in this game.	NEU	POS	POS	NEU
Product Quality	Cozy place to drink your coffee!	POS	NEU	NEU	POS
Product Quality	loudly_crying_face It literally tasted like coffee.	NEG	POS	POS	NEG

Table 16: Qualitative Error Analysis: Correct Predictions by XLM-RoBERTa Across Discrepant Cases (Part 2: Rows 12–21)

Target Aspect	**Review Text**	**True**	**VADER**	**SVM**	**XLM-R**
Product Quality	The drinks are not bad either.	NEU	POS	NEG	NEU
Product Quality	Hot latte upsize then the serving is small compared to other branches that I always buy from, then the hot drink is not hot.	NEG	NEU	POS	NEG
Product Quality	The matcha cookie was everything I hoped it would be; soft and flavorful and not too sweet.	POS	NEU	NEG	POS
Product Quality	The matcha had a strange blue layer on top that had already hardened when it was served.	NEG	NEU	POS	NEG
Store Operations and Accessibility	Small parking place loudly_crying_face	NEG	NEU	POS	NEG
Product Quality	I'm paying for it, so I don't see the issue with how I want my coffee.	NEU	POS	POS	NEU
Store Operations and Accessibility	It was refreshing to have a snack here especially after hours of walking under the sun.	POS	NEU	NEU	POS
Customer Service	Accommodating staff and owner.	POS	NEU	NEG	POS
Store Operations and Accessibility	Parking was a minor setback though.	NEG	NEU	POS	NEG
Facilities and Amenities	They have free wifi but they only have one power outlet (as per their staffs) available if you want to charge your phone batteries (and you can't bring extensions for safety purposes).	NEG	POS	NEU	NEG

Table 17: Error Analysis: Instances Misclassified by XLM-RoBERTa (Weighted) (Part 1: Rows 1–20)

Target Aspect	Review Text	True	VADER	SVM	XLM-R
Product Quality	My daughters bestfriend is the niece of the owner of this place and I'm a friend of both of my daughters bestfriends parents I have came here 2 times to eat, mainly because I live abroad I have came here with my wife's side of the family once once with just my wife and 3 of my wonderful daughters everytime I was there I was greeted by the owner and the food is amazing there my broth and sister in-law and there children are regulars there, sometimes their grandchildren go ther when there in town, all I have to say it's a great place to eat if you had a long day, to catch up with friends or just because it's nice	POS	POS	NEU	NEU
Product Quality	Food needs to be piping hot when served for a better experience.	NEG	POS	NEG	NEU
Product Quality	Tack this on to your list of matcha places when you find yourself in the South.	POS	NEU	POS	NEU
Product Quality	Love the matcha in cheese and salt, but most of the drinks are too sweet, thats why i request for no syrup/sugar or half when ordering.	NEU	POS	NEU	NEG
Product Quality	Both drinks were surprisingly not sweet which was actually just divine.	POS	POS	NEG	NEU
Product Quality	Settled on P179 12oz Hot Malunggay latte.	NEU	NEU	NEU	NEG
Price and Value	Worth the price tou pay for and so much more.	POS	POS	NEG	NEG
Product Quality	Tried their matcha, I did not even taste the matcha and was overpowered by sweetness and milk.	NEG	POS	NEG	NEU
Product Quality	This coffee tastes so bad!	NEG	NEG	POS	NEU
Customer Service	The manager was so accommodating giving me more information that I needed about the consummables and if how much would be the staycation, entrance and if letting the kids swim the pool...	POS	POS	POS	NEG
Store Operations and Accessibility	it was a cute cafe with limited seating but they are still under renovation and I think their main area isn't open yet.	POS	POS	NEG	NEG
Ambiance and Atmosphere	noisy staff	NEG	NEG	NEG	POS
Facilities and Amenities	Food is great, fast ang wifi and valid for four hours each voucher pagyud.	POS	POS	POS	NEG
Facilities and Amenities	Not ideal for work or study.	NEG	NEG	NEG	NEU
Customer Service	Goods Kai naii duol.. their staff krezel is ok too beaming_face_with_smiling_eyes .. wish they were open 24/7.. hehehe..	POS	POS	NEG	NEG
Product Quality	Best matcha in town white_heart I would definitely drink their matcha everyday.	POS	POS	POS	NEU
Store Operations and Accessibility	They really open at 7am.	POS	NEU	POS	NEU
Product Quality	Their coffee tastes great as well although some of them may be an acquired taste.	POS	POS	POS	NEU
Price and Value	Sure gyud nga high quality and wa gi tipid ilang ingredients.	POS	POS	NEG	NEG
Facilities and Amenities	No tables.	NEG	NEG	NEG	POS

Table 18: Error Analysis: Instances Misclassified by XLM-RoBERTa (Weighted) (Part 2: Rows 21–40)

Target Aspect	Review Text	True	VADER	SVM	XLM-R
Product Quality	Drinks are even better.	POS	POS	NEG	NEU
Product Quality	Cheers, purge coffee!	POS	POS	NEU	NEG
Product Quality	Im a coffee lover here and tries different shops and this is the shop where me and my wife will definitely buy if we pass by.	POS	POS	POS	NEU
Product Quality	Food, Drinks, Ambiance check_mark_button hundred_points	POS	NEU	POS	NEG
Product Quality	matcha also tastes good.	POS	POS	POS	NEU
Facilities and Amenities	Very limited parking space though.	NEG	NEG	POS	POS
Customer Service	but i wanted a glass cup instead of a paper cup to be eco-friendly.	NEU	NEU	NEU	NEG
Product Quality	Mas mukha ciang library kaysa coffee shop	NEU	NEU	NEU	NEG
Facilities and Amenities	Space inside is a bit small but there is enough room for everyone.	NEU	NEU	POS	POS
Product Quality	loudly_crying_face Medyo ma init na din since late na nga kami pumunta.	NEU	NEU	NEG	NEG
Ambiance and Atmosphere	Crowded most of the time, place is aesthetically pleasing, food and beverages are on the pricey side.	NEU	POS	NEU	POS
Product Quality	Not aromatic, but I support local coffee growers.	NEG	NEG	POS	NEU
Product Quality	Waiting for almost an hour now for a small latte and a slice of cake.	NEG	NEU	NEG	NEU
Customer Service	There were only two staff members at the counter, and one of them was handling both hot and iced drinks, so our orders took a while—understandable since the café was nearly full.	NEU	NEU	NEG	NEG
Price and Value	The best part is, it is higly affordable.	POS	POS	NEG	NEG
Facilities and Amenities	You don't have to fight for a seat near the wall or worry about your battery running out in the middle of work, which is something I appreciate.	POS	POS	NEU	NEU
Ambiance and Atmosphere	I don't know if it's because the customers base changed but it's just too noisy now (+ the loud music/bass).	NEG	NEG	NEG	POS
Product Quality	Anyway back to Cafe Demitasse....it is obvious to me that the Chief (s) have PASSION about food.	POS	POS	NEU	NEU
Store Operations and Accessibility	Real estate is a bit small and feels cramped, but there is a second floor open air area.	NEU	NEU	NEG	POS
Store Operations and Accessibility	Parking slots at the sides.	NEU	NEU	POS	POS

Strengths: Resolving Mixed Sentiments and Context
Tables 15 and 16 demonstrate the transformer's ability to decouple conflicting sentiments within the same review.

- **Targeted Aspect Extraction:** In the review "Food is really Good 10\10 but the staff looks like they are having holly week", both VADER and SVM predicted Positive because they anchored on the phrase "Good 10\10". XLM-RoBERTa correctly isolated the "Customer Service" aspect and flagged the localized complaint as Negative.
- **Contextual Negation:** The model accurately processes implicit complaints and emojis. It correctly labeled "Waited two hours" and "Small parking place loudly_crying_face" as Negative, situations where VADER defaulted to Neutral due to a lack of explicit negative keywords.

Weaknesses: Idioms, Sarcasm, and Neutral Forcing
Tables 17 and 18 highlight the model's remaining limitations when handling highly ambiguous or localized text.

- **Idiomatic Code-Switching:** The model struggles with regional double-negatives or slang. In "Sure gyud nga high quality and wa gi tipid ilang ingredients" (they did not skimp on ingredients), the model anchored on the Bisaya negation marker "wa" and incorrectly predicted Negative, missing the culturally positive context.
- **Forcing Polarity on Mixed Text:** While the weighted loss improved the Neutral F1-score, the model still struggles with perfectly balanced observations. In "Space inside is a bit small but there is enough room for everyone" or "Love the matcha... but most of the drinks are too sweet", the model fails to predict Neutral, instead tipping into Positive or Negative.
- **Sarcasm and Blunt Phrasing:** The model misinterprets blunt structural complaints and sarcasm. Short, objective negatives like "No tables.", "noisy staff", and "Not ideal for work or study" were inexplicably classified as Positive or Neutral, indicating that the model sometimes requires longer contextual padding to confidently trigger a Negative prediction.

4.5 Granular Inference and Insight Aggregation Sample

place_name	Shop_Type	rating	TargetAspect	sentence	Predicted_Sentiment
Establishment 1	Coffee	5	Price and Value	Sulit na siya!!!!	NEGATIVE
Establishment 1	Coffee	5	Store Operations and Accessibility	++ big parking sooo 5	POSITIVE
Establishment 1	Coffee	5	Product Quality	Great place and great food.	POSITIVE
Establishment 1	Coffee	5	Product Quality	The ambiance is nice, friendly staff a...	POSITIVE
Establishment 1	Coffee	5	Customer Service	The ambiance is nice, friendly staff a...	POSITIVE

Table 19: Sample Aspect-Based Sentiment Predictions

place_name	Shop_Type	Avg_Star_Rating	Ambiance	Service	Facilities	Price	Quality	Operations
Establishment 1	Coffee	4.8 Stars	100% Pos	84% Pos	100% Pos	60% Neg	93% Pos	100% Pos
Establishment 2	Coffee	4.4 Stars	82% Pos	88% Pos	100% Pos	100% Pos	81% Pos	100% Pos
Establishment 3	Coffee	4.5 Stars	86% Pos	88% Pos	67% Neg	No Mentions	81% Pos	100% Pos
Establishment 4	Coffee	4.3 Stars	100% Pos	88% Pos	50% Pos	100% Neg	64% Pos	50% Pos
Establishment 5	Coffee	4.6 Stars	100% Pos	92% Pos	100% Pos	67% Pos	88% Pos	No Mentions

Table 20: Sample Aspect-Based Sentiment Summary by Establishment

Tables 19 and 20 present a sample of the system's final output to illustrate the translation of unstructured text into operational market intelligence, while also highlighting edge cases in localized sentiment classification. As indicated in the sample predictions (Table 19), the model misclassified the phrase "Sulit na siya!!!!" as a negative sentiment under the Price and Value dimension. In Philippine vernacular, "sulit" is an explicitly positive descriptor denoting high value or worth for the money spent. The error distorted the metrics for Establishment 1 in Table 20, driving the Negative proportion for Price and Value to 60% despite the source review presenting a 5-star rating. While the transformer architecture generalizes across general code-switched syntax, this sample demonstrates how specific lexical items with localized semantic polarity can still trigger false negatives that skew business-level aggregations. The summary applies a "No Mentions" label (observed under Price for Establishment 3 and Operations for Establishment 5) to indicate dimensions where the extraction algorithm detected zero relevant textual instances in the corpus. This convention explicitly accounts for missing data rather than assigning arbitrary default scores to unmentioned aspects.

Chapter 5

Conclusions and Recommendations

5.1 Conclusions

This study successfully addresses all five stated objectives that traditional rating systems and monolithic sentiment analysis methods fail to do. Due to the unstructured, multilingual, and heavily code-switched nature of Davao City's coffee and matcha shop consumer reviews, the study developed and validated a fine-tuned XLM-RoBERTa-based Aspect-Based Sentiment Analysis (ABSA) pipeline specifically tailored for Davao City's growing coffee and matcha market. The specific objectives outlined in Chapter 1 were achieved through the following findings:

1. **Identification of Aspect Categories and Impact of Code-Switching:** Through generative aspect taxonomy induction, the study formalized a localized master dictionary spanning six core operational dimensions: Product Quality, Customer Service, Ambiance and Atmosphere, Facilities and Amenities, Store Operations and Accessibility, and Price and Value. Structural analysis revealed that over 54% of local reviews evaluate two or more aspects simultaneously while blending regional Bislish and Taglish syntax (e.g., pairing Bisaya modifiers like *lami*, *dugay*, or *wa gi tipid* with English operational nouns). This high aspect density and mid-sentence language switching cause severe polarity cancellation in document-level classifiers, confirming the necessity of sentence-pair decomposition and cross-lingual subword tokenization, even as regional double-negatives and blunt complaints remain challenging edge cases.

2. **Performance Comparison between XLM-RoBERTa and Lexicon Baselines:** The fine-tuned XLM-RoBERTa model achieved a Macro-F1 score of 0.5967 and an overall accuracy of 79.75% on the 400-instance human-annotated test set. It significantly outperformed both the static rule-based VADER lexicon (Macro-F1: 0.4985, Accuracy: 68.75%) and the tuned SVM with TF-IDF baseline (Macro-F1: 0.5187, Accuracy: 73.00%). McNemar's test confirmed that XLM-RoBERTa's performance superiority over VADER ($\chi^2 = 17.93, p = 2.30 \times 10^{-5}$) and SVM ($\chi^2 = 10.57, p = 1.15 \times 10^{-3}$) was statistically significant at $\alpha = 0.05$. This proves that contextual cross-lingual representations significantly outperform static English

dictionaries and rigid n-gram baselines on informal, code-switched consumer text.

3. **Discovered Dimensions and Prevailing Consumer Sentiments:** Corpus-wide aspect-level analysis revealed distinct consumer priorities across Davao City cafés: Product Quality dominated overall discussion (47.88%), followed by Customer Service (20.00%), Ambiance and Atmosphere (13.38%), Facilities and Amenities (8.32%), Store Operations and Accessibility (5.24%), and Price and Value (5.18%). Prevailing sentiment trends showed that consumers were most consistently satisfied with Ambiance and Atmosphere (4.463 mean rating, SD = 0.911), whereas Customer Service exhibited the lowest satisfaction and highest polarization (4.005 mean rating, SD = 1.444), driven by complaints regarding wait times and staff attentiveness. Furthermore, segmented analysis demonstrated that matcha shop patrons prioritize Product Quality significantly more (59.30%) than coffee shop patrons (46.64%).

4. **Active Learning and Synthetic Data Augmentation:** Uncertainty and diversity sampling via Active Learning reduced manual labeling effort to 500 high-value instances while improving inter-annotator reliability from $\kappa = 0.6550$ on the initial seed batch to $\kappa = 0.7428$ on the queried batch. Targeted synthetic augmentation via Gemini 3 generated 214 TF-IDF-filtered instances to resolve severe minority deficits in rare complaint categories (Price and Value, Store Operations, Facilities, and Ambiance). Capping synthetic integration at 29.97% of the 714-row training baseline successfully stabilized minority decision boundaries while preventing generative model collapse.

5. **Model Evaluation and System Integration:** The validated pipeline was evaluated on a strictly isolated 400-instance gold-standard test firewall and integrated into a directory-style web interface. By translating granular aspect-level predictions into aspect-specific satisfaction percentages (e.g., contrasting Product Quality approval against Parking or Wi-Fi complaints), the system replaces ambiguous aggregate star ratings with actionable market intelligence for both Davao City consumers and café owners.

5.2 Future Work

To establish a rigorous roadmap for further research in low-resource, code-switched Aspect-Based Sentiment Analysis (ABSA), this section outlines the empirical, architectural, and methodological constraints encountered in the current pipeline, followed by concrete technical extensions designed to resolve them.

5.2.1 Limitations of the Study

1. **Data Scarcity and the Annotation Bottleneck:** There are no pre-existing gold-standard datasets for localized Davao City consumer reviews. The pipeline relied entirely on a heavily constrained pool of 500 authentic human-annotated instances. Manually labeling complex, code-switched Taglish and Bislish text is highly resource-intensive. This severe data bottleneck inherently restricted the

transformer's ability to map rare edge cases and localized slang without relying on aggressive synthetic augmentation.

2. **Severe Class Imbalance and the Minority Boundary Trade-Off:** Real-world consumer feedback skews heavily toward positive endorsements. Negative complaints are sparse. Neutral observations are exceedingly rare. During unweighted fine-tuning, the transformer suffered representation collapse on the neutral class, failing to delineate the neutral boundary and forcing ambiguous feedback into polarized categories. Applying inverse balanced class weights forced the optimizer to recognize the neutral decision space, improving macro-averaged performance. This approach, however, penalized borderline predictions and inflated validation loss. It demonstrated an unavoidable calibration trade-off when optimizing for minority class recall on sparse data.

3. **Parameter Capacity vs. Low-Resource Domain Scale:** Adapting a large multilingual transformer to a compact domain-specific dataset risked parameter memorization. Freezing the embedding layer and lower encoder blocks prevented catastrophic overfitting and stabilized gradient updates. This structural restriction limited lower-level syntactic adaptation. The network remains vulnerable to misinterpreting regional Bisaya negation structures, double negatives, and terse operational complaints that rely on localized syntax.

4. **Lexical Regex Anchoring in Candidate Extraction (Phase 3):** Scaling candidate extraction across the unannotated corpus relied on dictionary-driven regular expressions with word boundary anchors. Flat lexical scanning lacks syntactic dependency awareness. Aspect keywords inside incidental clauses, store policies, or comparative narratives were incorrectly extracted as candidate pairs. Human adjudication filtered this noise from the training and evaluation partitions. Unannotated reviews processed during bulk inference remain vulnerable to false-positive aspect extraction.

5. **Static Lexicon Filtering During Synthetic Augmentation (Phase 5):** Sparse lexical vector similarity filtering blocked exact word repetition and verbatim plagiarism. It completely failed to detect structural or syntactic homogenization. Synthetic instances using different regional synonyms within identical sentence templates bypassed the filter. This severely constrained the true syntactic diversity of the augmented minority classes.

6. **Single-Seed Evaluation:** Single random seed (seed = 42) was fixed across PyTorch, NumPy, and Python for optimization and checkpoint selection due to memory constraints on the 8 GB RTX 3050 GPU. Fine-tuning a 270M-parameter transformer on a modest 714-row training dataset creates acute sensitivity to classification head initialization and mini-batch ordering. Without multi-seed evaluations or k-fold cross-validation, the reported test metrics (0.5967 Macro-F1; 79.75% accuracy) lack empirical confidence intervals and variance estimates. This limitation is particularly pronounced for the underrepresented Neutral class ($N = 33$ in the test partition).

5.2.2 Recommendations and Technical Extensions

1. **Programmatic Labeling and Few-Shot LLM Annotation:** To overcome the human annotation bottleneck and scale the training corpus, future frameworks should integrate data programming paradigms. Utilizing strong generative language models as few-shot oracle annotators can programmatically label thousands of scraped reviews at a fraction of the cost. Human effort can then be strictly redirected to auditing low-confidence generative outputs, expanding the authentic training corpus from hundreds to tens of thousands of rows.

2. **Hierarchical Classification and Multi-Class Focal Loss:** To resolve neutral class instability without introducing calibration drift from extreme static weights, pipelines should decouple subjectivity detection from polarity classification. A primary binary classifier isolates neutral and objective statements, routing only polarized reviews to a secondary positive-negative classifier.

   Single-stage architectures should replace static class-weighted cross-entropy with multi-class focal loss and temperature scaling. This dynamically penalizes hard minority errors without destabilizing prediction probabilities on well-classified examples.

3. **Efficient Fine-Tuning and Augmentation:** Instead of freezing lower encoder blocks, implementations should apply Low-Rank Adaptation across attention projection matrices in all encoder layers. This constrains the trainable parameter footprint while allowing full-depth contextual representations to adapt to regional code-switching. To resolve classification errors on localized double negatives, synthetic pipelines must generate contrastive minimal pairs. Inverting sentiment through single regional negation markers forces attention heads to directly map negation scope.

4. **Span-Based Aspect-Sentiment Extraction:** An end-to-end aspect-sentiment triplet extraction framework or an instruction-tuned generative extractor should replace the dictionary scanner to eliminate regex-induced false positives. Conditioning extraction on syntactic dependency paths ensures aspect categories are identified only when tied directly to an active opinion modifier.

5. **Dense Semantic Deduplication via Multilingual Embeddings:** Augmentation workflows must implement dense semantic deduplication to overcome the structural limitations of sparse lexical filtering. Encoding synthetic candidates and authentic seed data using cross-lingual sentence embeddings allows systems to evaluate similarity in latent semantic space. Discarding structurally redundant generations enforces authentic syntactic diversity.

6. **Multi-Seed Ensembling:** Subsequent evaluations should substitute single-seed trials with multi-seed runs across 5 to 10 distinct initializations to report mean performance alongside standard deviations.

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Appendix A

Appendices

```
const cleanText = (raw) => (raw || '')
  .replace(/[\r\n]+/g, ' ')
  .replace(/[\t\v\f\u200B-\u200D\uFEFF]/g, ' ')
  .replace(/[\x00-\x1F\x7F-\x9F]/g, '')
  .replace(/[\uE000-\uF8FF]/g, '')
  .replace(/\s+/g, ' ')
  .trim();

const OWNER_REPLY_PATTERN = /response from (the )?owner/i;
const isOwnerReplyText = (el) => {
  let node = el;
  for (let i = 0; i < 5 && node; i++) {
    const label = node.getAttribute?.('aria-label') || '';
    const prevSiblingText = node.previousElementSibling?.innerText || '';
    if (OWNER_REPLY_PATTERN.test(label) || OWNER_REPLY_PATTERN.test(prevSiblingText)) return true;
    node = node.parentElement;
  }
  return false;
};
```

Figure 17: *DOM Scraping Script*

```
sv_path = Path(r"D:\Ateneo de Davao\Thesis-Project-Aspect-Based-Sentiment-Analysis\Dataset\Thesis_master_reviews_combined.
f = pd.read_csv(csv_path, encoding="utf-8-sig")

Use review_text directly
ef convert_emojis_to_text(text):
    if pd.isna(text):
        return ""
    return emoji.demojize(str(text), delimiters=(" ", " "))

Generate clean_review from review_text
f["clean_review"] = df["review_text"].apply(convert_emojis_to_text)

Save back to Dataset
f.to_csv(csv_path, index=False, encoding="utf-8-sig")
rint("Emoji conversion complete. Generated 'clean_review'.")
```

Figure 18: *Emoji Normalization and Text Demojization Pipeline*

```
pattern = r'\b(' + '|'.join(local_markers) + r')\b'
df['code_switched'] = df['clean_review'].str.contains(pattern, case=False, na=False)

coffee_local = df[(df['shop_type'] == 'coffee') & (df['code_switched'] == True)]
coffee_eng = df[(df['shop_type'] == 'coffee') & (df['code_switched'] == False)]
matcha_local = df[(df['shop_type'] == 'matcha') & (df['code_switched'] == True)]
matcha_eng = df[(df['shop_type'] == 'matcha') & (df['code_switched'] == False)]

# Stratified Dynamic Sampling
target_per_type = 250

n_coffee_local = min(125, len(coffee_local))
n_coffee_eng = target_per_type - n_coffee_local
samp_coffee_local = coffee_local.sample(n=n_coffee_local, random_state=42)
samp_coffee_eng = coffee_eng.sample(n=min(n_coffee_eng, len(coffee_eng)), random_state=42)

n_matcha_local = min(125, len(matcha_local))
n_matcha_eng = target_per_type - n_matcha_local
samp_matcha_local = matcha_local.sample(n=n_matcha_local, random_state=42)
samp_matcha_eng = matcha_eng.sample(n=min(n_matcha_eng, len(matcha_eng)), random_state=42)

samples = [samp_coffee_local, samp_coffee_eng, samp_matcha_local, samp_matcha_eng]
stratified_sample = pd.concat(samples).sample(frac=1, random_state=42).reset_index(drop=True)
```

Figure 19: *Stratified Sampling*

```python
#Verified
aspect_dictionary = {
    "Product Quality": [
        "coffee", "matcha", "latte", "espresso", "sikwate", "barako",
        "pandesal", "pastry", "cake", "taste", "lami", "masarap", "lasa",
        "pait", "tam-is", "umay", "lamiay", "food", "drinks", "isog",
        "lasaw", "tab-ang", "luod", "puol", "init", "bugnaw"
    ],
    "Customer Service": [
        "barista", "staff", "crew", "cashier", "service", "mabait", "sungit",
        "accommodating", "suplada", "paspas", "dugay", "maayos", "attitude",
        "hinay", "buotan", "snob", "taray", "friendly"
    ],
    "Ambiance and Atmosphere": [
        "ambiance", "vibe", "aesthetic", "chill", "quiet", "noisy", "saba",
        "sabaan", "mingaw", "music", "tugtog", "muni-muni", "chika", "tropa",
        "atmosphere", "crowded", "samok", "ig-worthy", "instagrammable",
        "ngitngit", "tambay"
    ],
    "Facilities and Amenities": [
        "wifi", "internet", "connection", "socket", "outlet", "plug", "cr",
        "restroom", "comfort room", "aircon", "tables", "chairs", "space",
        "study", "tuon", "work", "laptop", "hina", "kusog", "tugnaw", "igang"
    ],
    "Price and Value": [
        "price", "value", "mahal", "barato", "sulit", "tipid", "overpriced",
        "affordable", "sayang", "budget", "expensive", "cheap", "kamahal", "pricy"
    ],
    "Store Operations and Accessibility": [
        "parking", "parkingan", "location", "accessible", "layo", "malayo",
        "duol", "hours", "open", "close", "traffic", "tago"
    ]
}
```

Figure 20: *Multilingual Aspect Lexicon Dictionary and Seed Keyword Definitions*

```python
#Phase 3.1: Targeted Scanning for Aspect Extraction

# prevent substring mismatches
compiled_patterns = {
    aspect: re.compile(r'\b(?:' + '|'.join(map(re.escape, keywords)) + r')\b')
    for aspect, keywords in aspect_dictionary.items()
}

def target_scan(text):
    if not isinstance(text, str):
        return []
    text = text.lower()
    return [aspect for aspect, pattern in compiled_patterns.items() if pattern.search(text)]

# 'df' and text is in 'clean_review'
# 1. Tokenize reviews into sentences (with strict string validation to bypass rogue floats/NaNs)
df['sentence'] = df['clean_review'].apply(lambda x: sent_tokenize(x) if isinstance(x, str) else [])

# 2. Explode the sentences into distinct rows
df_sentences = df.explode('sentence').reset_index(drop=True)

# 3. Execute Step 3.1: Targeted Scanning
df_sentences['TargetAspect'] = df_sentences['sentence'].apply(target_scan)

# 4. Clean up: Drop sentences where no aspect was detected
df_scanned = df_sentences[df_sentences['TargetAspect'].astype(bool)].copy()

# Step 3.2: Data Explosion (Isolating Review-Aspect Pairs)
df_exploded = df_scanned.explode('TargetAspect').reset_index(drop=True)
```

Figure 21: *Targeted Lexical Aspect Scanning and Sentence-Aspect Pair Explosion*

```python
def assign_rating_bucket(rating):
    r = int(rating)
    if r <= 2:
        return '1-2_stars'
    elif r == 3:
        return '3_stars'
    return '4-5_stars'

def draw_stratified_evaluation_sample(input_csv, sample_size=400, seed=SEED):
    df = pd.read_csv(input_csv)

    # Create joint stratification key
    df['rating_bucket'] = df['rating'].apply(assign_rating_bucket)
    df['stratum'] = df['TargetAspect'].astype(str) + " | " + df['rating_bucket']

    # Exact stratified draw
    sample_df, _ = train_test_split(
        df,
        train_size=sample_size,
        stratify=df['stratum'],
        random_state=seed,
        shuffle=True
    )

    # Prepare annotation columns for 3 independent raters
    base_cols = ['place_name', 'rating', 'shop_type', 'sentence', 'TargetAspect']
    export_df = sample_df[base_cols].copy().reset_index(drop=True)
    export_df['annotator_1'] = np.nan
    export_df['annotator_2'] = np.nan
    export_df['annotator_3'] = np.nan

    return export_df
```

Figure 22: *Stratified Evaluation Sampling*

```python
# N = 400 instances, k = 3 categories (POSITIVE, NEUTRAL, NEGATIVE)
kappa_matrix = np.zeros((len(df), len(CATEGORIES)), dtype=int)
for i, cat in enumerate(CATEGORIES):
    kappa_matrix[:, i] = (df[rater_cols] == cat).sum(axis=1)

kappa_score = fleiss_kappa(kappa_matrix, method='fleiss')


if kappa_score > 0.80:
    interpretation = "Almost Perfect Agreement"
elif kappa_score > 0.60:
    interpretation = "Substantial Agreement"
elif kappa_score > 0.40:
    interpretation = "Moderate Agreement"
else:
    interpretation = "Fair / Poor Agreement"


def get_consensus(row):
    votes = [row['annotator_1'], row['annotator_2'], row['annotator_3']]
    val_counts = pd.Series(votes).value_counts()
    if val_counts.iloc[0] >= 2:
        return pd.Series([val_counts.index[0], False, f"{val_counts.iloc[0]}/3"])
    else:
        # 3-way tie fallback (1 POS, 1 NEU, 1 NEG)
        return pd.Series([row['annotator_1'], True, "1/1/1_tie"])

df[['ground_truth', 'is_tie', 'vote_breakdown']] = df.apply(get_consensus, axis=1)
```

Figure 23: *Fleiss' Kappa Computation and Majority Voting Consensus*

```python
# 3. Define the Plotting Function
def plot_learning_curve(df_train, df_val, target_class, batch_sizes):
    f1_scores = []

    y_train_binary = (df_train['TargetClass'] == target_class).astype(int)
    y_val_binary = (df_val['TargetClass'] == target_class).astype(int)

    print(f"Total augmented instances for '{target_class}': {y_train_binary.sum()}")

    proxy_model = make_pipeline(
        TfidfVectorizer(max_features=5000, ngram_range=(1,2)),
        LogisticRegression(max_iter=1000, class_weight='balanced', random_state=42)
    )

    valid_batch_sizes = [size for size in batch_sizes if size <= len(df_train)]

    for size in valid_batch_sizes:
        df_subset = resample(df_train, n_samples=size, replace=False, stratify=y_train_binary, random_state=42)
        X_subset = df_subset['sentence'].fillna('')
        y_subset = (df_subset['TargetClass'] == target_class).astype(int)

        proxy_model.fit(X_subset, y_subset)
        y_pred = proxy_model.predict(df_val['sentence'].fillna(''))

        score = f1_score(y_val_binary, y_pred, zero_division=0)
        f1_scores.append(score)

# 4. Evaluate the Augmented Minority Classes
batch_sizes = [200, 500, 1000, 2000, 4000, 6000, 8000, 10000]

plot_learning_curve(df_train_aug, df_val, 'Ambiance and Atmosphere - Negative', batch_sizes)
plot_learning_curve(df_train_aug, df_val, 'Store Operations and Accessibility - Negative', batch_sizes)
plot_learning_curve(df_train_aug, df_val, 'Price and Value - Negative', batch_sizes)
plot_learning_curve(df_train_aug, df_val, 'Facilities and Amenities - Negative', batch_sizes)
```

Figure 24: *Binary Proxy Learning Curve Generator*

```
# Step 5.3: Lexical Redundancy Filtering (cosine >= 0.85 discarded)
vectorizer = TfidfVectorizer(max_features=5000, ngram_range=(1, 2))
vectorizer.fit(pd.concat([sentences_auth, sentences_synth]))
tfidf_auth  = vectorizer.transform(sentences_auth)
tfidf_synth = vectorizer.transform(sentences_synth)

# Filter 1: synthetic vs authentic
max_sim_auth = cosine_similarity(tfidf_synth, tfidf_auth).max(axis=1)
mask_keep_auth = max_sim_auth < 0.85
df_synth_filtered = df_synth[mask_keep_auth].reset_index(drop=True)
tfidf_synth_filtered = tfidf_synth[mask_keep_auth]

# Filter 2: synthetic vs synthetic (internal dedup)
sim_synth_synth = cosine_similarity(tfidf_synth_filtered)
upper_tri = np.triu(sim_synth_synth, k=1)
to_drop_internal = upper_tri.max(axis=0) >= 0.85
df_synth_final = df_synth_filtered[~to_drop_internal].reset_index(drop=True)
```

Figure 25: *TF-IDF Cosine Similarity Thresholding*

```python
df_synth_clean['is_synthetic'] = True

# 3. Align columns (fills missing synthetic columns with blanks)
for col in df_auth.columns:
    if col not in df_synth_clean.columns:
        df_synth_clean[col] = ''

# Force matching column order
df_synth_clean = df_synth_clean[df_auth.columns]

# 4. Concatenate and strictly shuffle the dataset
df_augmented = pd.concat([df_auth, df_synth_clean], ignore_index=True)
df_augmented = df_augmented.sample(frac=1, random_state=42).reset_index(drop=True)

# 5. Export the final augmented training dataset
df_augmented.to_csv('Train_Set_70_Augmented.csv', index=False, encoding='utf-8')

print(f"Authentic Train Set: {len(df_auth)} rows")
print(f"Synthetic Rows Injected: {len(df_synth_clean)} rows")
print(f"Final Augmented Train Set: {len(df_augmented)} rows")
```

Figure 26: *Column Alignment, Dataset Concatenation, and Augmented Set Export*

```python
df_gt = pd.read_csv('Ground_Truth_500.csv')
df_train = pd.read_csv('Train_Set_70_Augmented.csv')


df_synthetic_pool = df_train[
    (df_train['is_synthetic'] == True) &
    (df_train['sentiment'] == 'NEGATIVE')
].copy()

# Define the absolute hard limits (Total = 214)
injection_targets = {
    "Ambiance and Atmosphere": 53,
    "Facilities and Amenities": 53,
    "Price and Value": 54,
    "Store Operations and Accessibility": 54
}


sampled_list = []
for aspect, quota in injection_targets.items():

    subset = df_synthetic_pool[
        (df_synthetic_pool['TargetAspect'] == aspect) |
        (df_synthetic_pool['TargetAspect'] == aspect.split()[0])
    ]
    assert len(subset) >= quota, f"Not enough synthetic samples for '{aspect}'. Found {len(subset)}, needed {quota}."
    sampled_list.append(subset.sample(n=quota, random_state=42))

df_synthetic_sampled = pd.concat(sampled_list, ignore_index=True)


df_synthetic_sampled['ground_truth'] = 'NEGATIVE'
df_synthetic_sampled['is_tie'] = False
df_synthetic_sampled['vote_breakdown'] = 'SYNTHETIC_GENERATION'
```

Figure 27: *Targeted Synthetic Data Injection and Negative Class Quota Balancing*

```python
# Strict Label Encoding
label2id = {'POSITIVE': 0, 'NEUTRAL': 1, 'NEGATIVE': 2}
id2label = {0: 'POSITIVE', 1: 'NEUTRAL', 2: 'NEGATIVE'}
train_df['encoded_label'] = train_df['sentiment'].astype(str).str.upper().map(label2id)

MODEL_NAME = 'xlm-roberta-base'
tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)

class ABSADataset(Dataset):
    def __init__(self, df, tokenizer, max_length=128):
        self.encodings = tokenizer(
            df['TargetAspect'].astype(str).tolist(),
            df['sentence'].astype(str).tolist(),
            truncation=True,
            padding=True,
            max_length=max_length,
            return_tensors='pt'
        )
        self.labels = torch.tensor(df['encoded_label'].tolist(), dtype=torch.long)

    def __len__(self):
        return len(self.labels)

    def __getitem__(self, idx):
        item = {key: val[idx] for key, val in self.encodings.items()}
        item['labels'] = self.labels[idx]
        return item

train_dataset = ABSADataset(train_df, tokenizer)
val_dataset = ABSADataset(val_df, tokenizer)
test_dataset = ABSADataset(test_df, tokenizer)
print("Datasets tokenized and converted to PyTorch Datasets.")
```

Figure 28: *PyTorch Dataset Class and XLM-RoBERTa Tokenization Pipeline*

```python
y_train = train_df['encoded_label'].values

class_weights = compute_class_weight(
    class_weight='balanced',
    classes=np.array([0, 1, 2]),
    y=y_train
)

weights_tensor = torch.tensor(class_weights, dtype=torch.float32).to(device)

print("Calculated Balanced Class Weights:")
for idx, name in id2label.items():
    print(f"  Class {idx} ({name}): {weights_tensor[idx].item():.4f}")

# Loss Function with Weighted Cross-Entropy and No Label Smoothing
loss_fn = nn.CrossEntropyLoss(weight=weights_tensor, label_smoothing=0.0)
print(f"Loss Function Initialized: {loss_fn} (label_smoothing=0.0)")
```

Figure 29: *Balanced Class Weight Computation*

```python
# Grid Search Parameters (As defined in Phase 7.4)
LEARNING_RATES = [2e-5, 3e-5, 5e-5]
BATCH_SIZES = [16, 32]
EPOCHS = 5
WEIGHT_DECAY = 0.1
LABEL_SMOOTHING = 0.1

grid_results = []

print("Starting Grid Search... Testing 6 configurations.\n")

for bs in BATCH_SIZES:
    for lr in LEARNING_RATES:
        print(f"--- Training: Batch Size {bs}, Learning Rate {lr} ---")

        # 1. Prepare DataLoaders
        set_seed(42) # Ensure reproducible shuffling for fair comparison
        gs_train_loader = DataLoader(train_dataset, batch_size=bs, shuffle=True)
        gs_val_loader = DataLoader(val_dataset, batch_size=bs, shuffle=False)

        # 2. Reset Model Weights (To avoid transferring knowledge between runs)
        model = XLMRobertaForSequenceClassification.from_pretrained(
            MODEL_NAME, num_labels=3, id2label=id2label, label2id=label2id,
            hidden_dropout_prob=0.1, attention_probs_dropout_prob=0.1
        ).to(device)

        # Apply Layer Freezing
        for param in model.roberta.embeddings.parameters():
            param.requires_grad = False
        for i in range(6):
            for param in model.roberta.encoder.layer[i].parameters():
                param.requires_grad = False
```

Figure 30: *Hyperparameter Grid Search Setup and Layer Freezing*

```python
# 3. Reset Optimizer & Scheduler
optimizer = AdamW(filter(lambda p: p.requires_grad, model.parameters()), lr=lr,weight_decay=WEIGHT_DECAY)
total_steps = len(gs_train_loader) * EPOCHS
warmup_steps = int(0.1 * total_steps)
scheduler = get_linear_schedule_with_warmup(optimizer, num_warmup_steps=warmup_steps,num_training_steps=total_steps)
loss_fn = CrossEntropyLoss(label_smoothing=LABEL_SMOOTHING)
best_val_f1 = 0.0

# 4. Training Loop
for epoch in range(EPOCHS):
    model.train()
    for batch in gs_train_loader:
        optimizer.zero_grad()
        outputs = model(batch['input_ids'].to(device), attention_mask=batch['attention_mask'].to(device))
        loss = loss_fn(outputs.logits, batch['labels'].to(device))
        loss.backward()
        torch.nn.utils.clip_grad_norm_(model.parameters(), max_norm=1.0)
        optimizer.step()
        scheduler.step()

    # Validation phase
    model.eval()
    xlmr_preds, true_labels_gs = [], []
    with torch.no_grad():
        for batch in gs_val_loader:
            outputs = model(batch['input_ids'].to(device), attention_mask=batch['attention_mask'].to(device))
            preds = torch.argmax(outputs.logits, dim=1).cpu().numpy()
            xlmr_preds.extend(preds)
            true_labels_gs.extend(batch['labels'].to(device).cpu().numpy())

    val_f1 = f1_score(true_labels_gs, xlmr_preds, average='macro', zero_division=0)
    if val_f1 > best_val_f1:
        best_val_f1 = val_f1

print(f"-> Peak Validation Macro-F1: {best_val_f1:.4f}\n")
grid_results.append({'Batch Size': bs, 'Learning Rate': lr, 'Peak Macro-F1': best_val_f1})
```

Figure 31: *Transformer Training Loop and Validation Macro-F1 Evaluation*

```python
# vader lexicon
nltk.download('vader_lexicon', quiet=True)
sia = SentimentIntensityAnalyzer()
def predict_vader(text):
    score = sia.polarity_scores(str(text))['compound']
    if score >= 0.05: return 0      # Positive
    elif score <= -0.05: return 2   # Negative
    else: return 1                  # Neutral
vader_preds = val_df['sentence'].apply(predict_vader).tolist()

# SVM with TF-IDF
vectorizer = TfidfVectorizer(max_features=5000)
X_train_svm = vectorizer.fit_transform(train_df['sentence'])
X_val_svm = vectorizer.transform(val_df['sentence'])

svm_model = SVC(kernel='linear')
svm_model.fit(X_train_svm, train_df['encoded_label'])
svm_preds = svm_model.predict(X_val_svm).tolist()
```

Figure 32: *VADER Lexicon and Unconditioned TF-IDF SVM Baseline Setup*

```python
#Retained ABSA pairin
train_inputs_svm = train_df['TargetAspect'].astype(str) + " " + train_df['sentence'].astype(str)
test_inputs_svm = test_df['TargetAspect'].astype(str) + " " + test_df['sentence'].astype(str)

#Hyperparameter Search Space
pipeline = Pipeline([
    ('tfidf', TfidfVectorizer()),
    ('svm', SVC(kernel='linear', random_state=42))
])

param_grid = {
    'tfidf__ngram_range': [(1, 1), (1, 2)],
    'tfidf__max_features': [3000, 5000],
    'svm__C': [0.1, 1.0, 10.0],
    'svm__class_weight': [None, 'balanced']
}

#5-Fold Cross-Validation on Macro-F1
print("Tuning Aspect-Conditioned SVM...")
grid_svm = GridSearchCV(pipeline, param_grid, cv=5, scoring='f1_macro', n_jobs=-1)
grid_svm.fit(train_inputs_svm, train_df['encoded_label'])

print(f"Optimal SVM Parameters: {grid_svm.best_params_}")
print(f"Best CV Macro-F1: {grid_svm.best_score_:.4f}\n")

# Test Set
test_df['svm_pred'] = grid_svm.predict(test_inputs_svm)

print("Tuned SVM Test Set Results:")
print(classification_report(test_df['encoded_label'], test_df['svm_pred'],
                            target_names=["POSITIVE", "NEUTRAL", "NEGATIVE"], digits=4))
```

Figure 33: *Aspect-Conditioned SVM Pipeline and Cross-Validation Hyperparameter Tuning*

```python
#Final Benchmark Table
models = {'VADER': test_df['vader_pred'], 'SVM (TF-IDF)': test_df['svm_pred'], 'XLM-RoBERTa':
test_df['xlmr_pred']}
results = []
for name, preds in models.items():
    f1_class = f1_score(true_labels, preds, average=None, zero_division=0)
    results.append({
        'Model': name,
        'Accuracy': accuracy_score(true_labels, preds),
        'Precision (Macro)': precision_score(true_labels, preds, average='macro', zero_division=0),
        'Recall (Macro)': recall_score(true_labels, preds, average='macro', zero_division=0),
        'Macro-F1': f1_score(true_labels, preds, average='macro', zero_division=0),
        'F1 (POS)': f1_class[0],
        'F1 (NEU)': f1_class[1],
        'F1 (NEG)': f1_class[2]
    })
benchmark_df = pd.DataFrame(results).round(4)
display(benchmark_df)

# McNemar's Test
def run_mcnemar(base_preds):
    b = sum((x == t) and (b != t) for x, b, t in zip(test_df['xlmr_pred'], base_preds, true_labels))
    c = sum((x != t) and (b == t) for x, b, t in zip(test_df['xlmr_pred'], base_preds, true_labels))
    table = [[0, c], [b, 0]] # We only care about b and c for McNemar calculation
    result = mcnemar(table, exact=False, correction=False)
    return b, c, result.statistic, result.pvalue
```

Figure 34: *Final Model Benchmark Evaluation and McNemar's Statistical Significance Test*

```
gold = test_df['encoded_label']
xlm_wins = test_df[
    (test_df['xlmr_pred'] == gold) &
    (test_df['vader_pred'] != gold) &
    (test_df['svm_pred'] != gold)
][['TargetAspect', 'sentence', 'sentiment', 'vader_pred', 'svm_pred', 'xlmr_pred']].copy()

for col in ['vader_pred', 'svm_pred', 'xlmr_pred']:
    xlm_wins[col] = xlm_wins[col].map(id2label)

print(f"\nFound {len(xlm_wins)} test cases where XLM-RoBERTa was correct and BOTH baselines failed.\n")
display(xlm_wins.head(10))
```

Figure 35: *XLM-RoBERTa Error Analysis*